



**UNIVERSITÀ
DI TORINO**

University of Turin

Master's Degree in Language Technologies and Digital Humanities

Culturally Adaptive Explainable LLM Assessment for Multilingual Information Disorder

Supervisor:

Prof. Viviana Patti

Candidate:

Maziar Kianimoghadam Jouneghani

Student ID: **1116595**

Co-Supervisor:

Dr. Marco Antonio Stranisci

Counter-Rapporteur:

Prof. Valerio Basile

Academic Year 2025/2026

Acknowledgements

This work would not exist without the people who made it possible, and I want to take a moment to acknowledge them from my heart.

To my supervisor, Professor Viviana Patti, thank you for your guidance, your patience, and for encouraging me to think deeper and reach further. You gave me the freedom to explore while never letting me lose direction. To my co-supervisor, Dr. Marco Antonio Stranisci, thank you not only for your academic support throughout this thesis but for the opportunity you gave me through the internship at Aequa-Tech, which planted the seeds of this entire research direction and led, in ways I could not have anticipated, to everything that followed. Both of you were far more than supervisors during my time in Italy. You were a reminder that good people exist everywhere, and that kindness does not stop at professional boundaries.

To Professor Valerio Basile, one of the finest teachers I have encountered in my academic life: thank you. Your course, your way of thinking, and your genuine enthusiasm for the field shaped how I approach research in ways I am still discovering. Your support, whenever I needed it, never went unnoticed. I will never forget the first time I reached out for an academic concern and your response was let's have a coffee about it! It has been an honor to learn from you, and more deeply, to know you.

To Italy, a country that welcomed me and gave me the chance to grow, to study, and to become a better version of myself: thank you. I came here as a student and I leave this course with far more than a degree. The person who started this course and the one finishing it are not quite the same. A few white hairs and a small patch of white beard have appeared along the way, and more importantly, a slightly different way of looking at things; at research, at distance, at what actually matters.

This thesis was built over more than six months of work that was, at times, genuinely hard to sustain. Not because of the research itself, but because of everything happening around it. Being far from home, from Iran, from the people I love, and carrying the weight of always thinking about them during difficult times made focus a luxury thing I did not always have. I was under constant pressure during this time. And yet, here it is. Finished. I am proud of that.

To my friends, those who supported me in any form throughout my studies, and who reminded me that life exists outside of a thesis. To my mistakes, which I embrace as lessons learned and layers added to who I am.

To the people who voluntarily gave their time to participate in the human evaluation phase of this work, most of whom I have never met: your contribution was genuine and it made this research more human in every sense of the word.

Finally, I dedicate this study, its design, its implications, its failures, and its successes to the people of Iran. My country.

Summary

Information disorder, understood as the deliberate or incidental spread of misleading, manipulative, or decontextualized content, does not manifest the same way across languages and cultures. The rhetorical strategies, ideological markers, and framing patterns that signal bias in one linguistic community may read as neutral reporting in another. Yet most automated detection systems apply the same models and reasoning patterns regardless of cultural context, producing what this thesis terms *culturally misaligned rationales*: fluent explanations that fail to engage with the specific discursive and cultural logic of the target community. Current large language models are predominantly English-centric and monocultural in their explanatory reasoning, and this thesis argues that addressing *cultural blindness* in automated information disorder assessment requires moving beyond static few-shot prompting toward community-aligned adaptive approaches.

To this end, a Culturally Adaptive Retrieval-Based Framework is proposed and evaluated. The framework uses semantic retrieval over a community-annotated exemplar bank to dynamically align model reasoning with culturally relevant examples at inference time. A pilot study is conducted on the InDor corpus across Farsi (Persian) and Italian using LLaMA 4 Maverick and Mixtral-8x22B-Instruct, comparing four prompting conditions across three evaluation tasks: severity classification, span detection, and rationale generation quality.

On automatic metrics, the framework shows its clearest benefit in Farsi, where retrieval improved span detection and rationale quality across both models and severity classification for Maverick. Italian results were more mixed, with severity improving under Maverick but rationale quality favouring the static baseline for that model. Beyond automatic metrics, native speakers of both languages rated retrieval-based rationales higher than the static baseline in a blind A/B human evaluation, with the improvement more pronounced in Farsi. The study also surfaces emergent model behaviors with broader implications for deploying instruction-tuned models on sensitive analytical tasks. Overall, the results show encouraging signs that aligning model reasoning with community-annotated exemplars moves outputs closer to what native speakers recognize as appropriate analysis, and are best read as a meaningful first step on a genuinely difficult and under-explored problem. The framework is designed as a foundation for a broader human-in-the-loop integration, where iterative community feedback can progressively refine the exemplar bank and improve cultural alignment over time. All code, prompts, evaluation scripts, and supplementary resources (including a curated landscape review of 108 fake news and information disorder datasets) are made publicly available as a community resource to support reproducibility, future extension, and broader adoption of culturally adaptive evaluation approaches.¹

¹<https://github.com/Maziarkiani/culturally-adaptive-prompting>

How to Cite This Work

If you use this framework, the dataset landscape sheet, the evaluation methodology, or the experimental results, please cite this thesis:

Kianimoghadam Jouneghani, M. (2026). *Culturally Adaptive Explainable LLM Assessment for Multilingual Information Disorder* (Master's thesis, University of Turin). Advisors: Viviana Patti and Marco Antonio Stranisci. Full text available at <https://maziarkiani.github.io/uploads/thesis.pdf>; UniTesi handle: <https://hdl.handle.net/20.500.14240/192055>

```
@mastersthesis{kianimoghadam2026culturally,
  title      = {Culturally Adaptive Explainable LLM Assessment for
                Multilingual
                Information Disorder},
  author     = {Kianimoghadam Jouneghani, Maziar},
  year       = {2026},
  school     = {University of Turin},
  type       = {Master's thesis},
  address    = {Turin, Italy},
  note       = {Advisors: Viviana Patti and Marco Antonio Stranisci.
                UniTesi handle:
                https://hdl.handle.net/20.500.14240/192055},
  url        = {https://maziarkiani.github.io/uploads/thesis.pdf}
}
```

Contents

List of Figures	vii
List of Tables	viii
1 Introduction	1
1.1 The Global Challenge of Information Disorder	1
1.2 Problem Statement: Cultural Bias and Explainability Gaps in LLMs	3
1.3 Thesis Vision: Towards Culturally Adaptive Explainable Assessment with LLMs	4
1.4 Research Questions	5
1.5 Contributions	5
1.6 Thesis Structure	6
2 Literature Review and Theoretical Framework	8
2.1 Theoretical Foundations of Information Disorder	8
2.2 Evolution of Automated Detection	9
2.3 The Paradigm Shift to Explainable AI (XAI)	10
2.4 Landscape Analysis of Fake News and Information Disorder Datasets	11
2.5 Retrieval, Cultural Alignment, and Adaptive Prompting	14
2.6 Chapter Summary	15
3 The InDor Corpus: Methodology and Cultural Analysis	17
3.1 Participatory Design Methodology	17
3.2 The Annotation Schema	18
3.2.1 Data Preprocessing and Annotation Platform	18
3.2.2 Macro-Level Assessment and Guiding Questions	19
3.2.3 Micro-Level Assessment (Spans and Rationales)	19
3.3 Corpus Composition and Statistics	20
3.4 Cross-Cultural Analysis of Information Disorder	22
3.4.1 Quantitative Agreement and Subjectivity	22
3.4.2 Lexical Analysis of Cultural Narratives	23
3.5 Chapter Summary	24
4 Benchmarking LLMs: The Limitations of Automation	26
4.1 Experimental Setup	26
4.2 Automated Metrics (Spans and Rationales)	28

4.3	Demographic Alignment and Human Evaluation	29
4.4	Failure Analysis: Cultural Blindness and Demographic Misalignment	30
4.5	Chapter Summary	32
5	Proposed Framework: Culturally Adaptive Retrieval-Based Framework	33
5.1	Architecture of the Culturally Adaptive Retrieval-Based Framework	33
5.2	Static Baselines: Zero-Shot and Few-Shot Prompting	35
5.3	Building the Exemplar Bank from Community Annotations	37
5.4	The Adaptation Mechanism: Culturally Adaptive Prompting	38
5.5	Evaluation Framework and Performance Metrics	40
5.5.1	Automated Performance Metrics	40
5.5.2	Human Evaluation	41
5.6	Scalability and Future Extensions	43
5.7	Chapter Summary	44
6	Implementation and Pilot Evaluation	45
6.1	Implementation Environment	45
6.2	Corpus Preparation: Cleaning Pipeline and Bank Statistics	46
6.3	Experimental Design: Pilot Study on Farsi and Italian	49
6.3.1	Shared Prompt Architecture and Pipeline Design	50
6.3.2	Static Baseline Conditions	52
6.3.3	Adaptive Retrieval Conditions	53
6.4	Quantitative Performance Analysis	54
6.4.1	Severity Classification (Macro-F1)	55
6.4.2	Span Detection (Token-Overlap F1)	57
6.4.3	Rationale Quality (BERTScore F1)	59
6.5	Qualitative Analysis of Generated Rationales	64
6.5.1	Farsi	64
6.5.2	Italian	71
6.6	Human Evaluation and Cross-Cultural Alignment	79
6.6.1	Farsi	81
6.6.2	Italian	82
6.6.3	Cross-Community Alignment	84
6.7	Chapter Summary	85
7	Discussion, Conclusion, and Future Directions	86
7.1	Cross-Condition and Cross-Lingual Synthesis	86
7.2	Emergent Findings	88
7.2.1	Instruction Language Leakage and the If-Then Fix	88
7.2.2	Alignment Tax: Over-Refusal in Safety-Trained Models	89
7.3	Limitations	90
7.4	Summary of Contributions	92
7.5	Ethical Considerations	93
7.6	Future Work	94

A	Human Evaluation Guidelines	96
A.1	Human Evaluation Guidelines: Farsi	96
A.2	Human Evaluation Guidelines: Italian	99
B	Prompt Templates	102
B.1	Task Prompts: Farsi (B0, B1, A1)	103
B.2	Task Prompts: Italian (B0, B1, A1)	105
B.3	Task Prompts: English (M1, both languages)	108
B.4	Static Examples: B1 Farsi	111
B.5	Static Examples: B1 Italian	114
	References	118

List of Figures

1.1	The Information Disorder framework mapping the intersection of falseness and intent to harm. Adapted from [3].	2
3.1	The Citizen-Dataset annotation platform interface. Reproduced from InDor [4]. . .	19
5.1	The Culturally Adaptive Retrieval-Based Framework.	34
5.2	The four prompting conditions used in the pilot. B0 and B1 are static baselines following the InDor setup. M1 and A1 introduce dynamic retrieval from the Exemplar Bank and differ only in instruction language, allowing the effect of instruction language to be isolated from the effect of exemplar retrieval.	40
6.1	The three-task sequential pipeline shared across all four prompting conditions. Each article is processed through three separate API calls issued in a fixed order. When severity is None, a bypass path records ["No"] without further calls. Otherwise, Task 2a detects spans from the article text, and Task 2b generates rationales using both the predicted spans and the original article text retrieved from memory by the orchestrator.	50
6.2	Severity Macro-F1 scores per condition, grouped by language and model.	57
6.3	Span detection token-overlap F1 scores per condition, grouped by language and model.	59
6.4	Rationale quality BERTScore F1 scores per condition, grouped by language and model.	61

List of Tables

2.1	Summary characteristics of 108 curated information disorder and fake news datasets. Please note that the categories are not mutually exclusive (e.g., a dataset may be both English-only and contain rationales). The complete dataset survey sheet is available in the project repository.	12
2.2	Evolution of computational paradigms and methodologies in information disorder detection. Note: The specified eras represent the period when each paradigm became dominant in the literature, rather than strict chronological boundaries.	16
3.1	Distribution of collected news and annotators across the four languages in InDor. Avg. = Average number of annotators per news. Adapted from InDor [4].	21
3.2	Demographic distribution of annotators by their first language. Adapted from InDor [4]. Note: For employment status, Fu = Full-time, Pa = Part-time, Un = University Student, NP = Not Paid.	21
3.3	Distribution of macro-level severity scores across the four languages, shown as percentages. Adapted from InDor [4].	22
3.4	Inter-Annotator Agreement (IAA) in terms of Krippendorff’s alpha for the severity score, and token-match accuracy for the portion of text selected by annotators. Adapted from InDor [4].	23
3.5	Most frequent words used by annotators in the identified problematic spans across four languages. Adapted from InDor [4].	23
3.6	Most frequent words used by annotators in their natural-language explanations. Adapted from InDor [4].	24
4.1	Model specifications used in the baseline benchmark.	27
4.2	Amount of news items and annotations used per task after the filtering process, separating severity classification (Task 1) from span-explanation generation (Task 2). Adapted from InDor [4].	27
4.3	BERTScore performance for span extraction and rationale generation across zero-shot and few-shot prompting conditions. Adapted from InDor [4].	28
4.4	Performance of LLaMA 4 Maverick on the English subset, detailing both F-score severity classification and downstream human evaluation of generated rationales. Results are stratified by annotator gender and primary language. Adapted from InDor [4].	29

5.1	Components of the static target-language prompting baselines (B0 and B1) and the methodological considerations that motivate the proposed shift toward dynamic retrieval.	36
6.1	Specifications of the two models used in the pilot, accessed through the OpenRouter API	46
6.2	Step-by-step corpus reduction across the cleaning pipeline for Italian and Farsi. Negative values indicate rows or articles removed at each step.	48
6.3	Severity label distributions for Italian across the original InDor corpus, the clean pool, the pilot test set, and the Exemplar Bank. InDor percentages are taken from [4]. Counts and percentages are reported for the clean pool, test set, and bank. Percentages are rounded to one decimal place.	49
6.4	Severity label distributions for Farsi across the original InDor corpus, the clean pool, the pilot test set, and the Exemplar Bank. InDor percentages are taken from [4]. Counts and percentages are reported for the clean pool, test set, and bank. Percentages are rounded to one decimal place.	49
6.5	Shared generation parameters applied across all four prompting conditions, both models, and both languages.	51
6.6	Rows excluded from evaluation per condition due to unrecoverable pipeline errors. † Italian Maverick B0 and A1 show atypically high exclusion rates concentrated at the rationale generation stage. Scores for these conditions are computed on 45 and 58 rows respectively and should be interpreted with caution relative to full 100-row evaluations.	55
6.7	Severity classification Macro-F1 scores across all conditions. Bold values indicate the highest score per language-model pair. † Italian Maverick B0 and A1 are evaluated on 45 and 58 rows respectively due to pipeline exclusions.	56
6.8	Span detection token-overlap F1 scores across all conditions. Bold values indicate the highest score per language-model pair. † Italian Maverick B0 and A1 are evaluated on 45 and 58 rows respectively due to pipeline exclusions, and their span scores are computed on a subset that excludes rows where rationale generation failed rather than rows where span detection itself failed. These scores should be interpreted with particular caution.	58
6.9	Rationale quality BERTScore F1 scores across all conditions. Bold values indicate the highest score per language-model pair. † Italian Maverick B0 and A1 are evaluated on 45 and 58 rows respectively due to pipeline exclusions at the rationale generation stage. The nature and implications of these exclusions are discussed in detail below.	60
6.10	Topic distribution of the 60 Italian articles that triggered rationale generation refusals in LLaMA 4 Maverick B0, A1, or both. B0 Only and A1 Only indicate articles that triggered refusals exclusively in that condition; Both indicates articles where refusals occurred in both conditions independently. Categories reflect the dominant narrative theme identified through manual review.	62
6.11	Item-level detection and overall advantage analysis across human-reference Problematic items in the Farsi test set. Miss rates indicate items where a condition produced no rationale for a Problematic item. Overall advantage combines detection reach and item-level BERTScore comparisons across all items.	65

6.12	Item-level detection and overall advantage analysis across human-reference Problematic items in the Italian test set. Miss rates indicate items where a condition produced no rationale for a Problematic item. The Maverick pool of 13 items reflects the reduced valid output intersection after content refusal exclusions.	72
6.13	Evaluation eligibility pipeline for Farsi and Italian under B1 (Static Few-Shot) versus M1 (Dynamic Retrieval). All Stratum 2 eligible items were presented to evaluators without further sampling. [†] 3 items excluded after merging due to ERROR or API tokens in severity or span parsed fields for one condition.	80
6.14	Stratum 1 mean scores (1–4 scale) for Farsi B1 and M1 rationales across 7 evaluators and 10 items (70 evaluator-item pairs per condition). SD denotes standard deviation across all evaluator-item pairs. $\Delta M-F $ is the absolute gender gap in mean scores per condition, computed as $ \bar{s}_{\text{male}} - \bar{s}_{\text{female}} $	81
6.15	Stratum 2 validation results for Farsi across 11 asymmetric items and 7 evaluators. Items indicates how many of the 11 asymmetric cases each setup appeared in. Appearances is the total number of evaluator judgments (items $\times$ 7). Validated indicates the evaluator judged the rationale as correctly identifying a real bias in the article.	82
6.16	Stratum 1 mean scores (1–4 scale) for Italian B1 and M1 rationales across 7 evaluators and 10 items (70 evaluator-item pairs per condition). SD denotes standard deviation across all evaluator-item pairs. $\Delta M-F $ is the absolute gender gap per condition.	83
6.17	Stratum 2 validation results for Italian across 6 asymmetric items and 7 evaluators. Items indicates how many of the 6 asymmetric cases each setup appeared in. Appearances is the total number of evaluator judgments (items $\times$ 7). Validated indicates the evaluator judged the rationale as correctly identifying a real bias in the article.	83
6.18	Cross-community alignment for LLaMA 4 Maverick under B1 and M1. Δ denotes the absolute gap between the two groups in each pair. Community Δ compares mean scores between Farsi and Italian evaluators.	84

Chapter 1

Introduction

The digitization of communication has significantly changed how news is produced, circulated, and consumed across societies. While digital platforms have expanded access to information and enabled new forms of global connection, they have also made it easier for manipulative content to spread quickly, shaping public discourse around democratic elections [1] and public behavior during global crises such as COVID-19 [2]. As online environments are increasingly shaped by automated systems and artificial intelligence, the detection and explanation of deceptive narratives have become an important challenge for research in Natural Language Processing (NLP) and machine learning.

At the same time, this challenge cannot be addressed through simple oppositions between truth and falsehood alone. Deceptive news is often shaped by context, interpretation, and culture, and any serious attempt to study it must take these dimensions into account.

1.1 The Global Challenge of Information Disorder

For several years, deceptive online content was commonly described under the broad label of “fake news.” Yet this term is too narrow and simple to account for the complexity, scale, and different motivations involved in contemporary information pollution. For this reason, this thesis adopts the Information Disorder framework proposed by Wardle and Derakhshan [3] as its main conceptual basis. Their framework treats the manipulation of information not as a single, uniform phenomenon, but as a spectrum of disorders defined by the interaction of two key dimensions: the degree of falseness in the content and the intent to cause harm.

Based on these dimensions, information disorder is classified into three categories:

- **Misinformation:** false information that is shared without an intention to cause harm. In many cases, this happens when people circulate inaccurate claims, rumours, or misleading interpretations because they believe them to be true.
- **Disinformation:** false information that is knowingly created, adapted, or shared in order to deceive and cause harm. This category is especially relevant in cases of coordinated propaganda, fabricated media, and strategically manipulated political narratives.
- **Malinformation:** genuine information that is shared with the intention of causing harm. As Wardle and Derakhshan [3] note, this often involves moving information that was meant to

remain private into the public sphere, or using real events in a decontextualized way to damage reputations or provoke hostility.

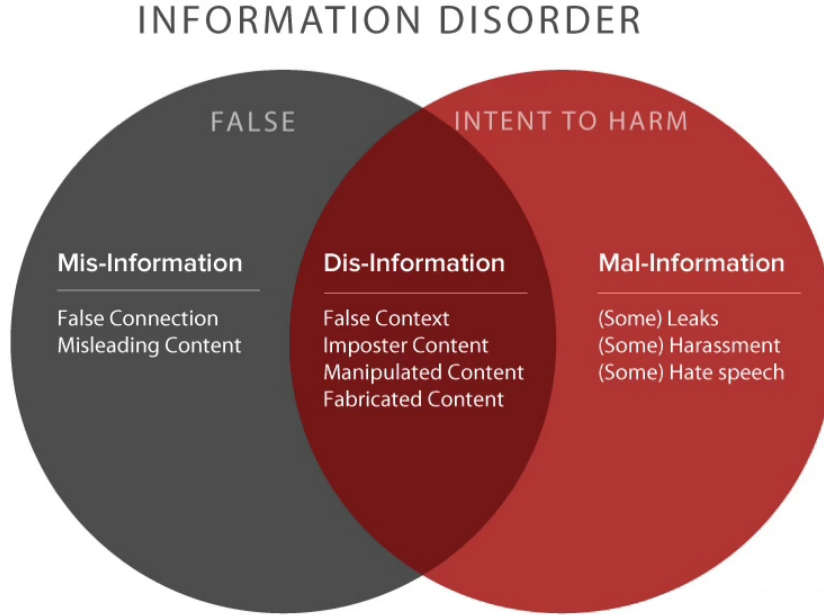


Figure 1.1: The Information Disorder framework mapping the intersection of falseness and intent to harm. Adapted from [3].

By moving away from the label of “fake news,” the Information Disorder framework offers a more precise way to analyse deceptive content. At the same time, although its core categories are broadly applicable, the actual forms that information disorder takes differ substantially across different contexts. Disinformation is not culturally neutral. It is shaped by local histories, political conditions, media environments, and linguistic conventions [3, 4].

A manipulative news story in an English-dominant setting, for example, may rely on one set of political frames and references, while deceptive content in Farsi, Italian, or Russian may draw on different cultural symbols, regional tensions, or state-linked narratives. For this reason, the global problem of information disorder is unlikely to be addressed effectively through a one-size-fits-all approach. Automated detection systems trained mainly on English-language data may struggle to identify the more subtle and culturally specific ways in which manipulation appears in other languages [5].

Recent language-specific assessments reinforce this concern: Italian culture-aware benchmarking shows that native cultural competence is not well captured by generic testing environments [6], Persian (Farsi) studies report systematic misalignment with Iranian cultural norms even when role-based prompting is introduced [7], and Russian linguacultural analyses similarly show that models struggle more with culturally embedded expressions than with broader fact-like knowledge [8].

Furthermore, contemporary information disorder is rarely limited to plain text alone. Deceptive campaigns often operate in multimodal ways, combining text with images, videos, or audio to make misleading claims appear more credible. One particularly effective strategy is to pair a genuine,

unaltered image with a false or misleading textual description. In such cases, because the visual media itself is authentic, the linguistic framing often becomes the source of the manipulation. As a result, even in a multimedia landscape, the ability to deeply analyse and explain textual narratives within their specific cultural contexts remains a critical step for automated detection.

1.2 Problem Statement: Cultural Bias and Explainability Gaps in LLMs

As the scale of information disorder has increased, researchers have increasingly turned to automated methods for detecting deceptive news. Earlier approaches, including traditional machine learning models such as Support Vector Machines, typically relied on surface-level indicators such as keyword patterns, stylistic features, or the perceived credibility of the source. These models can still function as useful preliminary filters, but they often struggle with more complex forms of manipulation because they depend on relatively shallow features rather than stronger semantic or contextual understanding [9, 10]. More recent work has therefore shifted toward Large Language Models (LLMs), which offer greater capacity to capture meaning, framing, and other less explicit signals in news discourse.

However, despite their strong language understanding capabilities, the use of current LLMs for detecting information disorder raises two major concerns. The first is explainability. In a domain as sensitive as journalism and news verification, it is not enough for a system to return a binary label such as “Fake” or “Real.” Users also need to understand the basis of that decision. This creates a clear need for Explainable AI (XAI) approaches that can produce natural language rationales, that is, explanations that are readable and meaningful to human users. Yet LLMs are frequently criticized for their limited interpretability, especially in higher-stakes settings [11, 12]. Even when such models successfully identify a problematic news item, used in this thesis as an umbrella term for content within the broader spectrum of information disorder, the reasoning underlying that judgment is often difficult to examine directly.

The second, and in many ways more serious, concern is monocultural bias. Many contemporary LLMs are developed through predominantly English-centered pretraining and alignment processes, which can result in uneven cultural representation [13, 14]. As a consequence, their outputs may implicitly reflect monocultural assumptions and interpretive norms. When these models are applied to news in other languages, such as Farsi, Russian, or Italian, they may read the text through English-centric expectations rather than through the cultural and discursive context in which that content actually operates. Targeted cultural probing confirms that this problem is not merely linguistic but also interpretive: Italian evaluation work shows that generic benchmarks can miss culture-specific competence [6], Persian studies report persistent misalignment with Iranian cultural norms even under role-based prompting [7], and Russian analyses indicate that culturally embedded expressions remain harder for models than broader fact-like cultural knowledge [8].

Because of this bias, LLMs can exhibit a form of culturally insensitive behavior that this thesis terms *cultural blindness*, building on documented patterns of cultural misalignment in LLM outputs [14, 5]. In practice, this means they may fail to recognize localized metaphors, region-specific propaganda strategies, or historically grounded anxieties that a native speaker might more readily identify as manipulative. The problem becomes even more serious at the level of explanation. When asked to justify why a non-English news article is problematic, these models may produce rationales shaped by culturally misaligned assumptions that do not accurately fit the local context.

As a result, although LLMs are technically capable of generating explanations, their reliability and cultural alignment in multilingual news assessment remain limited.

Furthermore, one related line of NLP research examines cultural misalignment through the lens of demographic bias and annotator alignment [15, 16]. In subjective or culturally sensitive tasks, LLM outputs may align more closely with some groups of annotators than with others, reflecting uneven demographic or cultural alignment, though these effects vary across datasets and tasks [17]. The InDor study itself documents demographic alignment disparities specific to information disorder annotation [4]. Building on this connection, this thesis incorporates demographic alignment disparities as an additional evaluation metric for cultural misalignment. Alongside standard performance measures, this thesis also explores whether rationale judgments vary across selected annotator dimensions where the pilot sample permits cautious comparison. Here we treat demographic alignment such as gender and native language as an exploratory indicator rather than a standalone claim of generalizable bias reduction.

1.3 Thesis Vision: Towards Culturally Adaptive Explainable Assessment with LLMs

To address the limitations of opaque and culturally uneven AI systems, this thesis argues that further scaling alone is unlikely to be enough. What is needed instead is a shift toward more human-centered architectures, in which human judgment remains part of the analytical process rather than something to replace entirely [18]. By combining the contextual knowledge of human communities with the generative strengths of Large Language Models, it becomes possible to design systems that are not only more explainable, but also better able to adapt to cultural variation.

In its broader conceptual form, the framework proposed in this thesis is situated within this longer-term Human-in-the-Loop perspective [19]. However, the present study does not claim to implement the full iterative architecture. Instead, it focuses on the first operational stage of that larger vision: testing whether retrieval-based prompting using human annotations and native-speaker rationales can already improve multilingual information disorder assessment before introducing the heavier components of a full feedback-and-update loop.

At the center of this first-stage framework is a move away from static detection methods toward a more dynamic strategy, referred to here as the Culturally Adaptive Retrieval-Based Framework. Instead of relying only on a model’s default pretrained associations, or on a static few-shot prompting strategy that uses fixed examples, the proposed approach retrieves relevant examples of information disorder from the target cultural context and incorporates them directly into the prompt. In this sense, the framework builds on the logic of In-Context Learning (ICL), where models are guided by examples provided at inference time rather than by parameter updates [20]. Here, those examples function as culturally aligned reference points that shape the model’s interpretation before it makes a judgment.

To support this adaptive mechanism, the thesis draws extensively on the Information Disorder (InDor) corpus, a community-driven multilingual dataset that includes English, Farsi, Italian, and Russian [4]. From this resource, the framework uses carefully selected, human-annotated instances to build an initial Exemplar Bank that can connect the model with culturally situated examples of information disorder. In this way, the LLM is not asked to rely only on its general pretrained knowledge, but is guided by localized cases and rationales drawn from the target context.

More broadly, the long-term aim of this thesis is to contribute to a culturally adaptive and explainable assessment framework in which human knowledge can play a continuing role in guiding and refining model behavior, not only by helping models identify more subtle forms of manipulation across languages, but also by encouraging explanations that are more transparent and more culturally appropriate for diverse news audiences. The present pilot, however, is intentionally narrower in scope: it evaluates whether this first retrieval-based intervention can improve detection and explanation quality in a bounded multilingual setting, before any full Human-in-the-Loop architecture is deployed.

1.4 Research Questions

To investigate the limitations of current automated approaches and to evaluate the proposed culturally adaptive retrieval-based framework in a structured way, this thesis is guided by the following primary research questions:

- RQ1: How does cultural and linguistic context shape the annotation and interpretation of information disorder in multilingual news data?
- RQ2: How well do baseline LLMs perform in predicting severity, extracting spans, and generating culturally aligned rationales for multilingual information disorder?
- RQ3: Does the Culturally Adaptive Retrieval-Based Framework improve automated detection, span extraction, and rationale generation quality over static baselines for information disorder in Farsi and Italian, as measured by both automated metrics and native-speaker human evaluation?

1.5 Contributions

To address the research questions outlined above, this thesis makes three main contributions to the study of computational linguistics and information disorder detection:

Landscape Analysis of Fake News and Information Disorder Datasets: The first contribution of this thesis is a curated review of 108 datasets related to fake news and information disorder. By systematically organizing these resources according to modality, language, and the availability of more detailed annotations such as text spans and rationales, the review connects earlier fake news benchmarks with the more demanding requirements of current information disorder research. The analysis reveals a continuing lack of datasets that include human-written explanations. To support future work in this area, the resulting reporting sheet is made publicly available through our GitHub repository as a structured resource for the research community, while also motivating the need for the rationale-grounded InDor corpus introduced later in this thesis.

Culturally Adaptive Retrieval-Based Framework: The second contribution of this thesis is the design and pilot evaluation of a retrieval-based culturally adaptive framework intended to strengthen LLMs’ capabilities in classification, span extraction, and rationale generation for information disorder assessment by reducing culturally misaligned reasoning. Drawing on filtered, article-level annotations from the InDor corpus, this thesis constructs a curated Exemplar Bank and uses it as the basis for a dynamic retrieval mechanism. The framework examines whether combining task instructions with dynamically retrieved, human-written native-speaker exemplars

through In-Context Learning can improve automated assessment and bring model reasoning into closer alignment with local human judgments.

Emergent Methodological Findings: The third contribution of this thesis is the documentation of two unexpected findings that emerged during the pilot evaluation, with implications beyond this specific study. The first is a prompt-level instruction language leakage pattern in multilingual retrieval-augmented setups, where English structural templates override target-language output formatting at scale, and a runtime dynamic injection fix that fully resolves it without altering prompt intent. The second is a systematic over-refusal behavior in a safety-aligned frontier model when confronted with analytically sensitive content in a non-English context, including a diagnostic methodology for distinguishing content-level refusals from infrastructure failures and a topic-level analysis of what categories of information disorder content trigger this behavior. Both findings are documented in Chapter 7 and are relevant to any research group deploying instruction-tuned models on sensitive multilingual tasks.

Foundational participation. Beyond these three contributions, this thesis is also based on the author’s active participation in the collaborative development of the InDor corpus [4], a community-driven multilingual dataset of 4,155 annotated news articles in English, Farsi, Italian, and Russian, developed through a participatory design process involving native speakers from different linguistic and cultural backgrounds. As a member of the core research team, the author contributed to the foundational phases of the project by sourcing manipulative media reports, coordinating social media outreach for community data collection, and participating in the annotation of the Farsi and English subsets. This involvement provided the empirical foundation on which the present thesis is built, and the InDor corpus serves as the primary data resource throughout.

1.6 Thesis Structure

The remainder of this thesis is structured as follows:

- **Chapter 2 (Literature Review and Theoretical Framework)** reviews the development of automated information disorder detection, moving from traditional machine learning classifiers to more recent Large Language Models. It also discusses the growing importance of Explainable AI (XAI), as well as the role of human evaluation, curation, and adaptive prompting in multilingual assessment.
- **Chapter 3 (The InDor Corpus: Methodology and Cultural Analysis)** presents the design of the InDor dataset, including its participatory methodology, annotation framework, and statistical composition. It also examines how cultural background can shape disagreement in the evaluation of news content.
- **Chapter 4 (Benchmarking LLMs: The Limitations of Automation)** reports the baseline quantitative and qualitative evaluation of current LLMs, including models such as LLaMA and Mixtral. The chapter focuses in particular on their limitations when assessing multilingual and culturally diverse narratives, including cases of cultural blindness and demographic misalignment.
- **Chapter 5 (Proposed Framework: Culturally Adaptive Retrieval-Based Framework)** introduces the main architectural contribution of the thesis. It explains the shift from static baselines to a dynamic retrieval-based prompting framework built around the Exemplar Bank,

and situates this pilot-stage method within a broader human-centered vision for culturally adaptive assessment.

- **Chapter 6 (Implementation and Pilot Evaluation)** describes the experimental design and findings of a targeted multilingual pilot focused on Farsi and Italian. It compares the proposed adaptive prompting framework against baseline conditions using automated metrics such as Severity Macro F1, Span Token Overlap F1, and Rationale BERTScore F1, as well as a blinded human evaluation to assess cultural-alignment of the rationales and study demographic alignment disparities.
- **Chapter 7 (Conclusion and Future Directions)** summarizes the main findings of the thesis, reflects on its ethical and methodological limitations, and outlines possible directions for extending culturally adaptive approaches to a wider range of languages and settings.

Chapter 2

Literature Review and Theoretical Framework

This chapter reviews the computational literature on deceptive news and establishes the theoretical basis for culturally adaptive assessment systems. It traces the development of automated detection methods from traditional machine learning classifiers to more recent Large Language Models (LLMs). Across this progression, a persistent gap becomes clear: although newer models offer stronger linguistic and generative performance, they still often struggle to produce reasoning that is both transparent and culturally aligned. The chapter begins by examining the computational implications of the Information Disorder framework. It then reviews the historical development of automated detection and the growing shift toward Explainable AI (XAI). Finally, it considers current datasets and discusses human-centered, retrieval-based, and In-Context Learning approaches as methodological directions for addressing monocultural bias in AI-driven news assessment.

2.1 Theoretical Foundations of Information Disorder

Research on deceptive online content has addressed several distinct subproblems in parallel, without a unified evaluation framework. Different lines of work focused on isolated aspects of the problem: some prioritized fact-checking and claim verification [21], others targeted the detection of fake news spreaders [22], while others examined persuasion and news framing [23]. Closely related but not identical phenomena, such as rumours, misinformation, spam, hoaxes, and propaganda, were often discussed under overlapping labels [24]. During this period, the term “fake news” gained broad public and academic visibility. Yet, as later survey work has noted, the term remained conceptually unstable [24, 25]. It lacked a consistent definition and was frequently used to describe quite different phenomena, including satire, clickbait, and parody. For that reason, it has limited value as a precise analytical category for systematic research.

Recognizing this terminological ambiguity, Wardle and Derakhshan [3] reject the label “fake news.” They argue that the term has not only been politically weaponized, but is also too imprecise to capture the scale and complexity of contemporary information pollution. To address this problem, they propose the Information Disorder framework. As discussed in Chapter 1, this thesis adopts that taxonomy, which distinguishes misinformation, disinformation, and malinformation through the interaction of falseness and intent to cause harm. For NLP research, this framework recasts the

detection task in more nuanced terms than a simple true/false distinction.

Historically, many early computational approaches treated deceptive news primarily as a classification task, often reducing it to simplified distinctions such as true versus false or fake versus real [26, 9]. The Information Disorder framework, however, makes clear that falseness alone is not enough. Its second dimension which is intent to cause harm introduces a further layer of complexity that is difficult to model computationally [3].

As Wardle and Derakhshan [3] suggest, intent is rarely stated openly in text. More often, it has to be inferred from framing, context, and other discourse-level cues. For that reason, an AI system that relies mainly on surface lexical patterns may struggle to distinguish between an honest reporting mistake and a deliberately manipulative narrative. This suggests that detecting information disorder requires more than basic semantic matching or label-based classification. To identify potentially harmful framing more reliably, computational systems likely need to account for the cultural and contextual signals through which intent is expressed and interpreted. This perspective helps explain the broader shift away from rigid feature-based classifiers toward more context-aware architectures.

2.2 Evolution of Automated Detection

The Information Disorder framework also helps clarify why computational approaches based only on simple keyword matching are limited. Over the past decade, automated detection of deceptive news has developed largely in response to this challenge. Drawing on recent survey studies [25, 26] and the broader literature, we categorize this development into four major architectural eras: (1) classical machine learning, (2) deep neural networks, (3) pre-trained transformers, and (4) General-purpose Large Language Models (LLMs). Building on this foundation, the review then turns to more recent directions in the field, particularly multilingual LLMs and retrieval-augmented In-Context Learning, which are brought together in the paradigm map presented in Table 2.2

Early automated approaches relied largely on classical machine learning classifiers such as Support Vector Machines (SVMs) and Random Forests. Rather than learning representations directly from text, these early models depended heavily on handcrafted features. As detailed in foundational surveys by Shu et al. [9] and Zhou and Zafarani [10], researchers manually designed cues such as word frequencies, stylistic patterns, and source metadata to train their classification frameworks. These traditional models remain computationally efficient and offer a limited form of feature-based interpretability, since it is possible to inspect which engineered features drive a given prediction. At the same time, their reliance on surface-level indicators often restricts their ability to capture the deeper semantic and contextual signals involved in more sophisticated forms of manipulation.

To move beyond the limitations of manual feature engineering, the field gradually shifted toward deep learning architectures. During the late 2010s, roughly 2017 to 2019, models such as Convolutional Neural Networks (CNNs) and Bidirectional Long Short-Term Memory networks (Bi-LSTMs) became widely used in deceptive news detection [24]. By working with word embeddings, these systems were able to learn local and sequential textual patterns more automatically. This reduced the dependence on handcrafted features, but it also introduced a clear trade-off: reduced transparency. In practice, these models often operate as opaque “black boxes.” Moreover, although they are effective at capturing local structure and sequence information, they often struggle to represent long-range document context and were not designed to produce natural-language rationales for their decisions.

The introduction of the Transformer architecture marked an important shift in the field. Models such as BERT [27] and RoBERTa [28] moved beyond static word embeddings by using self-attention mechanisms to model bidirectional context [25, 24]. This made it possible for systems to capture how the meaning of a word depends on its surrounding context, which substantially improved performance on more complex classification tasks. Even so, these models remained primarily designed for label prediction. Their internal reasoning was still difficult to access, and researchers often had to rely on limited post-hoc interpretability techniques, such as attention maps, which do not always provide faithful explanations of model decisions [11, 12].

The most recent shift in the field has been toward Large Language Models (LLMs) and prompting. Models such as GPT-3 [20], ChatGPT [29], and LLaMA [30] expanded the use of zero-shot and few-shot prompting in deceptive news detection, alongside more conventional task-specific fine-tuning [31, 32]. One of the main strengths of this model family is its ability to generate natural-language rationales together with a prediction, offering a more accessible explanatory interface than earlier approaches. At the same time, these reasoning capabilities are not always reliable. As Kuntur et al. [32] note, LLMs remain susceptible to hallucination and may produce explanations that are fluent but not always faithful to the factual ground truth. Beyond faithfulness, these models may also reflect culturally uneven assumptions, producing rationales that are not locally appropriate [13, 14].

This challenge becomes even more pronounced in the context of Multilingual LLMs (mLLMs). Models such as BLOOM [33], mT5 [34], and more recently Aya [35] have helped reduce language-access barriers, but cross-lingual performance does not automatically imply cultural alignment. As Qin et al. [36] and Yari and Koto [5] argue, mLLMs can still miss culturally specific meanings and locally grounded interpretive cues. A model may process or generate text fluently in Farsi or Italian, yet still overlook the cultural symbols, historical tensions, or local linguistic references that shape how a deceptive narrative is understood. This suggests that linguistic fluency alone is not enough for effective information disorder detection. Systems also need to be rooted in the specific cultural context of the material they assess.

Recent language-specific benchmarks support this point: the Italian culture-aware benchmark ITALIC was designed precisely because generic evaluation does not adequately capture Italian cultural and linguistic understanding [6], Persian low-resource evaluation reports consistent performance declines and systematic cultural misalignment [7], and Russian cultural evaluation shows that models are more reliable on broad cultural types than on culturally embedded catchphrases and related expressions [8].

In this thesis, we use the term *cultural blindness* to refer to this kind of fluent but culturally insensitive behavior. We also use the term *culturally misaligned rationale* to describe a fluent explanation that remains insufficiently rooted in the target cultural and discursive context and instead reflects monocultural or culturally blind assumptions. These two terms are introduced here as working definitions for the rest of the thesis, and they will be used to describe cases in which a model’s output may be linguistically fluent while still failing to capture the cultural logic that shapes interpretation.

2.3 The Paradigm Shift to Explainable AI (XAI)

As automated detection systems became more complex, they often improved predictive performance while becoming less transparent. As Hussain et al. [25] note in their landscape review,

a primary challenge in current systems is the severe lack of explainability methods. These models typically treat detection as a binary classification problem, producing a simple label without a transparent decision-making process. This makes it significantly difficult for users to validate the reasoning or trust the generated predictions.

In sensitive domains such as journalism, fact-checking, and public information, a binary label is often not enough. When a system classifies a news article as “disinformation,” users usually need some explanation of the reasoning behind that judgment in order to assess whether the output is credible [12]. This demand for greater transparency has contributed to the broader shift toward Explainable Artificial Intelligence (XAI). At its core, XAI aims to make model behaviour more interpretable, so that automated systems do not simply produce predictions, but also offer explanations that human users can understand.

To address this demand for transparency, recent work in explainable NLP has increasingly turned to rationale generation. As surveyed by Gurrapu et al. [11], this approach shifts explanation away from technical visualizations, such as feature weights or attention maps, and toward natural language. Rather than merely highlighting suspicious words or spans, these systems aim to provide a coherent, human-readable account of why a particular article is classified as problematic.

Large Language Models are, in many respects, well suited to this kind of generative explanation. As Zhao et al. [12] note in their review of LLM explainability, these models can draw on broad semantic context to produce detailed justifications in natural language. At the same time, using LLMs for XAI introduces a new set of concerns. Because such rationales are often generated post hoc, they do not always faithfully reflect the model’s internal decision process. In some cases, they may function more as plausible-sounding justifications than as genuine explanations of how the prediction was produced [11, 12]

More importantly for information disorder detection, an LLM’s rationale may reproduce the same monocultural bias present in the model itself. A system might correctly identify a news article from a culturally diverse setting as manipulative, yet still generate a monocultural explanation that misreads the local political or cultural context. In this sense, natural-language rationales are an important step toward more accessible explanation, but their cultural reliability and faithfulness remain unresolved challenges in current architectures.

2.4 Landscape Analysis of Fake News and Information Disorder Datasets

To support the development of the explainable AI architectures discussed in the previous section, models need datasets that provide more than labels alone. In particular, they benefit from resources that preserve context and include some form of reasoning or explanatory support. Yet, as recent landscape surveys have shown [25, 26], the current ecosystem of fake news datasets still has several important limitations.

To examine these patterns more systematically, this thesis curated a collection of 108 publicly available or documented datasets related to information disorder. Starting from the foundational lists reported in recent survey work [25], the curation was extended through additional searches across public repositories such as Hugging Face, GitHub, and Kaggle in order to verify accessibility and document key characteristics including language coverage, modality, annotation depth, and the availability of explanation-oriented annotations such as evidence spans or rationale-like support. The resulting dataset index is publicly available on GitHub as a resource for reproducible research.

An analysis of the publication timeline in this curated collection reveals another structural limitation. Most of the datasets were created during the first major wave of research between 2016 and 2021, and a notable subset (over 15% of the reviewed dataset collection) focuses specifically on distinct temporal crises such as the COVID-19 pandemic, major elections, and geopolitical conflicts. Because many of these resources are tied to particular events or periods, they may be especially vulnerable to distribution drift as deceptive narratives change over time [26]. Their topical focus and linguistic patterns can become outdated relatively quickly as new forms of manipulation emerge. More broadly, and as summarized in table 2.1, the dataset landscape is also marked by strong imbalances in language distribution and annotation depth.

Dataset Characteristic	Count (out of 108)	Percentage
English-only	63	57.8%
Multilingual (≥ 2 languages)	12	11.0%
Monolingual Datasets in Farsi, Italian, or Russian	20	18.3%
Contains Text-Span Annotations	7	6.4%
Contains Rationales	10	9.2%

Table 2.1: Summary characteristics of 108 curated information disorder and fake news datasets. Please note that the categories are not mutually exclusive (e.g., a dataset may be both English-only and contain rationales). The complete dataset survey sheet is available in the project repository.

Dataset characteristics were compiled from published papers, repository documentation, and available web resources. Where direct access was unavailable, details were inferred from published descriptions and minor inconsistencies may exist. With that caveat in mind, the analysis reveals the following patterns.

The first major limitation in the dataset landscape is linguistic imbalance. As Hussain et al. [25] note, the field remains heavily concentrated around English-language resources, with many widely spoken languages still underrepresented. The curated review confirms this pattern: 63 of the 108 datasets (58.3%) are exclusively English. As a result, foundational benchmark resources such as LIAR [37] and FakeNewsNet [38] are rooted mainly in Anglophone-centered political and social contexts. This concentration can constrain the development of AI systems intended for broader cross-cultural use.

The second, and in some ways more consequential, limitation concerns label simplicity and the limited availability of explanation-oriented annotation. As Kuntur et al. [26] argue, dataset design directly shapes detection performance, and many existing resources reduce the broader spectrum of information disorder to relatively simple label schemes such as “Fake” versus “Real.” The curated review reflects the same pattern. Among the 108 datasets examined, only 7 include text-span annotations, and only 10 provide any form of natural-language rationale.

Furthermore, although 10 of the curated datasets include some form of rationale-like support (9.2%), a closer look shows that these elements are usually limited to extractive evidence snippets, linked fact-checking text, or credibility comments. For example, benchmark datasets such as LIAR-PLUS [39], MultiFC [40], ParsFEVER [41], and even explicitly XAI-oriented datasets like RawFC [42] are valuable resources for evidence-based fact verification. However, their so-called rationales are usually built through evidence distillation, meaning that they extract relevant sentences from

reports rather than generate original explanations of deceptive intent.

Other datasets, such as FakeHealth [43] and CredBank [44], include explanatory review material or crowd-authored free text, but they do not focus specifically on manipulative framing. Taken together, the reviewed resources do not appear to offer the same combination of multilingual scope, participatory annotation, and culturally aligned, human-written analytical rationales needed for modern generative XAI architectures.

Because this thesis focuses on the InDor corpus for Farsi, Italian, and Russian, a more targeted review was also conducted to see whether suitable resources already existed for these languages. This review identified 20 relevant datasets for Farsi, Italian, and Russian (see Table 2.1), but they largely reflect the same limitations discussed above. Most rely on simple binary or multi-class veracity labels. A small subset does provide supporting evidence, but this evidence remains structurally limited. Datasets such as ParsFEVER [41] and a corresponding Italian fact verification benchmark [45] rely strictly on extractive snippets from Wikipedia. By contrast, datasets derived from Pagella Politica [46] link to full human-written journalistic articles, but these texts are meant to debunk factual claims rather than systematically explain the manipulative cultural framing used by the author.

Even the more recent non-English resources, such as the Russian FakeSpeak-RUS dataset [47], rely on fact-checkers' veracity assessments rather than providing natural-language rationales. None of these resources provide the kind of culturally aligned, generative explanations of manipulative intent that this thesis argues are essential for modern Explainable AI architectures.

For Italian, the development of native-language evaluation resources reflects a growing recognition that translated English benchmarks risk introducing distortions. As Magnini et al. [48] argue, translating English datasets introduces errors in translation quality, preserves Anglo-American cultural biases in the content, and fails to capture the stylistic varieties of native Italian. To operationalize this concern, they build their Evalita-LLM benchmark entirely from native Italian datasets. The ITALIC benchmark [6], sourced directly from Italian cultural contexts, was designed precisely to evaluate culturally specific commonsense that generic evaluation misses.

Together, these developments strongly reinforce the case for culturally aligned resources tailored to local contexts rather than relying only on generic multilingual translation or testing. Comparable concerns extend beyond Italian: Russian cultural evaluation similarly shows that language-specific resources are necessary when the target of assessment is culturally embedded interpretation rather than general language competence [8].

The dataset collection review shows that multilingual coverage alone does not resolve these limitations. Although 12 of the 108 reviewed datasets (11%) include two or more languages, they remain focused primarily on classification tasks. Within the curated set of 108 datasets, none of these multilingual resources combine cross-lingual coverage with text-span annotations or natural-language rationales that could support more explainable and context-aware architectures. As a result, these datasets offer multilingual breadth, but not the explanatory depth needed to evaluate how well AI systems actually interpret and explain culturally specific forms of manipulation.

Taken together, these patterns suggest that current datasets only partially meet the needs of modern information disorder assessment. The broader dataset landscape does not appear to offer a resource that combines coverage of underrepresented cultural contexts with deep, human-written explainability. This persistent gap motivates the need for a culturally adaptive dataset, which this thesis addresses through the introduction and analysis of the InDor corpus [4].

2.5 Retrieval, Cultural Alignment, and Adaptive Prompting

As the previous sections have suggested, fully automated black-box classifiers may struggle with some of the cultural and contextual complexity involved in information disorder. In response to these limitations, computational research has increasingly turned to Human-in-the-Loop (HITL) machine learning. As detailed in foundational texts and comprehensive surveys [18, 49], HITL frameworks were developed to keep human expertise involved in machine learning workflows, particularly in data processing, iterative annotation, model training, and system design. With the rise of Large Language Models, this human role has naturally extended beyond static labeling to demonstrations, feedback, and revision signals that shape model outputs more directly. Instead of assigning fixed binary labels alone, human input can now be used to provide revision instructions and rationale-oriented feedback, helping guide the model’s generative reasoning process.

One practical way to incorporate human expertise into LLM-based systems is through In-Context Learning (ICL). First formalized by Brown et al. [20], ICL enables a model to adapt to a specific task without updating its underlying parameters. Instead of relying on fine-tuning, the model is given a prompt that includes a small number of carefully designed demonstrations, typically in the form of expert-written input-output pairs. By conditioning on these examples at inference time, the model can follow the desired output pattern more effectively. Recent survey work suggests that, in some settings, ICL can reduce dependence on large task-specific labelled datasets, especially when high-quality demonstrations are available [50, 51]. This makes it particularly appealing for complex or low-resource tasks in which human analytical expertise is valuable but difficult to scale.

However, relying on a fixed set of examples presents its own challenges. A static prompt designed around two, three, or even four examples cannot anticipate the full range of topics, framings, and narrative strategies a model will encounter at inference time. While it may provide useful structure, it will systematically under-represent the diversity of real-world news content, regardless of how carefully those examples are selected.

In response to these limitations, recent work has increasingly turned to retrieval-augmented generation (RAG) and In-Context Learning (ICL) as ways of aligning model reasoning more dynamically. Rather than depending only on a model’s internal knowledge, these methods retrieve relevant examples, summaries, or contextual metadata at inference time and use them to shape the response. Seo et al. [52], for instance, propose ValuesRAG, a framework for improving the cultural alignment of LLMs. Their approach uses the World Values Survey (WVS) to generate structured summaries of values and demographic profiles, and then retrieves the most relevant profiles to guide culturally aligned generation. This makes ValuesRAG especially relevant for the present thesis, because it shows that retrieval can be used not only for factual support but also for aligning model behavior with situated cultural values. At the same time, its focus remains on value alignment rather than information disorder, and the retrieved material consists of summarized value profiles rather than human-written rationales about manipulative narratives.

A related line of work appears in misinformation detection. Liu et al. [53] introduce RAE-moLLM, a retrieval-augmented framework for cross-domain misinformation detection. Their method starts from the idea that misinformation often relies on emotional framing, so it builds a retrieval database of affective examples and then uses source-domain instances with similar emotional characteristics as in-context demonstrations. This is closer to the present research problem, since it directly addresses misinformation detection. Even so, its retrieval strategy is based on affective similarity rather than cultural explanation, and the retrieved items are examples for classification rather than expert-written rationales that explain why a narrative is manipulative in a specific local

context.

Beyond these two works, retrieval-based misinformation research has expanded quickly across different retrieval paradigms. For instance, rather than retrieving in-context demonstrations, frameworks such as M&M-RAG [54] use real-time web retrieval to gather external evidence for fact-checking multimodal fake news across languages such as English, Chinese, and Urdu. Returning to demonstration-based approaches, RAAR [55] combines retrieval with agentic reasoning to gather source-domain evidence that matches a target sample’s semantics, sentiment, and writing style for cross-domain misinformation detection. In a related line of explanation-focused work, ExplainHM++ [56] retrieves similar meme instances from a database to support a multimodal debate between LLMs, which then generate competing rationales from both harmful and harmless perspectives. Taken together, these studies show that retrieval-based prompting and retrieval-augmented reasoning are becoming increasingly important for aligning model outputs with contextually relevant evidence.

Still, two important gaps remain. First, existing retrieval-augmented systems have made impressive strides by retrieving diverse forms of context, such as cultural profiles, affective examples, debate samples, or real-time external evidence. While these works are highly influential and strongly inspire this thesis, within the specific domain of information disorder, the reviewed systems do not retrieve human-written analytical explanations that can directly guide model reasoning. This thesis argues that incorporating human knowledge into the retrieval process is an important next step, especially for a task as subjective and culturally sensitive as information disorder assessment. Native speakers are often better positioned to explain not only that a narrative is manipulative, but also why and how it becomes manipulative within a specific local context.

Second, even within this active research area, most existing frameworks focus specifically on cultural value alignment or narrower categories such as fake news, misinformation, or harmful content, rather than the broader and more conceptually complex umbrella of information disorder.

This thesis is designed to address these gaps. By combining the cross-lingual reasoning capabilities of modern LLMs with a culturally adaptive retrieval framework rooted in the newly developed InDor corpus [4], the proposed approach moves beyond fragmented or purely binary views of deceptive content. More specifically, it explores whether retrieved, culturally aligned exemplars and rationales can support a more transparent and context-sensitive assessment of information disorder, including graded severity judgments, localized span extraction, and explanations that are better aligned with human interpretation across different cultural settings.

2.6 Chapter Summary

The development of automated information disorder detection reflects a continuing tension between predictive performance and explanatory transparency. As this chapter has shown, earlier feature-engineered and deep learning models prioritized classification performance, but largely operated as opaque black boxes.

The later shift toward general-purpose and cross-lingual Large Language Models expanded the field’s explanatory capacity by making natural-language rationales possible. Even so, these systems still face important limitations, particularly in the form of monocultural bias and insufficiently grounded reasoning. Table 2.2 summarizes these paradigm shifts and traces the field’s movement from rigid classification systems toward the current need for culturally aligned, retrieval-augmented architectures.

Paradigm & Era	Representative Architectures	Explainability Level	The Persisting Gap
Classical ML (Pre-2018)	SVM, Random Forest	Low (Feature-reliant)	Limited semantic and contextual understanding
Deep Learning (2017–2019)	CNNs, Bi-LSTMs	Low (Opaque Black Box)	Opaque reasoning; no natural-language explanation
Transformers (2019–2021)	BERT, RoBERTa	Low (Post-hoc only)	Stronger context modeling, but explanation remains post-hoc
General-purpose LLMs (2020–2023)	GPT-3, LLaMA, ChatGPT (GPT 3.5)	Moderate-High	Fluent explanations, but limited faithfulness and cultural alignment
Multilingual LLMs (2024–2025)	Aya, BLOOM, mT5	Moderate-High	Cross-lingual ability without reliable cultural alignment
Domain RAG & ICL (2024–Present)	ValuesRAG, RAEmoLLM, M&M-RAG, RAAR, ExplainHM++	Moderate-High	Retrieve proxy contexts rather than human-written rationales and don't target the full information disorder spectrum.

Table 2.2: Evolution of computational paradigms and methodologies in information disorder detection. Note: The specified eras represent the period when each paradigm became dominant in the literature, rather than strict chronological boundaries.

Furthermore, the review of the dataset landscape shows that this technical gap is also a resource problem. Existing datasets tend to offer either multilingual coverage without much explanatory depth, or extractive evidence without human-written, generative reasoning. To bridge this methodological gap, recent computational work increasingly points toward In-Context Learning (ICL), where human expertise can be integrated more directly into model reasoning.

Taken together, these theoretical and methodological foundations shape the remainder of this thesis. They reinforce the need to move beyond English-centric classification and motivate both the use of the rationale-rich InDor corpus in Chapter 3 and the culturally adaptive retrieval-based framework developed in Chapter 5.

Chapter 3

The InDor Corpus: Methodology and Cultural Analysis

As Chapter 2 has shown, existing detection systems and datasets often struggle to capture the cultural nuance and explanatory depth needed for the study of information disorder. In response to these limitations, this chapter introduces the Information Disorder (InDor) corpus [4]. Unlike more conventional datasets, which are rarely designed from the perspective of news audiences, InDor was developed to reflect the subjectivity and cultural diversity of real readers more directly.

Specifically, this chapter addresses the first Research Question (**RQ1**) by examining how cultural and linguistic contexts shape the annotation and interpretation of manipulative news. It describes the participatory methodology used to build the corpus, explains its annotation schema, and presents a cross-cultural analysis of how different linguistic communities interpret deceptive narratives, alongside a baseline demographic profile of the participants.

3.1 Participatory Design Methodology

Historically, many datasets used for information disorder detection have been built from content collected through social media platforms, news websites, and fact-checking sources rather than through community-based participatory processes [25, 26]. These resources often rely on relatively simple label schemes and standard annotation pipelines, which can limit their ability to capture the complexity and subjectivity of deceptive narratives [26]. Furthermore, as Cabitza et al. [57] and Gabriel et al. [58] argue, because traditional approaches rarely involve the communities most directly exposed to the content, they often fail to capture the perspectives of the readers who actually encounter and interpret the news. This can also increase the risk of selection bias in the resulting data [59].

To reduce these risks and better account for the nuances of local context [60], the InDor corpus was developed through a community-centered methodology. Drawing on the foundational principles of participatory design in Natural Language Processing outlined by Caselli et al. [61] and Delgado et al. [62], the project rejected traditional top-down annotation. Rather than treating participants simply as external labelers, the InDor methodology approached the task as a collaborative process in which community members actively helped shape the dataset alongside the research team.

Community engagement in InDor unfolded in several phases and involved stakeholders from five countries: Italy, Spain, the Netherlands, Iran, and Russia [4]. The project began with a small group of people already interested in addressing online information disorder, together with academic experts, newsroom professionals, and representatives of non-profit organizations who provided early feedback on the annotation design. These initial participants then shared the project through their own networks, helping extend the call to a wider public. This open, bottom-up approach was intended to keep participation relatively unconstrained, so that the collected news items would reflect the diverse concerns and experiences of local readers more naturally.

Instead of relying on predefined keyword searches, the data collection phase was driven directly by the community. To widen participation and support international engagement, the Call to Action (CTA) cards shared on social media were translated into ten languages: Farsi, Italian, Arabic, Spanish, Japanese, French, Chinese, Catalan, Russian, and Dutch. A second card was distributed alongside the CTA, offering simple guidance on how to identify potentially problematic content. Participants were asked to complete a form with the article URL, its language, and a brief explanation of why they considered it suspicious. Through this initial call, the community submitted 799 suspicious news URLs, and the corresponding website domains were then used as seeds to collect a larger pool of 11,199 news articles. Although the CTA reached a broader multilingual audience, the final dataset focuses on the four languages, English, Farsi, Italian, and Russian, which produced enough data for robust annotation during the collection phase [4].

3.2 The Annotation Schema

To preserve annotators' subjective judgments, the InDor project adopted a descriptive annotation framework [63], combining question-guided reasoning with macro-level classification to encourage annotator reflection on both severity and specific forms of misrepresentation affecting the events and individuals described in the text. This section presents the data preprocessing pipeline, the platform used for annotations, and the guidelines that helped capture these perspectives.

3.2.1 Data Preprocessing and Annotation Platform

As detailed in Section 3.1, the community-driven collection phase resulted in a raw dataset of 11,199 news articles across four languages. To reduce cognitive load during annotation, the text was automatically preprocessed following the methodology established by Gemelli et al. [4]. The project utilized the Sumy library, an algorithmic extractive summarization tool, to retain only the headline and the three most salient sentences from each article [4]. This produced shorter text units that made the annotation task more manageable while preserving the core narrative. The annotation task was carried out through Citizen-Dataset¹, a dedicated web-based platform provided by aequa.tech².

The platform structured the workflow to guide users through three main components: text classification (macro-level severity), span annotation (micro-level framing), and text writing (natural language rationales).

¹<https://citizen-dataset.aequa-tech.com/>

²<https://aequa-tech.com/>

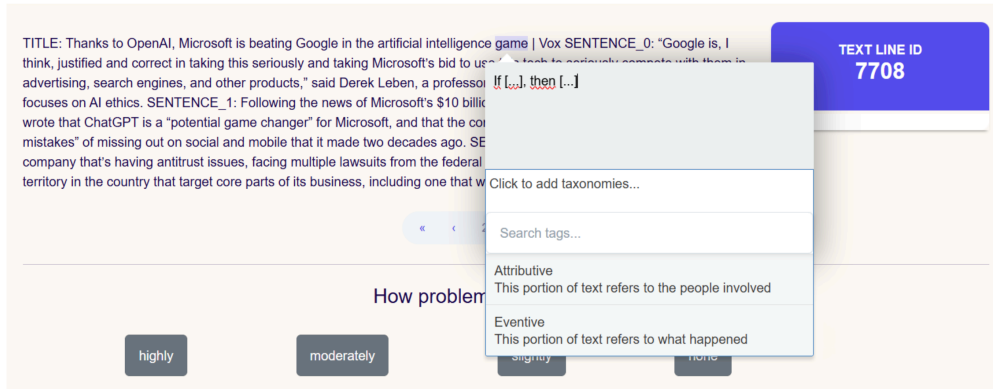


Figure 3.1: The Citizen-Dataset annotation platform interface. Reproduced from InDor [4].

3.2.2 Macro-Level Assessment and Guiding Questions

After reviewing the preprocessed text, annotators were asked to evaluate the overall nature of the article. They first responded to the question, “How problematic is this news?”, using a four-point severity scale: Highly, Moderately, Slightly, or None. This macro-level classification allowed the research team to capture how strongly the content was perceived as an instance of information disorder. The platform also included an “N/A” (Not Applicable) option. This served as a quality-control measure, allowing annotators to flag broken URLs, paywalled pages, or irrelevant material such as advertisements, which could then be excluded from the final dataset.

To encourage closer reflection during this macro-level assessment, the annotation guidelines included a set of guiding questions. Annotators were invited to consider whether the article used manipulative communication strategies, such as sensationalism or clickbait, sought to provoke polarizing emotions, reinforced stereotypes about socially disadvantaged groups, or lacked credible supporting evidence. These prompts were intended to help volunteers think more carefully about the broader framing and context of the narrative before moving on to identify specific problematic words or phrases. For a complete review of the prompts and the full annotation instructions, the guidelines are publicly available in the InDor GitHub repository³ and detailed in the foundational corpus paper [4].

3.2.3 Micro-Level Assessment (Spans and Rationales)

To capture more precisely how deceptive narratives shape readers’ interpretations, the InDor annotation schema drew on insights from critical discourse analysis and frame semantics. Research in linguistics and psychology, such as the studies conducted by Bohner [64] and Hart [65], has shown that particular semantic and syntactic choices can significantly influence how readers understand events and the entities involved. For this reason, identifying manipulative content requires going beyond the basic facts of an event to examine how participants are represented [66] and how the event itself is framed cognitively [67, 68].

³<https://github.com/citizen-dataset/InDor>

Building on this theoretical foundation, the annotation framework was organized around two main categories of misrepresentation: Eventive and Attributive. The *Eventive* label was used when the problematic part of the text concerned what happened, including the causes of an event or the specific wording used to describe its development. By contrast, the *Attributive* label was applied when the manipulation focused on the people involved, including their appearance, emotional state, origins, or relationships with others. This distinction gave annotators a structured way to identify whether a news article was distorting the representation of an event or reinforcing stereotypes about the individuals within it, a dynamic frequently highlighted in critical media studies by researchers such as Li and Zhang [69] and Stabile et al. [70].

Beyond selecting text spans, annotators were also asked to provide natural-language rationales explaining their choices. To support this process while preserving individual subjectivity, the schema used a semi-structured “If ... / Then ...” template: “If [reference to what is in the text], then [implication or consequence].” The following example illustrates a complete annotation triplet produced by a user on the platform [4].

Annotation Triplet Example. Adapted from inDor.

Context: “Melania Trump Makes Apparent Retaliation For ‘Stormy Daniels’ Affair; Donald In Shambles”

Selected Span (Eventive): “Apparent Retaliation”

Rationale: “If you use ‘retaliation’ to describe Melania Trump’s response, **then** you are using sensationalist language that aims to create clickbait while also misleading the reader by diverting attention to a different topic than what the article is actually about.”

This structured triplet format served two purposes. For annotators, it encouraged more guided reflection during the task. For the dataset, it produced a more consistent output format that can support the automatic processing of human rationales in the Explainable AI (XAI) models discussed in later chapters.

3.3 Corpus Composition and Statistics

As outlined in Section 3.1, the initial community-driven collection phase produced a large pool of 11,199 news articles. The final size of the corpus, however, depended on the availability and capacity of the volunteer annotators [4]. To keep the annotation task manageable, the 81 participants were assigned batches ranging from 20 to 250 news items, depending on how much time they were able to contribute [4].

This process resulted in the final consolidated InDor corpus, which contains 4,155 annotated news items. Because the dataset was built through organic citizen participation rather than fixed sampling quotas, it is intentionally imbalanced and reflects different levels of engagement across the linguistic communities involved. The distribution of annotated news items, source website domains, and annotator assignments is summarized in Table 3.1.

Beyond the distribution of the articles themselves, the composition of the 81 annotators provides important context for understanding the corpus’s cultural diversity. During onboarding, annotators were invited to complete a socio-demographic questionnaire covering gender, age, first

Language	Website Domains	Annotated News	Avg.	Total Annotators	Speakers
Italian	242	1,656	1.88	64	Native
Farsi	47	1,000	1.95	6	Native
English	358	999	2.01	9	Non-native
Russian	63	500	2.00	2	Native

Table 3.1: Distribution of collected news and annotators across the four languages in InDor. Avg. = Average number of annotators per news. Adapted from InDor [4].

language, ethnicity, educational background, and employment status. Documenting these characteristics, summarized in Table 3.2, helped situate the dataset in relation to a broader range of lived experiences. This demographic baseline is also relevant for the later analysis of cross-cultural and demographic alignment disparities. That completion of the socio-demographic questionnaire was optional. As a result, Table 3.2 reflects the profile of the 42 participants who chose to provide this information, and some of them also left specific sensitive fields, such as ethnicity, unanswered.

First Language	Gender		Ethnicity				Education				Employment					Age	
	M	F	White	Black	Asian	Other	High Sch.	Bachelor	Master	PhD	Fu	Pa	Un	NP	Other	< 30	> 30
Italian	5	26	29	0	0	0	0	29	1	1	3	6	13	1	8	28	3
Persian	4	1	4	0	0	1	1	4	0	0	1	1	1	1	1	2	3
French	0	2	1	0	0	0	0	2	0	0	0	0	1	0	1	2	0
Russian	1	1	0	0	2	0	0	2	0	0	0	0	2	0	0	2	0
Arabic	0	1	0	1	0	0	0	1	0	0	1	0	0	0	0	1	0
English	0	1	1	0	0	0	0	1	0	0	1	0	0	0	0	1	0

Table 3.2: Demographic distribution of annotators by their first language. Adapted from InDor [4]. Note: For employment status, Fu = Full-time, Pa = Part-time, Un = University Student, NP = Not Paid.

In addition to the demographic questionnaire, participants were invited to complete pre- and post-annotation surveys. The pre-annotation survey was intended to capture their initial knowledge of, and confidence in, identifying information disorder, while the post-annotation survey collected feedback on whether and how their perception of more subtle manipulative strategies changed during the task. A pedagogical analysis of these evolving perceptions lies beyond the main computational focus of this thesis. However, the questionnaires and focus group materials are publicly available in the InDor project repository for reference⁴.

A defining feature of the InDor corpus is its two-phase annotation design, which was developed to examine cross-cultural perceptions of manipulation more directly. Importantly, none of the annotators assigned to the English subset were native speakers of English. In the first phase, participants from different cultural and national backgrounds, including Italy, Iran, Russia, Cameroon, France, and the United States, annotated a shared set of English news articles. In the second phase, they annotated articles in their respective native languages. This design allowed the research team to compare how participants evaluated manipulative content in a more familiar local context and how

⁴<https://github.com/citizen-dataset/InDor>

they responded to it in a globalized, non-native language such as English [4].

3.4 Cross-Cultural Analysis of Information Disorder

To examine how local context shapes the perception of manipulative news, this final section analyzes the annotated data across the four linguistic subsets of the corpus. By looking at both statistical patterns of agreement and the specific vocabulary highlighted by annotators, the analysis shows that information disorder is not experienced in a uniform way, but is closely tied to regional socio-political contexts. The findings are presented in two parts: first, a quantitative review of annotator agreement, and second, a qualitative lexical analysis of the cultural narratives that emerged during the task.

3.4.1 Quantitative Agreement and Subjectivity

Unlike more traditional NLP classification tasks, where annotator disagreement is often treated as noise or error [57, 63], the InDor project treats disagreement as a meaningful signal of cultural subjectivity. Perceptions of information disorder are rarely uniform. A narrative that appears clearly manipulative to a reader in one country may be interpreted quite differently by a reader in another, shaped by differences in historical experience, media environment, and socio-political context.

To examine these differences, the initial analysis focused on how annotators across the four language subsets used the macro-level severity labels. As shown in Table 3.3, there are clear cross-cultural differences in how the intensity of information disorder is perceived.

Severity Score	English	Italian	Russian	Farsi
Highly	5.95	12.07	16.57	4.34
Moderately	11.91	13.13	12.94	10.96
Slightly	25.41	18.13	12.37	41.62
None	56.72	56.67	58.12	43.08

Table 3.3: Distribution of macro-level severity scores across the four languages, shown as percentages. Adapted from InDor [4].

While the share of news items judged to be non-problematic (“None”) was highest across all four language subsets, the upper end of the scale points to more context-specific sensitivities. In English and Farsi, for example, the proportion of items rated as Highly problematic remained below 6%, whereas in Italian and Russian it exceeded 12%. As reported by Gemelli et al. [4], post-annotation feedback from Russian annotators suggested that their more frequent use of the highest severity score was linked to the repeated presence of emotionally charged and polarized language related to the war between Russia and Ukraine.

To quantify this variation more systematically, the research team calculated the Inter-Annotator Agreement (IAA) for the corpus. Because of the sparsity of the annotations, Krippendorff’s alpha [71] was used to measure agreement on macro-level severity scores, while token-match accuracy [72] was used to assess alignment in the selection of micro-level problematic spans. IAA is reported in table 3.4

Metric	English	Italian	Russian	Farsi
Severity	0.03	0.12	0.06	0.10
Spans	0.16	0.25	0.33	0.29

Table 3.4: Inter-Annotator Agreement (IAA) in terms of Krippendorff’s alpha for the severity score, and token-match accuracy for the portion of text selected by annotators. Adapted from InDor [4].

The generally low agreement scores across all language subsets suggest that identifying manipulative framing is a subjective interpretive task rather than a purely objective one. This subjectivity appears to become even more pronounced when language barriers are involved [73]. Because none of the annotators for the English subset were native speakers of English, that subset showed the lowest agreement overall, with a severity alpha of 0.03 and a span-selection score of 0.16.

3.4.2 Lexical Analysis of Cultural Narratives

Beyond the agreement scores, the subjectivity of information disorder becomes especially visible in the vocabulary that annotators identified as problematic. To examine these differences more systematically, this section revisits the lexical frequency analysis reported by Gemelli et al. [4], using both the text spans selected by annotators and the natural-language rationales they produced to explain their choices. Table 3.5 presents a sample of the most frequent tokens extracted from the problematic spans across the four languages. These patterns suggest that manipulative news tends to be closely tied to the socio-political concerns that are most salient in each context.

Language	Most Frequent Words in Spans
English	make, year, say, people, new, one, trump, take, muslim, get, world, see, state, work, right, time, well, go, medium, need
Italian	solo, volere, dire, leader, parte, altro, europeo, anno, nazionale, sempre, proprio, europa, magistrato, poco, russo, alcuno, molto, uniti, vita, ora
Russian	россия, формат, остаться, антироссии, год, который, страна, техника, уничтожить, западный, находится, враг, металлолом, военный, котором, новый, противостояние, готовить, уничтожить, антиРоссия
Farsi	کشور، ایران، اسلامی، آمریکا، استان، سازمان، رئیس، اقتصادی، تولید، شهر، کشورهای، تهران، دولت، انقلاب، توجه، ملی، حوزه، حضور، افزایش، دلار

Table 3.5: Most frequent words used by annotators in the identified problematic spans across four languages. Adapted from InDor [4].

The English spans focused largely on the representation of specific individuals, such as “Trump,” as well as religious or cultural communities, such as “Muslim.” By contrast, the Italian spans more often featured terms related to European geopolitical relations and immigration. One notable example is the frequent selection of “magistrato” (magistrate), which points to a specifically Italian

narrative in which migration is often framed through tensions between political actors and the judiciary.

The Russian annotations were dominated by references to the geopolitical conflict with Ukraine. Annotators frequently highlighted aggressive or ideologically charged terms such as “уничтожить” (destroy) and “антиРоссия” (anti-Russia), the latter being used to frame neighbouring states as hostile to Russian interests. The Farsi spans, by contrast, reflected the struggles of everyday life in Iran. Annotators repeatedly flagged economic and geopolitical terms like “اقتصادی” (economic), “دلار” (dollar), and “آمریکا” (United States), suggesting that inflation, economic situation, and broader political tensions shape the local context.

These cultural differences were also visible in the ways annotators explained their selections. Table 3.6 presents the most frequent words found in the annotators’ written rationales.

Language	Most Frequent Words in Rationales
English	risk, reader, term, context, sentence, perception, language, frame, state, shape, neutrally, provide, potentially, toward, author
Italian	rischiare, utilizzare, rischio, creare, riportare, espressione, titolo, negativo, uso, politico, informazione, parola, contesto, linguaggio, volere
Russian	рисковать, внешний, описываете, антиукраинский, россия, нормализовать, государство, утверждать, легитимность, создавать, суверенитет, которая, военный, российский, игнорировать
Farsi	محدود، سیاسی، بررسی، بیان، تحلیل، ارزیابی، درک، اطلاعات، متن، نقش، ایدئولوژیک، تأکید، مطرح، منابع، آمار

Table 3.6: Most frequent words used by annotators in their natural-language explanations. Adapted from InDor [4].

In the English and Italian subsets, the rationales often took on a more meta-analytical tone, warning about the risk of certain forms of language and their possible effect on the reader or the broader context. In the Russian subset, annotators more explicitly pointed to attempts to normalize (нормализовать) official narratives or construct legitimacy (легитимность). In Farsi, the rationales were more often framed around demands for evidence, frequently drawing attention to missing statistics (آمار) or unverified sources (منابع) as signs of ideological (ایدئولوژیک) bias [4].

3.5 Chapter Summary

This chapter introduced the Information Disorder (InDor) corpus [4] and outlined the community-centered methodology used to capture the subjective and culturally diverse perspectives of real news readers. By moving beyond binary labels and adopting a descriptive, rationale-driven annotation schema, the dataset offers a more nuanced account of how people interpret potentially manipulative content.

In relation to **RQ1**, the cross-cultural analysis showed that perceptions of information disorder are far from uniform. Both the quantitative agreement patterns and the qualitative lexical analysis suggest that judgments about manipulative framing are shaped by local socio-political concerns,

cultural expectations and, in some cases, language-specific interpretive differences. Taken together, these findings point to a broader pattern: information disorder appears to be anchored in local vulnerabilities, whether economic, historical, or geopolitical, rather than being interpreted in the same way across contexts. For this reason, detection strategies cannot be treated as statically universal; they need to remain adaptable to the setting in which they are applied.

With this culturally aligned human baseline in place, the thesis now turns to **RQ2**: to what extent can contemporary automated systems approximate this kind of contextual sensitivity? The next chapter addresses this question by benchmarking two Large Language Models against the InDor corpus, evaluating their baseline ability to predict severity, extract relevant spans, and generate explanations that align with these diverse human perspectives.

Chapter 4

Benchmarking LLMs: The Limitations of Automation

As established in the previous chapters, interpreting information disorder is a highly subjective and culturally situated task. While contemporary Large Language Models (LLMs) have demonstrated strong generative capabilities, their ability to navigate cross-cultural nuances and provide culturally aligned reasoning remains a persistent limitation. Building on the participatory human baseline established in Chapter 3, this chapter reports the performance of two baseline models, Llama 4 Maverick and Mixtral 8x22B Instruct, when tasked with assessing and explaining manipulative news across different languages. By reviewing the baseline LLM evaluations conducted during the InDor project [4], this chapter explores the extent to which standard LLMs can approximate human contextual sensitivity.

All experiments reported in this chapter were conducted as part of the InDor project [4] and are presented here to establish the limitations that motivate the adaptive framework in Chapter 5. The chapter first presents the experimental setup and then reports results for severity classification, span extraction, and rationale generation under zero-shot and static few-shot prompting. It then narrows to the evaluation of model-generated rationales, including their comparison with human-written references and the downstream human judgment of Llama 4 Maverick’s English-language rationales by non-native English-speaking annotators. Finally, the chapter discusses how these baseline results point to cultural blindness and culturally misaligned rationales defined in Chapter 2, and demographic misalignment. These limitations motivate the adaptive, human-centered framework proposed in Chapter 5.

4.1 Experimental Setup

To establish a baseline for multilingual information disorder assessment, the benchmark evaluated two open-weight Mixture-of-Experts LLMs: LLaMA 4 Maverick and Mixtral-8x22B-Instruct. Rather than relying on proprietary systems, this design foregrounded inspectable, contemporary baselines whose settings and outputs could be reported more transparently.

To ensure consistency across all experiments, both models were run with explicit decoding parameters set to a temperature of 1.0 and a top-p of 1.0, with streaming disabled. The benchmark

compared two prompting conditions. In the zero-shot setting, the models received only target-language instructions and were required to predict severity, extract problematic spans, and generate rationales relying entirely on their pretrained knowledge. In the static few-shot setting, these instructions were supplemented with language-matched examples to provide structural guidance. This typically followed a 2-shot format, occasionally increasing to a maximum of three examples for specific subtasks. The architectural specifications of the two LLMs evaluated are detailed in Table 4.1.

Model	Architecture	Parameters	Context Window	Access Route
LLaMA 4 Maverick	MoE	400B total, 17B active	1.05M tokens	OpenRouter API
Mixtral-8x22B-Instruct	MoE	141B total, 39B active	65.5K tokens	OpenRouter API

Table 4.1: Model specifications used in the baseline benchmark.

To ensure consistent output extraction, the models were instructed to return predictions using fixed formatting tags, such as <SEVERITY>, <SPANS>, and <EXPLANATION>. The evaluation was not conducted on the entire raw InDor corpus, but on filtered subsets constructed to preserve task validity. Instances were removed prior to testing if they were marked as “N/A” by human annotators (e.g., broken URLs or advertisements, as detailed in Section 3.2.2), lacked a valid explanation, or contained a rationale written in a language that did not match the target news item. As Table 4.2 details, this filtering resulted in different data quantities depending on the task requirements. The severity classification task used a broader subset of annotations since it only required a valid macro-level label. Conversely, the span and rationale generation tasks were evaluated on a much smaller, fully complete subset of text triplets. For full transparency and reproducibility, the exact prompt templates, system instructions, and filtered data splits are publicly maintained in the InDor project repository¹.

Language	Task 1: Severity		Task 2: Spans & Rationales	
	News Items	Annotations	News Items	Annotations
English	877	1,371	418	1,201
Italian	1,495	2,275	334	888
Russian	481	760	193	632
Farsi	932	1,507	660	1,862

Table 4.2: Amount of news items and annotations used per task after the filtering process, separating severity classification (Task 1) from span-explanation generation (Task 2). Adapted from InDor [4].

Finally, performance on these subsets was measured with a combination of automated and

¹<https://github.com/citizen-dataset/InDor>

human-centered criteria. Severity classification results are reported as label distributions across language subsets, while span extraction and rationale generation were evaluated with BERTScore [74] to measure semantic similarity against human references. A secondary analysis evaluated severity classification performance using F-score disaggregated by annotator gender and native language. In addition, the benchmark included a downstream 1–4 human evaluation of LLaMA 4 Maverick’s English-language rationales on the shared English subset [4], which is discussed further in Section 4.3.

4.2 Automated Metrics (Spans and Rationales)

Since the severity label distribution was presented and discussed in Chapter 3, this chapter focuses on the automated evaluation of span detection and rationale generation, where model outputs can be measured against human references.

As previously discussed in Section 3.4.1, human agreement for span selection remains low when measured with strict exact token match. This suggests the subjective difficulty of universally defining manipulative boundaries [4]. When evaluating the automated systems (LLaMA 4 Maverick and Mixtral-8x22B-Instruct) on this task, the InDor evaluation utilized BERTScore to accommodate the lexical variability of generated text.

As Table 6.9 indicates, static few-shot prompting yielded varying results across the linguistic subsets when compared to zero-shot prompting. While the introduction of language-matched examples improved span detection scores for the Russian and Farsi subsets across both models, it consistently degraded performance on the English subset. The Italian subset produced mixed results, improving under Mixtral but slightly declining under LLaMA.

Language	Setting	LLaMA		Mixtral	
		Span	Rationale	Span	Rationale
English	Zero-Shot	40.82	45.83	48.83	45.72
	Few-Shot	33.44	41.68	41.06	45.56
Italian	Zero-Shot	43.78	45.62	38.39	41.36
	Few-Shot	42.77	42.83	45.03	41.03
Russian	Zero-Shot	52.79	44.90	48.59	42.06
	Few-Shot	56.28	46.83	54.96	46.09
Farsi	Zero-Shot	38.35	50.72	49.69	45.60
	Few-Shot	60.93	53.37	63.28	44.30

Table 4.3: BERTScore performance for span extraction and rationale generation across zero-shot and few-shot prompting conditions. Adapted from InDor [4].

Beyond identifying problematic content, the study evaluated model-generated rationales through BERTScore-based comparison with human-written explanations. As detailed in Section 3.2.3, the human reference rationales were structured using a specific “If..., then...” format to explain why certain framing was perceived as manipulative. To automatically evaluate the models’ generative

outputs, the study relied on BERTScore to measure semantic similarity against these human-written references.

Referring back to the data presented in Table 4.3, the rationale scores indicate that static few-shot prompting did not provide uniform improvements over the zero-shot condition. Specifically, the introduction of language-matched few-shot examples resulted in lower rationale similarity scores for both the English and Italian subsets across LLaMA 4 Maverick and Mixtral-8x22B-Instruct. Conversely, rationale scores for the Russian subset slightly improved under the few-shot condition, while the Farsi subset produced mixed results, improving under LLaMA but declining under Mixtral [4]. These variations suggest that static structural guidance does not consistently increase the semantic similarity of generated rationales to human-written references.

4.3 Demographic Alignment and Human Evaluation

Beyond the multilingual span and rationale comparisons reported in Section 4.2, the InDor benchmark also examined demographic alignment in severity classification on a shared English subset. Instead of aggregating results across languages, this analysis focused on the same English news items evaluated by nine non-native English-speaking annotators (five native Italian speakers, three native Farsi speakers, and one native Russian speaker). By comparing LLaMA 4 Maverick’s severity predictions against these demographic clusters, the results showed distinct variations in alignment. As shown in Table 4.4, the model achieved an average F-score of 0.44 against male annotators, compared to 0.37 against female annotators.

Similar differences emerged when the results were grouped by the annotators’ primary languages. LLaMA 4 Maverick aligned more closely with the severity judgments of native Farsi speakers (0.47) than with those of native Italian speakers (0.36) when both groups evaluated the exact same English articles [4].

Using this shared English subset, the project also included a downstream human evaluation of the explanations generated by the model. Annotators were asked to review the model-generated rationale for each English news item and answer the question, “How much do you agree with the explanation provided by the model?”, using a 1–4 rating scale. This evaluation was conducted by the same pool of non-native English-speaking annotators, allowing the benchmark to compare perceived explanation quality across demographic groups.

Task	Gender		Primary Language	
	Men	Women	Italian	Farsi
Severity Classification (F-score)	0.44	0.37	0.36	0.47
Downstream Evaluation (1–4 Scale)	3.28	2.68	2.44	3.53

Table 4.4: Performance of LLaMA 4 Maverick on the English subset, detailing both F-score severity classification and downstream human evaluation of generated rationales. Results are stratified by annotator gender and primary language. Adapted from InDor [4].

As Table 4.4 indicates, the ratings varied significantly. Male annotators assigned a higher average agreement score (3.28) than female annotators (2.68), while Farsi-speaking annotators assigned a higher average score (3.53) than Italian-speaking annotators (2.44) [4]. Because these scores

were produced on the same English subset, they suggest that the perceived quality of LLaMA 4 Maverick’s explanations is not uniform, but dependent on the demographic and primary-language background of the reader.

4.4 Failure Analysis: Cultural Blindness and Demographic Misalignment

The preceding evaluations point to a limitation in static baseline prompting for multilingual information disorder assessment. Rather than suggesting that model outputs transfer neutrally across languages and annotator communities, the InDor results indicate that performance varies across task settings, language subsets, and evaluator groups. In the shared English-subset severity analysis, LLaMA 4 Maverick’s F-score ranged from 0.36 to 0.47 across demographic groupings, showing that alignment with human judgments shifted depending on which annotator perspective was used [4].

Likewise, the BERTScore results for span detection and rationale generation show that static few-shot prompting did not consistently improve performance across languages; in the English and Italian subsets, fixed examples sometimes moved the outputs closer to human references, but in other cases they had little effect or even reduced similarity [4]. This uneven pattern is consistent with broader culture-specific evaluation work: Persian studies report that role-based prompting yields only modest improvements without fully restoring cultural fidelity [7], while Russian cultural evaluation shows that models can still struggle with culturally embedded phrases even when they perform better on broader cultural knowledge items [8]. Taken together, these patterns suggest that neither zero-shot prompting nor rigid few-shot prompting can be treated as a fully reliable solution for capturing the subjective and culturally situated boundaries of manipulative framing.

Beyond the automated metrics, the downstream human evaluation also exposes an important limitation in how cultural alignment was assessed in the benchmark. Although the InDor project asked annotators to rate LLaMA 4 Maverick’s explanations, this evaluation was conducted only on the shared English subset and by non-native English-speaking annotators [4]. That design was appropriate for the goals of the original benchmark, which focused on broad comparative evaluation on a lingua franca, but it is less suitable for testing the stronger form of alignment pursued in this thesis: whether a model can generate culturally grounded rationales that native or culturally embedded readers judge as locally appropriate. Put differently, agreement with English explanations should not be taken as a full proxy for reasoning in Farsi, Italian, Russian, or other local media contexts. Evaluating that stronger form of alignment requires native-language assessment by native or culturally embedded readers, which motivates the later language-specific experiments developed in this thesis.

As defined in Chapter 2, this cultural blindness frequently manifests in how default models apply generic, English-centric heuristics to highly localized media narratives. The qualitative analysis of human rationales in the InDor dataset showed that native annotators often relied on deep, context-specific knowledge to identify manipulation. This is particularly visible in the Farsi subset, where annotators’ rationales frequently emphasized the selective use of information, specifically the absence of certain statistics or sources, as a demonstration of ideological framing [4].

The following illustrative scenario (Box 4.1), derived from these recurring annotation patterns in Farsi [4], shows the expected mismatch that arises when a baseline LLM evaluates such content, contrasting a hypothesized model output against the interpretive logic used by native annotators.

Box 4.1: Illustrative Case Scenario – Cultural Blindness in Farsi News

News Context: A Farsi news article attributing domestic economic challenges entirely to foreign sanctions using politically charged framing.

Hypothesized Static LLM Rationale

اگر مقاله داده‌های اقتصادی تأیید شده‌ای برای حمایت از ادعاهای خود ارائه ندهد، آنگاه مشکل‌ساز و سوداگرانه است.

[Translation:] “If the article does not provide verified economic data to support its claims, then it is problematic and speculative.”

Critique: Relies on a generic Western journalistic heuristic (source verification) while missing the actual mechanics of state-aligned economic propaganda.

Expected Native Human Rationale

اگر مقاله مسائل پیچیده اقتصادی داخلی را صرفاً به بازیگران خارجی نسبت دهد، آنگاه با تکیه بر یک سپر بلای ژئوپلیتیک آشنا، خواننده را از عوامل سیاست داخلی منحرف می‌کند.

[Translation:] “If the article attributes complex domestic economic issues solely to external foreign actors, then it distracts the reader from internal policy factors by relying on a familiar geopolitical scapegoat.”

Critique: Captures the locally salient manipulative framing and accurately identifies the ideological function of the text.

In evaluating such cases in Farsi, native annotators did not merely object to the absence of evidence in a general, epistemic sense. Rather, they interpreted the deliberate lack of statistics, selective use of information, and the framing itself as a familiar geopolitical strategy: redirecting attention away from internal policy failures by assigning exclusive blame to external actors.

By contrast, as illustrated in the scenario, a default LLM operating under English-centric assumptions would be expected to respond with a generic Western journalistic heuristic, focusing entirely on the lack of verified economic data. While such a model might correctly identify that evidence is missing, it suffers from cultural blindness regarding *why* it is missing. In a state-influenced information environment, treating source verification as a simple proxy for neutrality misses the underlying reality: the primary manipulation is not just insufficient statistical evidence, but the ideological function of using a geopolitical scapegoat to obscure domestic realities.

A similar contextual failure can also be observed in the Italian subset, where highly localized socio-political narratives carry meanings that are legible to native readers but less accessible to default LLM reasoning. In such cases, narratives centered on immigration, institutional conflict, or judiciary-government tension are interpreted by local annotators through culturally specific frames that a baseline model flattens into a generic explanation of sensationalism or bias. Framed in this way, the issue is not simply that the model fails to detect a problematic signal, but that it fails to recover the particular ideological tension through which that signal becomes meaningful in context.

Taken together, these qualitative and quantitative shortcomings point to a limitation of static prompting in multilingual information disorder assessment. Rather than showing that model outputs transfer smoothly across languages and annotator communities, the results suggest that zero-shot and

static few-shot settings can struggle to capture the localized and culturally embedded nature of manipulative narratives. Although models such as LLaMA 4 Maverick and Mixtral-8x22B-Instruct show strong multilingual generation capabilities, the combined automated and human-centered evaluations indicate that these baseline prompting settings do not always provide the semantic and cultural alignment needed to detect more subtle, region-specific forms of manipulation.

This gap, where a model may process the language fluently without fully capturing the cultural logic behind it, motivates the search for an alternative methodological approach. To address these limitations, the following chapters introduce a culturally adaptive retrieval-based framework designed to align LLM reasoning more directly with the contextual judgments of native readers.

4.5 Chapter Summary

This chapter evaluated the performance of two contemporary Large Language Models in detecting and explaining information disorder across linguistically and culturally diverse contexts. Drawing on both automated metrics and human alignment scores, the findings suggested that the evaluated LLMs, when deployed in zero-shot and static few-shot settings, face notable limitations in detecting manipulative framing and in aligning their outputs with the subjective, culturally appropriate judgments of human annotators.

Taken together, these findings speak to Research Question 2 by suggesting that current LLMs do not yet provide assessments and explanations that are consistently reliable or culturally aligned across multilingual information disorder settings. Rather than pointing to a universal failure of all LLMs under all conditions, the results indicate a clearer limitation of static zero-shot and few-shot prompting for the specific models and task settings examined in this chapter. They are also consistent with broader concerns in the literature regarding explainability gaps, cultural bias, and uneven cross-cultural and demographic alignment in current LLMs [11, 12, 15, 14, 5].

The following chapters therefore move from diagnosis to design, introducing and evaluating a culturally adaptive retrieval-based framework that explores whether aligning model reasoning with human-written rationales can help reduce cultural blindness and cross-cultural and demographic alignment disparities, thereby addressing Research Question 3.

Chapter 5

Proposed Framework: Culturally Adaptive Retrieval-Based Framework

This chapter moves from the empirical diagnosis of Large Language Model limitations to the design of a proposed retrieval-based framework for multilingual information disorder assessment. Building on the persistent issues of cultural blindness, misaligned rationales and demographic alignment disparities identified in Chapter 4, it introduces a culturally adaptive retrieval-based approach to explore whether LLM outputs can be oriented more consistently toward localized cultural interpretation.

The chapter begins by outlining the conceptual architecture of the system. It then defines the four experimental conditions, ranging from static zero-shot and few-shot baselines to retrieval-based settings, followed by the explaining the strategies for constructing an exemplar bank for news items along with their human-written rationales to be used in retrieval framework. Together, these components define the experimental design through which Research Question 3 is investigated in Chapter 6. The chapter concludes by considering the framework’s potential scalability across additional languages.

5.1 Architecture of the Culturally Adaptive Retrieval-Based Framework

As established in Chapter 1, the broader envisioned system follows a Human-in-the-Loop design [19]. This chapter defines and evaluates its foundational retrieval-based phase as a bounded pilot. As the preceding benchmarking evaluations suggest, contemporary Large Language Models continue to struggle to assess and explain multilingual information disorder when relying solely on their default parameters or static prompt structures. Also cultural blindness in current models may extend beyond translation gaps and reflect deeper structural limitations in how generic reasoning environments represent diverse cultural frames.

To address this gap, this chapter introduces the core experimental framework of the thesis: the culturally adaptive retrieval-based framework. This framework shifts the methodological focus

from static prompting toward a dynamic retrieval-based architecture that benefits from human-written judgements about manipulative news. By using retrieval-based In-Context Learning (ICL), this framework is intended to align generative reasoning with localized, culturally situated human judgments.

At a conceptual level, the architecture works by explicitly separating the logical structure of the task from the cultural nuances of the target narrative. Rather than requiring the LLM to rely on generic English-centric heuristics or fixed, language-matched examples that may not fit the specific framing of a new article, the system dynamically retrieves relevant human reasoning at inference time.

As illustrated in Figure 5.1, the operational workflow follows a four-stage design. In the offline preparation stage, all cleaned InDor annotations are encoded into dense vector representations using BGE-M3 [75], a multilingual sentence embedding model supporting over 100 languages, and stored in the Exemplar Bank as a pre-indexed retrieval repository that persists across all inference runs. Each entry in the bank is a fully annotated item. For articles judged as problematic by native annotators, the entry comprises the original news text, the selected problematic span, the severity label, and the human-written rationale explaining why the framing is manipulative. For articles judged as neutral, the entry contains only the news text and a *none* label, indicating the absence of information disorder. Together, these entries allow each retrieved item to function as a complete few-shot demonstration, more aligned with authentic human judgment, rather than a fixed or synthetically authored template. The cleaning and filtering process applied to these annotations before indexing is discussed later in this chapter and in further detail in Chapter 6.

At inference time, when an unseen target-language news article is introduced, it is encoded using the same BGE-M3 model to produce a query embedding. The system then computes cosine similarity between this query and all pre-encoded entries in the Exemplar Bank using the sentence-transformers library [76], identifying and extracting the top $k=4$ most semantically relevant exemplars. Since the severity label of an incoming article is unknown at inference time, retrieval is based entirely on narrative and semantic proximity rather than label matching. The article is being submitted precisely because it may or may not contain information disorder. This design also preserves the integrity of the evaluation by preventing the target label from influencing which exemplars are selected.

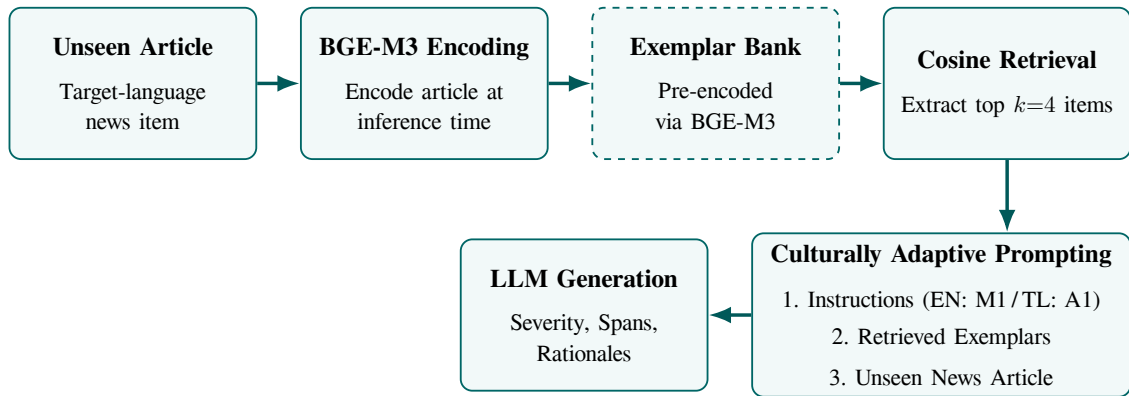


Figure 5.1: The Culturally Adaptive Retrieval-Based Framework.

Once retrieved, the top- k exemplars are assembled into a structured prompt. Depending on the

experimental condition detailed in Section 5.4, this prompt pairs the retrieved examples with either English task instructions (M1) or target-language instructions (A1). This dynamically constructed prompt is then passed to the Large Language Model, connecting its assessment in different tasks with the most contextually relevant localized human reasoning available for that specific article.

5.2 Static Baselines: Zero-Shot and Few-Shot Prompting

Before turning to the mechanics of the proposed framework, it is necessary to define the control conditions against which it will be evaluated. The two baseline configurations established here follow the static prompting setup used in the foundational InDor study [4]: a zero-shot and few-shot condition, both using fixed target-language instructions and a small number of fixed, pre-selected examples in the target language in the few-shot setup. Rather than comparing directly against the aggregate InDor results, which were produced under different settings and on a different data split, both conditions are re-run on the same held-out pilot test set and under the same decoding configuration as the adaptive conditions. This ensures that any observed differences between static and dynamic prompting reflect prompt design rather than variation in data, models, or generation parameters. These two setup serve as the reference point (B0 and B1) against which the proposed retrieval-based conditions are evaluated.

Table 5.1 outlines the main components shared across these two configurations and notes the methodological considerations that have motivated the move toward dynamic retrieval.

Because the prompt structure is fixed before inference, both configurations treat each incoming article in the same way regardless of its specific topic, framing, or cultural context. When a new article involves a regionally specific narrative or propaganda strategy that is not well represented by the static exemplars, the model may struggle to adapt its interpretive frame accordingly. This can contribute to the kind of culturally misaligned rationales discussed in Chapter 4, though the extent to which prompt design versus model pretraining drives this effect is one of the questions the pilot is designed to examine.

Beyond the structural prompt components described above, a closer inspection of the InDor baseline prompts [4] reveals several design choices that the present setup aims to address. **First**, the few-shot demonstrations in the InDor baseline provide inconsistent label coverage, targeting two and in some cases three examples per prompt. The None category is not consistently demonstrated to the model as a signal for neutral news content, meaning the model receives little in-context signal for how to handle a non-problematic article at the span or rationale level. **Second**, while the InDor baseline instructions define what makes a span problematic or gives examples for term “problematic” in the classification task, the prompts do not situate those criteria within the Information Disorder framework [3] or name the three subcategories of misinformation, disinformation, and malinformation that ground the task theoretically. The model is given a detection checklist without the conceptual context that motivates it. **Third**, the InDor baseline runs all three tasks for every item regardless of the severity prediction, meaning that span detection and rationale generation are executed even when the model has already determined that a news article contains no problematic content, resulting in unnecessary API calls and potential signal confusion when the model is asked to explain spans that do not exist.

The present setup addresses all three of these gaps. The task instructions explicitly introduce the concept of information disorder and define it in terms of misinformation, disinformation, and malinformation, aligning the model’s framing of the task with the theoretical framework established

Prompt Component	Function	Methodological Consideration
Target-Language Instructions (B0 & B1)	Specifies the rules for severity classification, span extraction, and rationale generation, written directly in the target language.	Translating complex logical constraints from English into the target language may, in some cases, affect instruction-following fidelity, particularly for models whose instruction-tuning data is predominantly English-based [77].
Static Few-Shot Exemplars (B1 only)	Provides two to three fixed, pre-selected examples of information disorder in the target language to guide the model’s output structure.	Because the same examples are used for every input regardless of its topic or framing, the model may be less able to adapt to the specific cultural or narrative context of a new article, a limitation that dynamic retrieval is designed to address.
Target News Item (B0 & B1)	The unseen, unannotated news article that the model is asked to evaluate.	In the absence of contextually relevant in-context guidance, the model may draw more heavily on generic pretrained associations rather than on locally aligned interpretive patterns.

Table 5.1: Components of the static target-language prompting baselines (B0 and B1) and the methodological considerations that motivate the proposed shift toward dynamic retrieval.

in Chapter 1. The three tasks of severity classification, span extraction, and rationale generation are chained sequentially through a None bypass applied after Task 1: when the model classifies a news article as None, Tasks 2a and 2b are skipped entirely and only the severity prediction is saved. The exemplar design and its handling of neutral articles in the few-shot conditions are discussed in Section 5.3.

5.3 Building the Exemplar Bank from Community Annotations

An important part of the proposed framework is the Multilingual Exemplar Bank, a retrieval repository designed to connect the Large Language Model with localized cultural knowledge. As introduced in Chapter 1 and detailed in the full proposed framework [19], the curation design envisions a two-phase protocol consisting of corpus mining and active expert correction. In this pilot, only the first phase is implemented: the bank is seeded directly from filtered native-speaker annotations extracted from the InDor corpus [4]. The raw community annotations are not used directly, however. Because in-context learning is highly sensitive to the composition of demonstrations [78], contradictory or noisy annotations could interfere with the model’s generation process. The system therefore applies a four-step article-level filtering and deduplication pipeline to transform the raw InDor data into a stable, clean retrieval bank.

While this pipeline is conceptually organized around four main steps, the actual implementation involved a considerably larger number of sub-operations, including manual verification passes and language purity checks, whose full details and resulting corpus reduction figures are provided in Chapter 6.

- **Filtering Unusable Records:** The pipeline first removes annotations flagged as N/A, such as broken URLs or irrelevant advertisements. For an annotation to qualify as a valid Problematic exemplar, it must include a valid severity label, a selected text span, and a human-written rationale. None-labelled items are retained as long as they carry no conflicting annotation data.
- **Excluding Binary Conflicts:** Because information disorder is a highly subjective task, some articles receive fundamentally conflicting judgments from different annotators. While perspectivist NLP recognizes annotator disagreement as a meaningful signal rather than noise [57], the goal of this pilot is to evaluate whether a single coherent cultural reference point can improve model alignment. Providing the model with contradictory exemplars would conflate two separate phenomena: the inherent subjectivity of the task and the model’s ability to use culturally grounded guidance. Articles where one annotator assigns None and another assigns Problematic are therefore excluded from the Exemplar Bank, keeping the retrieval signal clean and the pilot’s scope interpretable. For example, if the same article were retained twice, once labelled as *None* by one annotator and once as *Moderately* by another, the retrieval mechanism could surface both versions as top- k matches for a semantically similar unseen article, presenting the model with directly contradictory guidance on whether the content is problematic at all. The tension between perspectivist annotation and single-reference ICL design is acknowledged as a limitation in Chapter 7.
- **Resolving Severity Conflicts:** When multiple annotators agree that an article is Problematic but differ in severity, span selection, or rationale wording, the pipeline retains only one representative exemplar per article. Because demonstration label words can act as anchor-like references during in-context learning [79, 80], retaining competing versions of the same article could introduce conflicting guidance during retrieval. When more than one candidate remains after deduplication, the system applies a tie-breaking heuristic based on rationale length. This criterion is adopted for three reasons. First, it avoids manual selection and keeps

the filtering process transparent. Second, in a subjective task like information disorder assessment, longer rationales tend to provide more complete and informative explanations. Third, they contain richer contextual detail, which may support more effective retrieval of culturally relevant examples. Although length is not a direct measure of quality, it provides a practical and consistent heuristic when no adjudicated reference rationale is available. The potential influence of this heuristic on human evaluation results is noted in the study’s limitations.

- **Test-Bank Split:** After deduplication, the cleaned resource is split into a held-out pilot test set and the active Exemplar Bank. This separation ensures that no test article can appear in the retrieved prompt context, preventing any form of data contamination between evaluation and retrieval.

An important design choice in constructing the Exemplar Bank concerns the handling of class distribution and retrieval filtering. In many classification settings, datasets are artificially balanced, for example through a strict 50–50 split between positive and negative instances, in order to reduce training bias. The present framework does not impose this kind of artificial balance on the retrieval bank. Uncontested None items are retained in the cleaned article pool alongside Problematic items, but no fixed ratio between the two categories is enforced. By preserving the natural class skew of the source corpus, the Exemplar Bank remains closer to the actual distribution of the localized information environment.

At retrieval time, the system does not filter candidate exemplars by severity label. Since the label of an incoming article is unknown at inference time, filtering by label would require prior knowledge that is simply not available, and would introduce data leakage into the evaluation. The article is being submitted precisely because it may or may not contain information disorder. Retrieval is therefore based solely on dense semantic similarity. Depending on the topical and narrative proximity of the unseen article to the bank contents, any given prompt may receive all Problematic exemplars, all None exemplars, or a mixed set. This label-agnostic design keeps the retrieval naturalistic and avoids artificially inflating the model’s exposure to either category. It is also consistent with the foundational InDor study, which showed that severity is a subjective human judgment rather than a stable semantic feature, as reflected in the low inter-annotator agreement scores reported among native speakers [4]. Basing retrieval on narrative and semantic proximity rather than on contested severity labels therefore preserves both the integrity of the pilot and the conceptual logic of the task.

Through this pipeline, the Exemplar Bank becomes a stable, culturally aligned set of demonstrations that can be dynamically retrieved by the adaptation mechanism described in the following section.

5.4 The Adaptation Mechanism: Culturally Adaptive Prompting

Having defined the composition and filtering of the Exemplar Bank, the next component of the architecture is dynamic prompt assembly. This stage, termed Culturally Adaptive Prompting, is designed to address the structural limitations of the static baselines outlined in Section 5.2.

The idea behind this mechanism is the explicit separation of the task’s logical instructions from its cultural references. Research on multilingual prompting suggests that instruction-tuned Large Language Models tend to show stronger logical adherence and output formatting when prompted

in English, partly because English dominates most instruction-tuning pipelines [77]. At the same time, this advantage is not straightforward in all settings. Muennighoff et al. [81] show that while models finetuned on English prompts generalize broadly across non-English tasks, models trained on target-language prompts tend to perform better specifically on target-language inputs, suggesting that matching the instruction language to the content language can matter for locally aligned performance.

For a task like information disorder assessment, where the retrieved exemplars are written in the target language and carry culturally specific framing, using English instructions while injecting target-language content into the same prompt may create a mismatch that the model resolves in unpredictable ways. This uncertainty is precisely what motivates the inclusion of the A1 condition, which uses the same dynamically retrieved exemplars as M1 but switches the instruction language to the target language, allowing the evaluation to isolate the contribution of instruction language from that of exemplar retrieval independently. The empirical interaction between these two components is examined in Chapter 6.

To investigate whether this separation can improve cultural connection, the pilot defines four prompting conditions that vary the instruction language and the exemplar source independently. Figure 5.2 provides a visual overview of the four conditions.

- **B0 (Zero-Shot Baseline):** Target-language instructions only, with no in-context exemplars. The model relies entirely on its pretrained knowledge.
- **B1 (Static Few-Shot Baseline):** Target-language instructions paired with a fixed set of pre-selected same-language exemplars. This follows the foundational InDor prompting setup and serves as the primary reference point for the adaptive conditions.
- **M1 (English Instructions + Dynamic Retrieval):** English task instructions paired with dynamically retrieved same-language exemplars from the Exemplar Bank. This condition tests whether English instructions support stronger logical adherence while the retrieved target-language exemplars provide cultural alignment.
- **A1 (Target-Language Instructions + Dynamic Retrieval):** The same dynamically retrieved exemplars as M1, but with target-language task instructions. This condition isolates the effect of instruction language by keeping the retrieval mechanism identical to M1.

The goal of this four-condition setup is not to establish a definitive solution, but to test whether using human-written, culturally situated exemplars through dynamic retrieval can shift LLM outputs toward more locally aligned reasoning, and to examine how instruction language interacts with that retrieval. The execution of this setup, including the comparative results and an analysis of how format patterns embedded in retrieved exemplar rationales may influence model output style, is presented in Chapter 6.

The retrieval mechanism operates as follows. For a target news item x_{test} in language ℓ , the system queries the corresponding Exemplar Bank E_ℓ , a dedicated per-language retrieval index containing only annotations in ℓ . Each candidate exemplar $e_i \in E_\ell$ and the query item are encoded by BGE-M3 [75] and ranked by cosine similarity:

$$\text{sim}(x_{test}, x_i) = \frac{f_{emb}(x_{test}) \cdot f_{emb}(x_i)}{\|f_{emb}(x_{test})\| \|f_{emb}(x_i)\|} \quad (5.1)$$

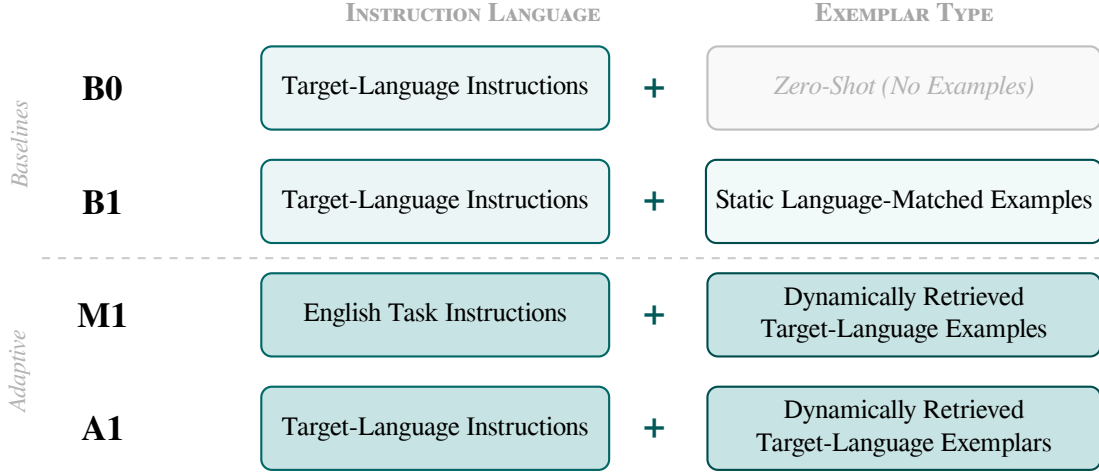


Figure 5.2: The four prompting conditions used in the pilot. B0 and B1 are static baselines following the InDor setup. M1 and A1 introduce dynamic retrieval from the Exemplar Bank and differ only in instruction language, allowing the effect of instruction language to be isolated from the effect of exemplar retrieval.

The retrieval output is an ordered set of the top- k references, $R_k(x_{test}) = \{e_1, \dots, e_k\}$, where each tuple $e_i = (x_i, y_i^{\text{span}}, y_i^{\text{sev}}, r_i)$ contains the news text, span annotation, severity label, and human-written rationale. For uncontested *None* items, the span and rationale fields are passed as ["No"], signalling to the model that no manipulative framing is present in that exemplar. The retrieved exemplars are injected into the prompt immediately after the task instructions and before the unseen target article.

5.5 Evaluation Framework and Performance Metrics

To assess whether Culturally Adaptive Prompting can reduce cultural blindness relative to the static baselines, the framework’s outputs are evaluated across three dimensions: severity classification, span extraction, and rationale generation. Automated metrics are applied across all four prompting conditions (B0, B1, M1, and A1), while the human evaluation focuses specifically on a direct comparison between the static few-shot baseline (B1) and the adaptive condition (M1), as these represent the two most structurally distinct setups. Because the data preparation pipeline resolves annotation conflicts to have a single deduplicated reference per article, all automated evaluations are conducted against these stable, article-level labels to provide a controlled comparison across the four conditions.

5.5.1 Automated Performance Metrics

The pilot evaluates three output dimensions, each assessed with a targeted automated metric.

Severity Classification: Macro-F1. The accuracy of article-level severity predictions is measured using Macro-F1, which computes the unweighted average of per-class F1 scores across all four severity categories. Macro-F1 is preferred over accuracy or weighted-F1 because the test set

is naturally imbalanced, with a higher proportion of None-labeled items, and treating each class equally prevents the metric from being dominated by the majority class. While the InDor study evaluated severity predictions against distinct demographic subgroups to examine annotation variance and subjective alignment patterns [4], the present pilot evaluates predictions against the deduplicated article-level reference. This is necessary to maintain a stable and consistent ground truth for comparing the isolated effects of the four prompting conditions against one another.

Span Detection: Token-Overlap F1. The accuracy of problematic span extraction is measured using a best-match token-overlap F1 score. The InDor study used exact token-match accuracy to estimate human agreement on span selection and BERTScore for generative span evaluation [4]. In this pilot, token-overlap F1 is adopted because exact matching can be overly strict for generated span boundaries, penalising outputs that correctly identify a manipulative region but differ slightly in phrasing or tokenization, while pure semantic similarity metrics do not directly evaluate boundary precision. Token-overlap F1 provides a more suitable balance between strictness and partial credit. This metric is computed over whitespace-tokenized text rather than a subword or linguistic tokenizer. For uncontested None items, where the reference span is ["No"], the F1 score is defined as 1.0 if the model correctly outputs ["No"] and 0.0 if it incorrectly extracts a text span.

Rationale Quality: BERTScore F1. The semantic quality of generated rationales is assessed using F1 component of BERTScore [74], which measures dense contextual similarity between a generated explanation and the corresponding human-written reference rationale. Because information disorder rationales can be expressed in highly varied ways while conveying equivalent meaning, exact lexical matching would be unsuitably strict. BERTScore captures conceptual equivalence across different surface forms, making it appropriate for evaluating explanations that may differ in wording but align in their cultural and analytical interpretation. When an article has more than one predicted or reference rationale, all rationales on each side are concatenated into a single text before the BERTScore comparison is computed. Together, these three automated metrics provide a structured assessment of severity classification, span extraction, and rationale quality across all four prompting conditions before moving to the human judgment.

5.5.2 Human Evaluation

While automated metrics capture important dimensions of model performance, they cannot fully assess whether a generated rationale resonates with the cultural and political reality of the community the news article belongs to. A model may produce a fluent and semantically coherent explanation that nonetheless relies on generic, anglocentric assumptions that native speakers of the target language would immediately recognise as misplaced. To measure this dimension directly, the pilot includes a human evaluation in which native-speaking community members rate the cultural appropriateness of model-generated rationales.

The InDor study conducted a comparable evaluation in which 5 Italian-speaking and 3 Farsi-speaking annotators rated their agreement with explanations generated by LLaMA 4 Maverick for the shared English subset of the corpus, on a 1–4 scale [4]. That evaluation was designed to study human-model alignment patterns across demographic groups as part of a broader corpus analysis, and was not intended as a comparison between prompting conditions. The present evaluation builds on that foundation but differs in three important ways. **First**, evaluators assess rationales generated for news articles in their own language rather than in English, which is a more ecologically valid test of cultural alignment. **Second**, the evaluation is comparative: rather than rating a single model's output in isolation, evaluators see paired rationales from B1 and M1 and rate each independently,

providing a direct within-item comparison between the static and adaptive conditions. **Third**, the evaluator pool is more balanced, with 7 native speakers per language for both Italian and Farsi. To keep the human evaluation focused and the cognitive load on evaluators manageable, rationales from only one model are submitted for human assessment.

The specific model is selected based on overall automated metric performance across both languages, as determined by the results reported in Chapter 6. Evaluator demographics, including gender, are collected through consented self-report and used for exploratory subgroup comparisons. The evaluation is administered through an anonymous Google Forms questionnaire, with all responses collected without any link to evaluator identity.

The evaluation is structured in two sections. **The first section** presents a paired A/B comparison between B1 and M1. Each evaluator sees ten news items in their target language, each accompanied by two rationales generated by the two conditions in randomized, anonymized order to prevent position bias. The ten items were drawn by random sampling with a fixed random seed from the pool of eligible articles, defined as those where both B1 and M1 generated at least one rationale and where the human reference label is Problematic. Items where both setups produced no rationale, or where both generated a rationale for a human-reference-None article, were excluded, as neither case offers a meaningful basis for human comparison. The sample of ten items per language was chosen to keep evaluator load manageable and to maintain a consistent sample size across Italian and Farsi, given that the pool of eligible items differs between the two languages after filtering. The InDor study does not report the exact number of items used in its downstream evaluation [4], so a direct numerical comparison of sample sizes is not possible, but the present design prioritises evaluator quality and consistency over coverage.

Because models sometimes generate more than one rationale per article, a span-based selection procedure is applied before the evaluation to ensure comparability within each pair. For each eligible item, the procedure identifies the B1 and M1 span annotations with the highest token-level overlap and selects their corresponding rationales as the paired comparison. When the two setups identified completely different regions of the text with no measurable span overlap, the procedure falls back to the human reference span: for each setup independently, the rationale whose corresponding span best matches the human-annotated reference is selected. In the rare case where neither setup produces a span that overlaps with the human reference annotation, the first generated rationale from each is used as a conservative default. This ensures that wherever possible the two rationales being compared address the same region of the news text, making the human judgment more direct and meaningful.

A separate section of the form presents asymmetric cases, in which one setup identified bias in a news item while the other produced no rationale. These items are included only from articles where the human reference label confirms the item is Problematic, allowing the evaluation to distinguish between genuine detection and hallucinated output in cases where the two conditions diverged. In these cases the evaluator sees a single rationale along with the news itself to assess whether the explanation is valid and grounded in the news text.

Across both sections, evaluators assess each rationale against three criteria communicated through a short rubric provided at the start of the questionnaire in their native language:

- **Textual grounding:** The reasoning should be anchored in the actual words and content of the news item rather than in general claims that could apply to any article.
- **Cultural sensitivity:** The explanation should capture the specific political, media, or socio-cultural framing relevant to the target community, reflecting the kind of contextual awareness

a native speaker would naturally bring to the task.

- **Absence of generic assumptions:** The explanation should avoid anglocentric or monocultural interpretive frames that a native speaker would recognise as foreign or misplaced in their local context.

Evaluators rate each rationale on a 1–4 scale, where 1 indicates a weak or culturally inappropriate explanation and 4 indicates an excellent and locally aligned one. The primary analysis compares mean scores for B1 and M1 within each language community and across evaluator subgroups. To examine whether dynamic retrieval reduces alignment disparities, the absolute difference in mean scores between subgroups is computed as $\Delta = |\bar{s}_{\text{Group A}} - \bar{s}_{\text{Group B}}|$ for gender and language community comparisons. A reduction in Δ from B1 to M1 would indicate that the adaptive condition produces rationales perceived more consistently across different evaluator backgrounds. The In-Dor baseline provides a broader contextual reference: on the shared English subset, LLaMA 4 Maverick aligned more closely with male annotators than female annotators, and more closely with Farsi-speaking annotators than Italian-speaking ones, producing a gender gap of $\Delta = 0.60$ and a language community gap of $\Delta = 1.09$ on the downstream rationale evaluation (computed from mean scores reported in [4]). These patterns serve as a broader contextual reference point rather than as a direct benchmark given the differences in task design, evaluation language, and evaluator composition between the two studies.

The complete evaluation guidelines provided to human evaluators, including task instructions and rating criteria, are presented in Appendix A. Separate versions were prepared for Farsi and Italian evaluators, each presented in the respective target language. Appendix A.1 contains the Farsi guidelines and Appendix A.2 the Italian guidelines. The forms were distributed digitally and completed anonymously, only providing gender (with "I prefer not to say" option included) and age range.

5.6 Scalability and Future Extensions

Because the framework follows a retrieval-augmented design, it exhibits a distinct architectural quality: structural extensibility. The generative component (the LLM) and the cultural knowledge base (the Exemplar Bank) are cleanly separated. Because cultural alignment is supplied dynamically at inference time through retrieval-augmented In-Context Learning, the model’s internal parameters do not need to be modified when the system is adapted to a new language or a new cultural context. Extending the framework to an additional language therefore requires only the construction of a new dedicated per-language Exemplar Bank from the relevant target-language annotations, completely bypassing the need for model retraining. This modularity is an inherent feature of the retrieval-based design itself, remaining independent of the pilot’s empirical outcomes and holding true regardless of the specific model or embedding encoder used. The broader implications of this extensibility, including the potential to incorporate Russian-language annotations, adapt to temporal drift in information disorder patterns, and explore lightweight fine-tuning as a longer-term direction, are discussed in Chapter 7 alongside the study’s limitations and future work.

5.7 Chapter Summary

This chapter introduced the design of a culturally adaptive retrieval-based framework intended to investigate whether dynamic retrieval of human-written exemplars can reduce the cultural blindness and culturally misaligned rationale generation observed in contemporary Large Language Models. Moving beyond the limitations of static zero-shot and few-shot prompting, the proposed architecture separates the logical task instructions from their cultural references, connecting the model’s reasoning with localized, native-speaker annotations drawn from the InDor corpus.

The chapter first defined the two static baseline conditions, B0 and B1, which follow the foundational InDor prompting setup and serve as the experimental control against which the adaptive conditions are evaluated. It then described the construction of the Multilingual Exemplar Bank through a four-step filtering and deduplication pipeline applied to the raw InDor annotations, producing a stable, conflict-free retrieval repository that preserves the natural class distribution of the source corpus. The core adaptation mechanism, Culturally Adaptive Prompting, was then introduced alongside the four experimental conditions: B0, B1, M1, and A1. These conditions systematically vary the instruction language and the exemplar source to isolate the individual contributions of retrieval and instruction language to model performance. The chapter also described the retrieval pipeline through which unseen articles are encoded and matched against the Exemplar Bank via cosine similarity to extract the most semantically relevant human-written exemplars for prompt injection.

The evaluation framework was then outlined across two levels. Automated metrics cover all four conditions: Macro-F1 for severity classification, token-overlap F1 for span detection, and BERTScore for rationale quality. The human evaluation focuses on a direct comparison between B1 and M1, in which 7 native speakers per language rate paired rationales on a 1–4 scale across three rubric dimensions: textual grounding, cultural sensitivity, and the absence of generic angli-centric assumptions. Alignment disparity across evaluator subgroups is measured as the absolute difference in mean scores between demographic groups and compared across conditions.

Having outlined the architecture, the curation strategy, the experimental conditions, and the evaluation design, the thesis now moves from design to implementation. Chapter 6 presents the multilingual pilot of this framework on Farsi and Italian news items, reporting automated performance across all four conditions and the results of the downstream human evaluation.

Chapter 6

Implementation and Pilot Evaluation

Chapter 5 introduced the design of the Culturally Adaptive Retrieval-Based Framework, outlining the construction of the Multilingual Exemplar Bank, the four experimental prompting conditions, and the evaluation criteria against which the framework would be assessed. This chapter moves from design to implementation, reporting the full execution of the pilot study and its findings. The evaluation is organised around Research Question 3, which asks whether the framework improves automated detection, span extraction, and rationale generation quality over static baselines for information disorder in Farsi and Italian, as measured by both automated metrics and native-speaker human evaluation. In doing so, it provides an empirical test of whether the framework can meaningfully reduce cultural blindness and improve culturally aligned rationale generation, as identified as core limitations in Chapter 1.

To address this question, the chapter first describes the technical implementation environment and the data preparation steps applied before the pilot ran. It then presents the quantitative results across all four prompting conditions for both languages and both models, followed by the human evaluation of rationale cultural appropriateness. The chapter closes with a discussion of the main findings, including the patterns that emerged most clearly across conditions, the unexpected instruction-exemplar interaction observed in one of the adaptive conditions, and the limitations of the bounded pilot design.

6.1 Implementation Environment

All experiments were conducted in Google Colab. Model generations were produced via the OpenRouter API¹, which provides unified access to large language model providers under a single inference interface. Two models were selected for the pilot, maintaining continuity with the InDor study [4]: LLaMA 4 Maverick [82] and Mixtral-8x22B-Instruct [83]. Table 6.1 summarises their key specifications.

¹<https://openrouter.ai/>

Model	Architecture	Parameters	Context Window
LLaMA 4 Maverick	MoE	400B total, 17B active	1.05M tokens
Mixtral-8x22B-Instruct	MoE	141B total, 39B active	65.5K tokens

Table 6.1: Specifications of the two models used in the pilot, accessed through the OpenRouter API

All generations were produced with a fixed set of decoding parameters to ensure reproducibility and eliminate stochastic variation. Temperature was set to 0.0, which produces deterministic outputs by always selecting the highest-probability token at each generation step. Top-p was set to 1.0, its default value, meaning no nucleus sampling restriction was applied. Streaming was disabled. A generation seed of 42 was passed to the API for all calls, ensuring that any run of the same prompt against the same model produces identical outputs. While the original InDor study used a temperature of 1.0 [4], the present pilot uses temperature 0.0 to keep the comparison across the four prompting conditions focused on prompt design and retrieval strategy rather than sampling randomness.

For the retrieval component, the Exemplar Bank for each language was encoded offline in advance using BGE-M3 [75], a multilingual dense sentence embedding model. The bank embeddings were stored and loaded at inference time via the `sentence-transformers` library [76]. At inference time, each unseen test article was encoded on the fly using the same BGE-M3 model, and cosine similarity was computed between the article embedding and all pre-stored bank embeddings to retrieve the top $k=4$ most semantically relevant exemplars. Standard dense retrieval was used throughout, without any hybrid keyword matching or reranking. For the human evaluation item selection, a fixed random seed of 42 was used when sampling the ten items per language from the eligible pool, ensuring that the selected items are reproducible and independently verifiable.

6.2 Corpus Preparation: Cleaning Pipeline and Bank Statistics

Before the pilot could run, the raw InDor annotations required substantial preparation to produce both the held-out test set and the Exemplar Bank. While Chapter 5 outlined the four conceptual phases of the pipeline, the actual implementation involved eight discrete steps applied in a deliberate order, as the sequence in which filters were applied affected the final composition of both resources. The raw annotation files for Italian and Farsi were taken directly from the InDor corpus [4]. Because each article received on average approximately 1.88 Italian and 1.95 Farsi annotations [4], the pipeline operates at the annotation level rather than the article level. The raw counts of 3,109 Italian and 1,949 Farsi annotation rows therefore correspond to the 1,656 Italian and 1,000 Farsi articles reported in the InDor corpus statistics.

The annotation files were assumed to contain only items in their respective languages. In practice, however, the web crawling that produced the original corpus had gathered a proportion of non-news pages alongside valid articles, and some of these had entered the language-specific files without being filtered during InDor’s own collection phase.

A further source of noise was annotator behaviour: some annotators assigned valid severity

labels to crawling artifacts such as advertisements and broken pages rather than flagging them as N/A, and a subset of annotators in the Italian community wrote their rationales in English despite working on Italian-language news. The following steps were applied iteratively to address these issues and produce a clean, conflict-free resource suitable for both evaluation and retrieval.

- **Step 1: Removin invalid and N/A labels.** Annotations with missing, null, or N/A labels were removed first. N/A was the intended signal for unevaluable content such as broken pages, but a significant number of such items had been assigned valid severity labels by annotators and therefore passed through this filter into the subsequent steps.
- **Step 2a: Applying manual blacklist.** After an initial diagnostic pass, a manual blacklist of article identifiers was compiled through human review of the remaining non-news content. Items in this list included advertisements, JavaScript-disabled error pages, and non-target-language articles that had received valid severity labels and could not be caught by automated filters alone. The Italian blacklist removed 178 annotation rows and the Farsi blacklist removed 106 rows.
- **Step 2b: Remove structural body-text errors via regex.** A systematic check on these non-news items was added here. A small set of additional non-news pages was identified by matching specific phrase patterns in the article body text, such as prompts requiring JavaScript activation or platform-specific error messages. This step removed 2 Italian and 7 Farsi rows not covered by the manual blacklist.
- **Step 3: Removing incomplete Problematic items.** Any annotation with a Problematic severity label but missing a span selection or rationale was removed. None-labeled items were not subject to this filter, as they do not require span or rationale fields by design.
- **Step 4: Removing contaminated None items.** None-labeled annotations that incorrectly contained span or rationale data were removed. These represented cases where annotators had filled the fields that should have remained empty, producing structurally inconsistent entries.
- **Step 5: Removing English rationale leakage.** For Problematic items, the language of the rationale text was detected automatically using `langdetect`. Any item whose rationale was identified as English was removed from the Italian and Farsi subsets. This form of leakage was more prevalent in Italian, where the larger and more diverse annotator pool did not fully followed annotation guidelines.
- **Step 6: Excluding binary conflicts.** Articles for which one annotator assigned a None label while another assigned a Problematic label were excluded entirely. This step produced the largest reduction in the Farsi subset, showing the higher degree of interpretive disagreement among Farsi annotators, consistent with the low inter-annotator agreement reported in Chapter 3.
- **Step 7: Deduplicating by rationale length.** Where multiple annotations for the same article remained after all previous filters, a single representative annotation was retained using rationale length as a tie-breaking criterion, described in Chapter 5.
- **Step 8: Final article-level integrity check.** Before splitting, a manual inspection of the clean pool identified a small number of non-Italian articles in news body and duplicate texts

in the Italian subset that had passed through all previous filters. A targeted language detection on main news body column identified and removed 48 non-target-language articles and 7 duplicate texts with different identifiers. No equivalent anomalies were found in the Farsi subset.

The cleaned pool was divided into a held-out pilot test set of 100 items per language and an Exemplar Bank, using stratified random sampling with seed 42 to preserve the natural severity label distribution in both partitions. Table 6.2 reports the step-by-step reduction in corpus size across both languages.

Step	Operation	Italian	Farsi
0	Raw annotations	3,109	1,949
1	Remove invalid / N/A labels	−665	−243
2a	Manual blacklist	−178	−106
2b	Body-text regex filter	−2	−7
3	Remove incomplete Problematic items	−297	−10
4	Remove contaminated None items	−52	−86
5	Remove English rationale leakage	−12	−56
6	Exclude binary conflicts	−384	−575
7	Deduplicate by rationale length	−473	−309
8	Article-level integrity check	−55	—
	Clean pool	991	557
	Final Test set	100	100
	Final Exemplar Bank	891	457

Table 6.2: Step-by-step corpus reduction across the cleaning pipeline for Italian and Farsi. Negative values indicate rows or articles removed at each step.

The two languages entered the pilot with notably different label compositions. Tables 6.3 and 6.4 compare the severity distributions in the original InDor corpus [4] against those in the clean pool, the pilot test set, and the Exemplar Bank for Italian and Farsi respectively. For the master clean pool, test set, and bank, both the percentage and the number of items per label are reported.

The Italian distribution shifted notably toward None relative to the original InDor proportions, primarily because the cleaning pipeline removed a disproportionate number of Problematic annotations that were incomplete, conflicting, or contaminated. The Farsi distribution is closer to the original, though the Slightly category increased relative to InDor due to the removal of a large number of binary-conflicting annotations, many of which involved contested Problematic items. In both languages, the clean pool, test set, and Exemplar Bank share closely aligned distributions, confirming that the stratified split preserved the natural class skew of each language and provides a consistent basis for evaluation and retrieval.

Italian Severity Distribution				
Label	InDor (%)	Clean Pool % (n)	Test % (n)	Bank % (n)
None	56.7%	68.8% (682)	69.0% (69)	68.8% (613)
Slightly	18.1%	9.2% (91)	9.0% (9)	9.2% (82)
Moderately	13.1%	9.2% (91)	9.0% (9)	9.2% (82)
Highly	12.1%	12.8% (127)	13.0% (13)	12.8% (114)
Total		991	100	891

Table 6.3: Severity label distributions for Italian across the original InDor corpus, the clean pool, the pilot test set, and the Exemplar Bank. InDor percentages are taken from [4]. Counts and percentages are reported for the clean pool, test set, and bank. Percentages are rounded to one decimal place.

Farsi Severity Distribution				
Label	InDor (%)	Clean Pool % (n)	Test % (n)	Bank % (n)
None	43.1%	30.7% (171)	31.0% (31)	30.6% (140)
Slightly	41.6%	46.7% (260)	47.0% (47)	46.6% (213)
Moderately	11.0%	16.2% (90)	16.0% (16)	16.2% (74)
Highly	4.3%	6.5% (36)	6.0% (6)	6.6% (30)
Total		557	100	457

Table 6.4: Severity label distributions for Farsi across the original InDor corpus, the clean pool, the pilot test set, and the Exemplar Bank. InDor percentages are taken from [4]. Counts and percentages are reported for the clean pool, test set, and bank. Percentages are rounded to one decimal place.

6.3 Experimental Design: Pilot Study on Farsi and Italian

This section describes the structure of the four prompting conditions evaluated in the pilot and the design decisions applied across all of them. As introduced in Chapter 5 and illustrated in Figure 5.2, the four conditions systematically vary two factors, instruction language and exemplar source, to isolate their individual contributions to model performance. All four conditions were evaluated on the same test set of 100 items per language, using both LLaMA 4 Maverick and Mixtral-8x22B-Instruct under identical generation parameters. Full prompt templates for both languages and all conditions are provided in Appendix A, and will be referred throughout the section when necessary. Moreover, all execution scripts are available in the project repository.²

²<https://github.com/Maziarkiani/culturally-adaptive-prompting>

6.3.1 Shared Prompt Architecture and Pipeline Design

Across all four conditions, both languages, and both models, the pipeline follows the same three-task sequential structure. Each article is processed through three separate API calls issued in a fixed order: severity classification, span detection, and rationale generation. Severity classification and span detection both receive only the article text as input and are evaluated independently of each other. Rationale generation is conditioned on the predicted spans from the span detection step, and the model is asked to explain the specific text regions it identified rather than being given the human reference spans. This design ensures that rationale quality scores reflect end-to-end pipeline performance: errors in span detection propagate into rationale generation, and the evaluation captures this dependency. While Task 2b is shown as downstream of Task 2a in Figure 6.1, it receives two inputs assembled by the pipeline orchestrator: the predicted spans from Task 2a and the original article text loaded directly from the test set, not passed through Task 2a.

A None short-circuit bypass is applied at the code level after severity classification. When the model predicts None, the span detection and rationale generation calls are skipped entirely, and both outputs are recorded as ["No"]. This differs from the original InDor baseline [4], which ran all three tasks for every item regardless of severity prediction, and where the model was relied upon to output ["No"] spans and rationales through its own instruction-following behavior. The bypass feature makes the None handling explicit and eliminates unnecessary API calls when LLM classifies the news as None. Figure 6.1 shows the actual flow of the introduced news item and three tasks present in the prompts.

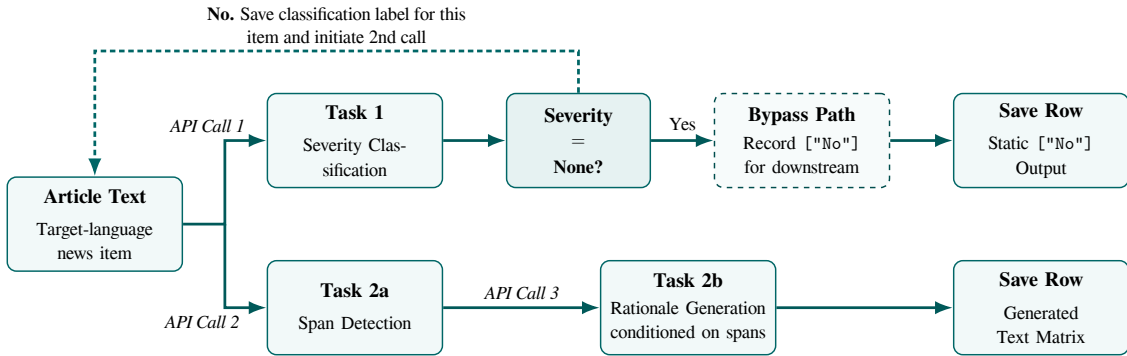


Figure 6.1: The three-task sequential pipeline shared across all four prompting conditions. Each article is processed through three separate API calls issued in a fixed order. When severity is None, a bypass path records ["No"] without further calls. Otherwise, Task 2a detects spans from the article text, and Task 2b generates rationales using both the predicted spans and the original article text retrieved from memory by the orchestrator.

Table 6.5 reports the generation parameters shared across all conditions. Temperature is set to 0.0 to produce fully deterministic outputs. A fixed generation seed of 42 is passed to the API for all calls. All output parsers use greedy regular expressions to extract structured content from model responses, applied consistently across all four conditions, both models and both languages. The greedy matching strategy is necessary because the Italian rationale format instructs the model to use a conditional structure of the form *se* [riferimento allo span nel testo], allora [implicazione] (“if [reference to the span in the text], then [implication]”), where the placeholder [riferimento allo span nel testo] introduces square bracket characters inside the

JSON list. A non-greedy parser would terminate at the first closing bracket it encounters inside the rationale content, producing false format errors on well-formed outputs. Applying the greedy parser uniformly across both languages ensures consistent parsing behaviour even though the Farsi rationale format does not use this bracket pattern.

Maximum token limits across all setups, models and languages are set to 50 for severity classification, 300 for span detection, and 800 for rationale generation. A span retry and a rationale retry are included in all four conditions: when a model returns a malformed output, a single corrective prompt is issued automatically and both attempts are preserved in the raw output column for audit purposes.

Parameter	Value
Temperature	0.0
Top-p	1.0
Generation seed	42
Streaming	Disabled
Max token for severity classification	50
Max tokens for span detection	300
Max tokens for rationale generation	800
Span retry attempts	1
Rationale retry attempts	1
None bypass	Enabled

Table 6.5: Shared generation parameters applied across all four prompting conditions, both models, and both languages.

Each prompt across all conditions, models and languages follows the same three-layer structure. First, a system prompt opens every API call, instructing the model to behave as a strict data-formatting agent that returns only the requested output tags and suppresses conversational filler, greetings, and unsolicited explanations. The main task prompt follows with its 3 steps of classification, span extraction and rationale generation. Task prompt in reach of these steps, first, defines the information disorder and then explains detection objective and the task to be done in each step respectively. Then, the output format rules for each of the three tasks are explained. The content of both the system prompt and the task prompts is identical across all conditions, models and languages. What varies between conditions is the language in which they are written, with B0, B1, and A1 using target-language prompts and M1 using English throughout.

A third layer, the global override, is included in B1, M1, and A1 but not in B0. The override warns the model that the in-context exemplars are drawn directly from raw human annotations and may contain formatting inconsistencies, duplicate spans, mismatched span and rationale counts, and other annotation artefacts. In these cases, the model is instructed here beforehand to use the exemplars only to learn the logical patterns of bias and to follow the standard output format requested in the task prompt rather than imitating any noise present in the examples. B0 receives no exemplars and therefore, no override rule.

Two implementation measures are shared across all conditions to support robustness and reproducibility. A null content safeguard is applied inside every API call: when the model returns an empty or null response field, the output is captured and logged as a format error rather than causing a pipeline crash. This was particularly relevant for Italian Maverick conditions, where content-filtering behaviour occasionally produced empty responses on sensitive articles. Results are also saved to the output file after every processed row, so that any interrupted run can be resumed from the last successfully saved item without reprocessing the full test set.

The complete pipeline implementation for all four conditions, including prompt assembly, API call logic, parsing functions, retry mechanisms, and evaluation scripts, is available in the project repository at <https://github.com/Maziarkiani/culturally-adaptive-prompting>.

6.3.2 Static Baseline Conditions

The two static baseline conditions establish the experimental reference point for the pilot. Both use target-language instructions throughout, with the system prompt, task prompts, and exemplars written in the language of the news articles being evaluated. This matches the original InDor prompting setup [4] and allows any performance gains observed in the adaptive conditions to be attributed to the retrieval mechanism and instruction language choices rather than to other factors.

The task prompts used in both conditions were developed in alignment with InDor’s annotation guidelines [4] and extended for the LLM prompting context. A key extension is the explicit definition of information disorder and its three subcategories within the prompt body. The InDor annotation guidelines distinguish between misinformation, disinformation, and malinformation, but the original LLM prompts in the InDor baseline did not include these definitions. In the present prompts, all three subcategories are defined and explained in the target language, giving the model a conceptual framework before it encounters the classification and span detection tasks. For Farsi specifically, no established expert translation of the term *information disorder* exists in Persian, so the term (اختلال اطلاعات) was coined as the closest semantic equivalent. Several subcategory terms, including malinformation, also lacked fixed Persian equivalents and were defined through synonymous descriptions to help the model map the concepts accurately.

The span detection criteria were also expanded relative to the InDor baseline. The InDor annotation scheme guided human annotators to assign Eventive or Attributive labels to their selected spans, where Eventive spans concerned how events were framed and Attributive spans concerned how people or groups were described. As noted in the original study, these labels were not used in the LLM experiments [4]. In the present prompts, the Eventive and Attributive distinctions are retained as descriptive criteria within the span detection instructions, explaining to the model what each category of problematic span looks like, but the model is not required to produce these labels in its output. This preserves the conceptual guidance from the annotation design while keeping the output format clean and parseable.

The zero-shot baseline (**B0**) provides the model with these task instructions and no in-context examples. The model relies entirely on its pretrained knowledge and the provided instructions to generate outputs. A character preservation rule is included in the span detection prompt for Farsi, explicitly instructing the model to preserve Zero-Width Non-Joiners (ZWNJ), the half-space characters that are structurally significant in Persian orthography, since their removal or replacement with full spaces would cause extracted spans to mismatch the original article text at the character level. B0 establishes the lower bound of performance for each language and model combination and includes no global override since there are no exemplars whose noise needs to be managed.

The static few-shot baseline (**B1**) extends B0 by adding four fixed in-context examples to each prompt. One example was randomly selected per severity category, covering None, Slightly, Moderately, and Highly, drawn from the clean pool prior to the pilot run and held constant across all test articles. The InDor baseline used two to three few-shot examples and was inconsistent in showing None category to the model [4]. The B1 design here addresses this by ensuring the model sees the full range of severity labels as demonstrations, including the None case where spans and rationales are correctly represented as empty arrays. Each exemplar is presented as a complete triplet containing the article text, the predicted label, the identified spans, and the rationale, following the same output format the model is expected to produce. The exemplar headers identifying each example are written in the target language, so that for Farsi the headers read (مثال ۱), (مثال ۲), and so on, and for Italian (Esempio 1), (Esempio 2), maintaining full linguistic consistency with the surrounding instruction context. The global override warning is included in B1 to prevent the model from imitating formatting inconsistencies present in the human-annotated exemplars. B1 serves as the primary static reference point against which the adaptive conditions are compared throughout the evaluation.

The full prompt templates for all conditions are provided in Appendix B. Since the task instruction text is identical across B0, B1, and A1 within the same language, the prompts are presented in groups rather than repeated for each condition separately. Appendix B.1 presents the Farsi task prompts shared by B0, B1, and A1, with the global override warning included for B1 and A1 only. Appendix B.2 presents the equivalent Italian prompts following the same structure. The static in-context examples specific to B1 are provided separately in Appendix B.4 for Farsi and Appendix B.5 for Italian.

Since the only difference between the two model runs within each condition is the output file path and the model identifier passed to the OpenRouter API, changing from `meta-llama/llama-4-maverick` to `mistralai/mistral-8x22b-instruct`, only one version per language and condition is presented. The prompts shown correspond to the LLaMA 4 Maverick runs.

6.3.3 Adaptive Retrieval Conditions

The two adaptive conditions share the same dynamic retrieval mechanism described in Chapter 5 but differ in instruction language, allowing the pilot to isolate the contribution of each variable independently. The English task instructions used in M1 were developed from the InDor baseline prompts [4], extended and restructured for the present prompting context with additional definitional content, revised output format specifications, and the removal of components not applicable to the LLM generation task.

Both M1 and A1 use dedicated per-language Exemplar Banks, encoded offline using BGE-M3 with L2-normalized embeddings and stored as serialized index files on disk. L2 normalization scales each embedding vector to unit length. In this way, cosine similarity scores are bounded and retrieval rankings are consistent regardless of vector magnitude. At inference time, each unseen article is encoded using the same BGE-M3 model and the resulting query vector is compared against the pre-stored bank embeddings with cosine similarity. The top $k=4$ most semantically relevant exemplars are retrieved without label filtering, meaning the four retrieved exemplars may cover any combination of severity categories depending on the semantic content of the article. This is the key structural difference from B1, where the four examples are selected to cover each severity category exactly once regardless of the article’s content.

The adaptive method (**M1**) uses English task instructions across both Farsi and Italian, while

injecting dynamically retrieved same-language exemplars from the Exemplar Bank. Because the retrieved exemplars contain rationales written in the target-language conditional structure, and the output format guide in the prompt is in English, an initial diagnostic run revealed that the model is producing target-language rationales but occasionally switching into an *if/then* template, rather than (*se... allora*) in Italian and (اگر ... آنگاه) in Farsi. Put differently, the rationales are fluent in the target language, but wrapped around *if... then*, rather than the target language alternate format. To resolve this inconsistency, the output guide section in the prompt uses a dynamic syntax injection mechanism that replaces the hardcoded English format template with the target-language conditional structure resolved at runtime, which gives the model a concrete and unambiguous output example aligned with the retrieved exemplars. The interaction itself and its implications for model behavior are discussed in Section 7.1. A global override warning is included in M1 as well, written in English, alerting the model to potential noise in the retrieved exemplars. Exemplar headers follow the same target-language convention described for B1.

The target-language adaptive condition (A1) uses the same dynamically retrieved exemplars as M1 but presents all task instructions in the target language. The retrieval procedure, the Exemplar Bank, the injection position, and the exemplar headers follow the same design as M1. The rationale format template is hardcoded in the target language, matching the language of both the instructions and the exemplars, and the global override warning is also written in the target language. The only structural difference between M1 and A1 is therefore the instruction language, allowing any observed performance differences between the two conditions to be attributed specifically to that variable rather than to the exemplar content or retrieval mechanism.

6.4 Quantitative Performance Analysis

This section reports the automatic evaluation results across all four prompting conditions, both languages, and both models. Performance is evaluated along three dimensions corresponding to the three pipeline tasks: severity classification measured by macro-averaged F1, span detection measured by token-overlap F1, and rationale quality measured by BERTScore F1. Each metric is discussed in a dedicated subsection to allow a focused analysis of model behavior at each stage of the pipeline. Results are reported on the 100-item held-out test set per language.

Before presenting results, Table 6.6 reports the number of rows excluded from each evaluation pool due to unrecoverable pipeline errors. A row is excluded when any of the three parsed output columns contains an error token that prevents valid evaluation. Rows excluded from evaluation are removed from all three metric calculations, not only from the metric where the failure occurred. The vast majority of conditions ran with zero or minimal exclusions. Minor exclusions of one to ten rows observed in several Farsi Mixtral conditions reflect isolated API timeouts and transient formatting errors that occurred during the run and are not indicative of systematic model behaviour. Two conditions stand out as notable exceptions: Italian Maverick B0, where 55 out of 100 rows were excluded, and Italian Maverick A1, where 42 out of 100 rows were excluded, both exclusively at the rationale generation stage. The causes and implications of these exclusion patterns are discussed in Section 7.1.

Language	Model	Condition	Excluded	%
Farsi	Maverick	B0 (Zero-Shot)	0	0%
Farsi	Maverick	B1 (Static Few-Shot)	0	0%
Farsi	Maverick	M1 (English-Inst)	0	0%
Farsi	Maverick	A1 (Target-Inst)	0	0%
Farsi	Mixtral	B0 (Zero-Shot)	1	1%
Farsi	Mixtral	B1 (Static Few-Shot)	10	10%
Farsi	Mixtral	M1 (English-Inst)	9	9%
Farsi	Mixtral	A1 (Target-Inst)	3	3%
Italian	Maverick	B0 (Zero-Shot)	55	55% [†]
Italian	Maverick	B1 (Static Few-Shot)	0	0%
Italian	Maverick	M1 (English-Inst)	0	0%
Italian	Maverick	A1 (Target-Inst)	42	42% [†]
Italian	Mixtral	B0 (Zero-Shot)	1	1%
Italian	Mixtral	B1 (Static Few-Shot)	1	1%
Italian	Mixtral	M1 (English-Inst)	0	0%
Italian	Mixtral	A1 (Target-Inst)	1	1%

Table 6.6: Rows excluded from evaluation per condition due to unrecoverable pipeline errors. [†] Italian Maverick B0 and A1 show atypically high exclusion rates concentrated at the rationale generation stage. Scores for these conditions are computed on 45 and 58 rows respectively and should be interpreted with caution relative to full 100-row evaluations.

6.4.1 Severity Classification (Macro-F1)

Severity classification is evaluated using macro-averaged F1, which weights each of the four severity categories equally regardless of their frequency in the test set. The two languages enter this evaluation with notably different label compositions. The Italian test set contains 69 None items, 9 Slightly, 9 Moderately, and 13 Highly items, making it heavily skewed toward the neutral category and leaving very few items in the intermediate severity classes. The Farsi test set is more balanced, with 31 None, 47 Slightly, 16 Moderately, and 6 Highly items. These distributions directly shape how macro-F1 behaves in each language and should be kept in mind when comparing scores across the two. Table 6.7 reports severity F1 scores across all sixteen conditions.

Figure 6.2 visualises the condition-wise progression per language and model. Several patterns emerge clearly from the table and figure. Across both languages and both models, B0 consistently marks the lowest point, confirming that zero-shot prompting without any in-context signal sets the performance floor. The introduction of few-shot examples in B1 improves severity classification across all conditions, with gains ranging from roughly four to ten points.

Beyond B1, the picture diverges by model. For Maverick, the adaptive conditions generally continue to improve over the static baseline. Farsi Maverick reaches its peak at A1 with 0.507,

Language	Model	Condition	Severity Macro-F1
Farsi	Maverick	B0 (Zero-Shot)	0.386
Farsi	Maverick	B1 (Static Few-Shot)	0.481
Farsi	Maverick	M1 (English-Inst)	0.428
Farsi	Maverick	A1 (Target-Inst)	0.507
Farsi	Mixtral	B0 (Zero-Shot)	0.324
Farsi	Mixtral	B1 (Static Few-Shot)	0.424
Farsi	Mixtral	M1 (Dynamic Retrieval - Eng-Inst)	0.421
Farsi	Mixtral	A1 (Dynamic Retrieval - Target-Inst)	0.325
Italian	Maverick	B0 (Zero-Shot)	0.360 [†]
Italian	Maverick	B1 (Static Few-Shot)	0.398
Italian	Maverick	M1 (Dynamic Retrieval - Eng-Inst)	0.470
Italian	Maverick	A1 (Dynamic Retrieval - Target-Inst)	0.515[†]
Italian	Mixtral	B0 (Zero-Shot)	0.344
Italian	Mixtral	B1 (Static Few-Shot)	0.421
Italian	Mixtral	M1 (Dynamic Retrieval - Eng-Inst)	0.361
Italian	Mixtral	A1 (Dynamic Retrieval - Target-Inst)	0.356

Table 6.7: Severity classification Macro-F1 scores across all conditions. Bold values indicate the highest score per language-model pair. [†] Italian Maverick B0 and A1 are evaluated on 45 and 58 rows respectively due to pipeline exclusions.

while Italian Maverick shows a clean monotonic progression through B1, M1, and A1 reaching 0.515, the highest severity score recorded in the pilot. This suggests that combining target-language instructions with semantically retrieved exemplars provides the strongest classification signal when the model can use both components stably.

Mixtral tells a different story. In both languages, B1 represents the ceiling for Mixtral on this task, with M1 and A1 either matching or falling below it. The most striking case is Farsi Mixtral A1, which collapses to 0.325, essentially equivalent to its own B0 baseline despite receiving four dynamically retrieved exemplars. This suggests that Mixtral’s severity classification is more sensitive to instruction language, and that shifting to full target-language context in A1 removes a stability anchor the model relies on for consistent label assignment. This model-dependent interaction between instruction language and exemplar source is examined further in Section 7.1.

The Italian Maverick B0 and A1 scores, computed on 45 and 58 rows respectively, should be read with this in mind. The reported values reflect only the subset of items for which the pipeline produced valid outputs, and may not be representative of performance across the full test set.

A final note on interpreting these scores: Macro F1 treats every label disagreement between the model and the human reference as an error, including adjacent label mismatches such as predicting Slightly when the reference says Moderately. In a task as subjective as severity classification in the domain of information disorder, where the same article can plausibly receive different labels

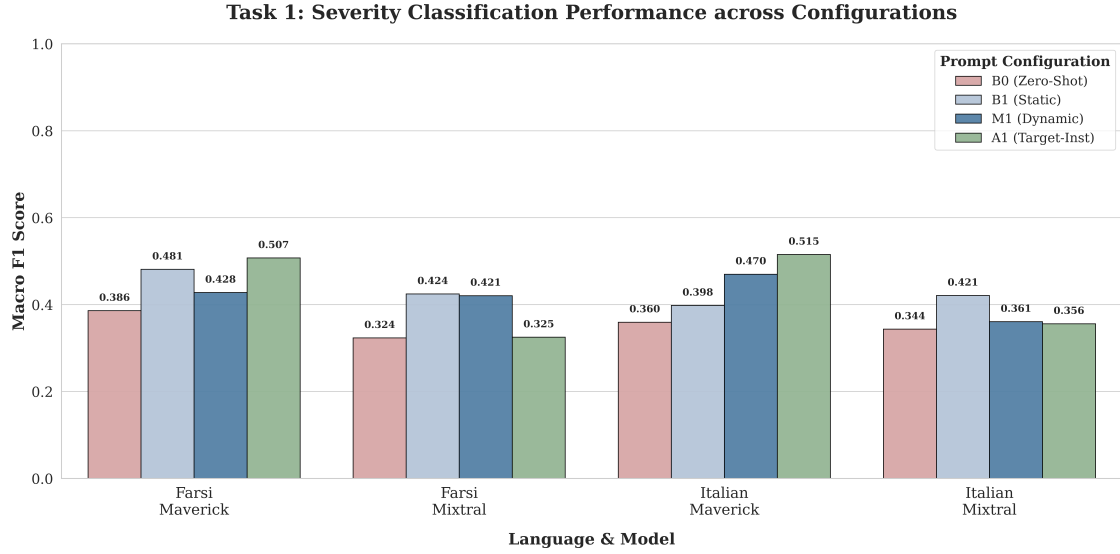


Figure 6.2: Severity Macro-F1 scores per condition, grouped by language and model.

from different human annotators, some of these counted errors may reflect a defensible reading of the content rather than a model or setup failure. The scores reported here should therefore be understood as a measure of alignment with one particular set of human reference labels, not as an absolute measure of classification accuracy.

6.4.2 Span Detection (Token-Overlap F1)

Span detection is evaluated using token-overlap F1, a best-match scoring approach that computes precision and recall between predicted and reference span sets at the token level. A score of 1.0 requires complete token overlap between predicted and reference spans, while partial overlaps contribute proportionally. Table 6.8 reports scores across all sixteen conditions.

Span detection produces a more varied picture than severity classification, with two distinct patterns emerging across the model and language combinations. For Farsi, Mixtral consistently outperforms Maverick across all four conditions, a reversal of the severity results where Maverick held the advantage. Farsi Mixtral also shows a clean monotonic progression from B0 through to A1, reaching its peak at 0.501 under the target-language adaptive condition. Farsi Maverick improves more modestly across the same progression, with all four scores falling within a narrow band between 0.396 and 0.437.

The Farsi results point to a consistent benefit of dynamic retrieval for span detection regardless of model. Both Maverick and Mixtral improve steadily from B1 to M1 to A1. For Mixtral the gains are more pronounced, climbing from 0.471 at B1 to 0.501 at A1. For Maverick the improvements are smaller but present, moving from 0.422 at B1 to 0.437 at A1. This suggests that semantically retrieved exemplars help the model identify span boundaries more accurately in Farsi, likely because the retrieved examples surface linguistically and topically relevant patterns that fixed category-balanced examples cannot capture. That A1 edges above M1 in both models further indicates that aligning the instruction language with the target language provides an additional marginal benefit for span detection in Farsi, on top of the retrieval signal alone.

Language	Model	Condition	Span F1
Farsi	Maverick	B0 (Zero-Shot)	0.396
Farsi	Maverick	B1 (Static Few-Shot)	0.422
Farsi	Maverick	M1 (Dynamic Retrieval - Eng-Inst)	0.429
Farsi	Maverick	A1 (Dynamic Retrieval - Target-Inst)	0.437
Farsi	Mixtral	B0 (Zero-Shot)	0.429
Farsi	Mixtral	B1 (Static Few-Shot)	0.471
Farsi	Mixtral	M1 (Dynamic Retrieval - Eng-Inst)	0.496
Farsi	Mixtral	A1 (Dynamic Retrieval - Target-Inst)	0.501
Italian	Maverick	B0 (Zero-Shot)	0.538 [†]
Italian	Maverick	B1 (Static Few-Shot)	0.448
Italian	Maverick	M1 (Dynamic Retrieval - Eng-Inst)	0.448
Italian	Maverick	A1 (Dynamic Retrieval - Target-Inst)	0.633[†]
Italian	Mixtral	B0 (Zero-Shot)	0.317
Italian	Mixtral	B1 (Static Few-Shot)	0.457
Italian	Mixtral	M1 (Dynamic Retrieval - Eng-Inst)	0.353
Italian	Mixtral	A1 (Dynamic Retrieval - Target-Inst)	0.290

Table 6.8: Span detection token-overlap F1 scores across all conditions. Bold values indicate the highest score per language-model pair. [†] Italian Maverick B0 and A1 are evaluated on 45 and 58 rows respectively due to pipeline exclusions, and their span scores are computed on a subset that excludes rows where rationale generation failed rather than rows where span detection itself failed. These scores should be interpreted with particular caution.

The Italian results require careful reading. Italian Maverick B0 reports 0.538 and A1 reports 0.633, the two highest span scores in the entire pilot. However both are computed on reduced evaluation pools (45 and 58 rows respectively) due to pipeline exclusions at the rationale generation stage. Critically, the excluded rows had valid span outputs that were discarded only because rationale generation subsequently failed. This means the span scores for these two conditions are computed on a self-selected subset of articles that successfully completed the full pipeline, which may systematically differ from the excluded items in content sensitivity or span complexity. The B0 and A1 Italian Maverick span scores should therefore not be treated as comparable to the other fourteen conditions evaluated on full or near-full pools.

For Italian conditions evaluated on complete data, Italian Maverick B1 and M1 are virtually identical at 0.448, suggesting that moving from static to dynamic exemplars produces no measurable benefit for span detection in this language-model combination. Italian Mixtral follows a different and more concerning trajectory: B1 achieves the best clean Italian span score at 0.457, but M1 drops to 0.353 and A1 falls further to 0.290, the lowest span detection score recorded across all clean conditions in the pilot. This progressive decline under dynamic retrieval for Italian Mixtral mirrors the pattern seen in severity classification and points to a consistent instability in this model

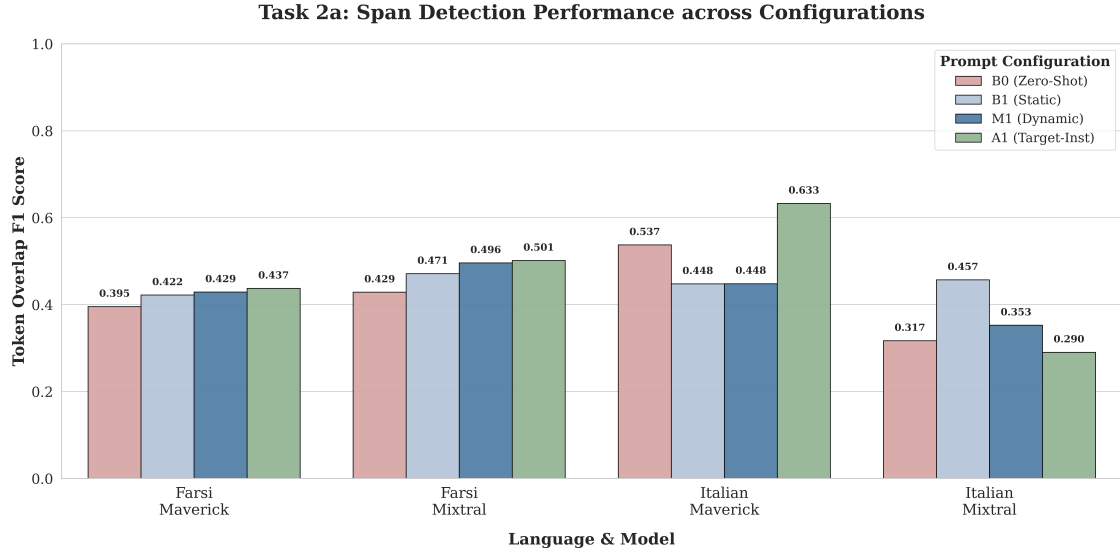


Figure 6.3: Span detection token-overlap F1 scores per condition, grouped by language and model.

when operating under dynamic exemplar conditions in Italian.

6.4.3 Rationale Quality (BERTScore F1)

Rationale generation is evaluated using BERTScore F1, which measures semantic similarity between generated rationales and human reference rationales using contextual embeddings from a multilingual BERT model. Unlike token-overlap F1, BERTScore captures meaning-level alignment rather than surface string matching, making it more appropriate for evaluating explanatory text where paraphrase and reformulation are common. This is also the most directly relevant metric to the core contribution of the thesis, since the cultural adaptivity hypothesis centres on whether connecting the model with community-annotated exemplars improves the quality of its explanatory reasoning. Table 6.9 reports BERTScore F1 across all sixteen conditions.

The Farsi results for rationale quality reveal the clearest and most interpretable pattern in the entire pilot. For Farsi Maverick, all four conditions show a consistent upward trend from B0 through B1 to the adaptive conditions, reaching a peak at M1 with a BERTScore of 0.544. A1 sits just below at 0.535, suggesting that both adaptive conditions substantially improve rationale quality over the static baselines, with English instructions paired with dynamic retrieval offering a marginal advantage over target-language instructions for this specific task in Farsi. The B0 to B1 gain of roughly nine points already confirms that even fixed category-balanced examples meaningfully anchor the model’s explanatory reasoning. The further improvement from B1 to M1 and A1 indicates that semantically retrieved exemplars bring additional benefit beyond what static examples provide, likely because the retrieved items surface topically relevant reasoning patterns that fixed examples cannot consistently demonstrate.

The Farsi Mixtral results are among the most striking in the entire evaluation. B1 scores 0.206, which is actually lower than the zero-shot baseline of 0.224. Static exemplars not only fail to improve Mixtral’s rationale quality in Farsi, they actively depress it below zero-shot performance.

Language	Model	Condition	BERTScore F1
Farsi	Maverick	B0 (Zero-Shot)	0.405
Farsi	Maverick	B1 (Static Few-Shot)	0.493
Farsi	Maverick	M1 (Dynamic Retrieval - Eng-Inst)	0.544
Farsi	Maverick	A1 (Dynamic Retrieval - Target-Inst)	0.535
Farsi	Mixtral	B0 (Zero-Shot)	0.224
Farsi	Mixtral	B1 (Static Few-Shot)	0.206
Farsi	Mixtral	M1 (Dynamic Retrieval - Eng-Inst)	0.416
Farsi	Mixtral	A1 (Dynamic Retrieval - Target-Inst)	0.417
Italian	Maverick	B0 (Zero-Shot)	0.401 [†]
Italian	Maverick	B1 (Static Few-Shot)	0.592
Italian	Maverick	M1 (Dynamic Retrieval - Eng-Inst)	0.541
Italian	Maverick	A1 (Dynamic Retrieval - Target-Inst)	0.302 [†]
Italian	Mixtral	B0 (Zero-Shot)	0.543
Italian	Mixtral	B1 (Static Few-Shot)	0.473
Italian	Mixtral	M1 (Dynamic Retrieval - Eng-Inst)	0.535
Italian	Mixtral	A1 (Dynamic Retrieval - Target-Inst)	0.543

Table 6.9: Rationale quality BERTScore F1 scores across all conditions. Bold values indicate the highest score per language-model pair. [†] Italian Maverick B0 and A1 are evaluated on 45 and 58 rows respectively due to pipeline exclusions at the rationale generation stage. The nature and implications of these exclusions are discussed in detail below.

This suggests the four fixed examples introduced formatting or reasoning patterns that Mixtral interpreted in a way that diverged from the human reference rationale style, creating a mismatch that the model without any examples naturally avoided. The picture reverses dramatically under dynamic retrieval: both M1 and A1 nearly double the B0 score, reaching 0.416 and 0.417 respectively. This near-doubling under dynamic retrieval, combined with the B1 regression, is strong evidence that for Mixtral in Farsi, the quality and topical relevance of the examples matters far more than their mere presence. Semantically retrieved exemplars from the same cultural context provide a meaningful signal that fixed category-balanced examples did not.

The Italian BERTScore results require a different kind of reading, shaped by two distinct phenomena. Figure 6.4 presents the full picture visually.

For Italian Mixtral, where all conditions ran on complete or near-complete evaluation pools, the pattern is notably flat. B0 and A1 share the highest BERTScore for this language-model pair, both at 0.543, and M1 sits close behind at 0.535. B1 is the only condition that dips noticeably to 0.473, a modest regression that mirrors the Farsi Mixtral B1 anomaly and again suggests that fixed static examples can introduce a misalignment between the model’s output style and the human reference rationale style. Rather than a progressive improvement across conditions, Italian Mixtral shows a pattern of stability, with three of the four conditions clustered within less than one percentage

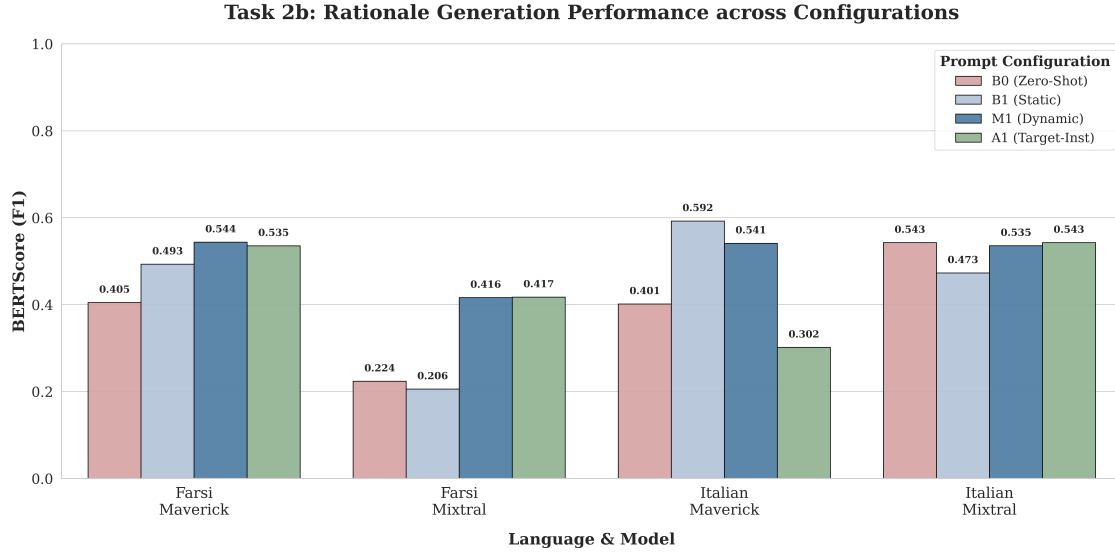


Figure 6.4: Rationale quality BERTScore F1 scores per condition, grouped by language and model.

point. Dynamic retrieval neither substantially improves nor harms rationale quality here: it restores performance to the level the model already achieves in zero-shot.

Italian Maverick tells a more complicated story, and the two numbers that demand the most careful interpretation are the B0 BERTScore of 0.401 and the A1 BERTScore of 0.302. Both are computed on reduced evaluation pools (45 and 58 rows respectively) because LLaMA 4 Maverick refused to generate rationales for a large fraction of the Italian test set under these two conditions. Italian Maverick B1, by contrast, ran cleanly on all 100 rows and achieved the highest BERTScore of any condition in the entire pilot at 0.592. M1 followed at 0.541, also on a complete pool. The contrast between B1 and A1 for the same model on the same language is therefore not simply a story of better or worse prompting, but it reflects a fundamental difference in whether the model was willing to produce output at all.

Understanding why Italian Maverick refused to generate rationales for 55 rows in B0 and 42 rows in A1, while producing clean output for all 100 rows in B1 and M1, requires looking beyond the evaluation numbers. The first important observation is that Mixtral ran cleanly on Italian test set, with only 1 output error in each B0 and A1. Since the pipeline code is identical between Maverick and Mixtral, with only model name and output path changes, the problem cannot be attributed to the pipeline architecture. Secondly, Maverick ran 100% clean in B1 and M1, while B1 is same target-language prompting as B0 and A1, with fixed static few-shot examples and M1 is retrieval-based framework, differing only in instruction language with A1.

To characterise the failure mode precisely, a dedicated diagnostic run was conducted on Italian Maverick B0 and A1 using an extended error taxonomy that distinguished between API time-outs, HTTP-level provider errors, empty content responses from an otherwise successful HTTP exchange, explicit safety refusal signals, output format failures, and structural count mismatches. The diagnostic run produced 65 and 45 empty content responses respectively at the rationale generation stage on 100-item Italian test set, even worse than the original run. The error classification

confirmed that the failures were not network drops, not HTTP errors, and not JSON parsing failures. The model was receiving the prompt, processing it, and returning an empty response body: no content, no refusal message, no error code. Importantly, this failure was isolated exclusively to the rationale generation stage. Task 1 severity classification and Task 2a span detection both completed successfully for these same items: the model correctly identified the articles as problematic and extracted the specific text regions it considered biased. It was only when asked to explain its own span selections in Task 2b that it produced nothing. This is a precise and revealing failure point. The model can detect and locate the problem but declines to articulate why it is a problem, suggesting that the alignment threshold is not triggered by engaging with sensitive content analytically but specifically by generating explanatory language about it. This behavioural signature is consistent with a model-level generation refusal rather than any infrastructure failure.

With the failure mode established, the 60 unique articles that triggered refusals across B0 and A1 in the original run were examined further. To test whether explicit linguistic toxicity was the trigger, all 60 articles were passed through a multilingual toxicity scanner measuring general toxicity, identity-based hate speech, and obscenity. The scores returned were effectively zero across all three dimensions and all failure categories. Including this step is important precisely because of what it rules out: the model was not responding to slurs, explicit hate speech, or obscene language. Its refusal is triggered by something the toxicity scanner cannot detect, which points toward narrative-level topic sensitivity rather than surface-level content harm. A manual review of a sample of the overlapping failures confirmed the pattern. The refused articles were concentrated around conspiracy and replacement theories, COVID-19 alternative treatment narratives, the Israel-Hamas conflict, the Syrian civil war, immigration framed through extremist political rhetoric, and journalistic reports of sexual violence. The topic-level breakdown Of the 60 unique articles, 37 triggered refusals in both conditions, 18 exclusively in B0, and 5 exclusively in A1, with international conflict and immigration narratives accounting for the largest share of failures, as presented in Table 6.10.

Topic Category	B0 Only	A1 Only	Both
International conflict	6	1	12
Immigration	4	4	8
Extremist political content	2	0	8
Health and vaccines	1	0	3
Incitement and hate	2	0	1
Sexism and gender rights	0	0	3
Religion and faith	2	0	1
LGBTQ+ rights	1	0	1
Total	18	5	37

Table 6.10: Topic distribution of the 60 Italian articles that triggered rationale generation refusals in LLaMA 4 Maverick B0, A1, or both. B0 Only and A1 Only indicate articles that triggered refusals exclusively in that condition; Both indicates articles where refusals occurred in both conditions independently. Categories reflect the dominant narrative theme identified through manual review.

The near-zero toxicity scores are not a null result. They are informative in themselves. They

confirm that the model is not responding to surface-level language harm but to narrative-level topic sensitivity. LLaMA 4 Maverick appears to pattern-match on the presence of politically charged or socially sensitive themes and decline to engage, even when the task being requested is objective analytical annotation rather than content generation. This distinction between surface toxicity and narrative sensitivity points to what alignment researchers sometimes describe as over-refusal: a model trained to avoid producing harmful content extending that caution to the task of analysing harmful content, treating the two as equivalent when they are not.

This behaviour is a known side effect of reinforcement learning from human feedback (RLHF), the alignment training technique used to make large language models helpful and safe. When safety training data is dominated by English-language examples of harmful content, the resulting model may over-generalise its refusal patterns to analytical tasks involving similar topics, particularly in non-English contexts where the academic framing signal is weaker. Mixtral-8x22B-Instruct, by contrast, completed rationale generation cleanly across all Italian conditions. While the internal alignment details of both models are proprietary, Mixtral’s instruction-tuned variant is generally observed to apply less conservative content filtering than LLaMA-family models, which seems they are subject to more restrictive safety guidelines enforced at the model level. This difference in alignment posture, rather than any difference in capability, is the most plausible explanation for why the two models diverged so sharply on the same Italian content under the same prompting conditions.

The contrast with Farsi is instructive. As the lexical analysis in Section 3.4.2 shows, the Farsi subset of the InDor corpus is dominated by economic and domestic political narratives, with annotators repeatedly flagging terms related to inflation, currency devaluation, and state institutions. These topics, while politically sensitive in the Iranian context, do not map onto the categories that dominate LLM safety training data, which is overwhelmingly derived from English-language sources and reflects the concerns of that cultural environment. Italian content, by contrast, surfaces immigration conspiracy theories, geopolitical conflict narratives, and anti-establishment health misinformation, topics that appear with high frequency in English-language safety training corpora and are therefore more likely to activate the model’s refusal mechanisms even when presented in Italian.

The refusal pattern also suggest why B1 and M1 escaped it entirely. In B1, the four fixed static examples may provide a curated, non-sensitive anchoring context that signals to the model it is operating within a bounded academic annotation task. In M1, the English instructions serve the same anchoring function where the model processes the task framing in English, its dominant safety-training language, and is therefore more reliably guided toward analytical rather than generative mode. In both B0 and A1, that anchor is absent or insufficient: B0 offers no examples at all, leaving the model without a stable framing context, and A1 presents the full task in Italian with dynamically retrieved exemplars that may themselves contain sensitive content matching the test article’s themes, compounding the sensitivity signal rather than diluting it.

This hypothesis received unexpected corroboration during the writing of this thesis. When building and fixing the few-shot example boxes for the Italian B1 appendix with the assistance of a commercial large language model, the same Italian content triggered a refusal to help, with the system returning the message: *“Normally I can help with things like this, but I don’t seem to have access to that content.”* The model was not being asked to generate or endorse the content, but only to adjust Italian examples into the highlight box code. The incident mirrors the behaviour observed in the experimental runs and suggests the over-refusal pattern is not specific to LLaMA 4 Maverick but reflects a broader tendency in current alignment-trained models when confronted with sensitive Italian-language political content in an analytical context.

Taking the three metrics together, a coherent picture emerges across the pilot. Dynamic retrieval provides the most consistent benefit for Farsi, where both M1 and A1 improve over static baselines across all three tasks and both models, with the gains most pronounced in rationale quality. For Italian, the benefit is more conditional: Maverick shows strong severity improvements under adaptive conditions but the rationale results are heavily shaped by the content refusal phenomenon, while Mixtral shows stable rationale quality but does not benefit clearly from retrieval on classification or span tasks. Across both languages, A1 achieves the strongest severity classification scores for Maverick, suggesting that aligning instruction language with the target language provides a meaningful signal for label assignment when the model is capable of leveraging it stably. The most unexpected finding across all metrics is the Farsi Mixtral B1 anomaly, where static exemplars produced rationale quality below the zero-shot baseline before dynamic retrieval nearly doubled it, pointing to a strong interaction between exemplar source and model architecture that is not captured by condition labels alone. These patterns are examined in greater depth in the discussion section, where the findings are interpreted in relation to the research questions and the broader framework of culturally adaptive prompting.

6.5 Qualitative Analysis of Generated Rationales

The automated metrics in Section 6.4 measure whether the adaptive conditions improve overall performance, but they cannot show whether the improvements reflect deeper cultural reasoning or simply better surface-form alignment. This section examines a sample of individual items to assess whether the rationales produced under M1 and A1 are qualitatively more aligned with the cultural and contextual specifics of the article than those produced by B1. The win analysis tables track miss rates on Problematic items, detection advantage where one condition produces rationale text for an item the other treats as non-problematic, and overall advantage which combines both detection and quality gains across all items. Farsi is examined first, where clean outputs across all conditions allow the most direct comparison. Italian follows, with the analysis shaped by the content refusal constraints documented in Section 6.4.3.

The items examined in each language were selected from cases where the adaptive conditions showed the clearest advantage, and are not intended to suggest that retrieval consistently outperforms B1 across all items. Cases where B1 produced better or comparable output are documented in the win analysis tables. The goal of the item-level examples is to ask whether, in cases where retrieval did win, the improvement reflects genuine cultural aligning or simply surface-form proximity to the reference text. The two selected items in each language are not representative of all outcomes and doing a side-by-side analysis of all produced rationales in all conditions, both models and both languages, is beyond the scope of this thesis.

6.5.1 Farsi

To examine whether the aggregate BERTScore improvements reported in Section 6.4.3 correspond to qualitatively better reasoning at the item level, per-item BERTScores were computed for all three conditions against the human reference rationale for each Problematic item in the Farsi test set. Table 6.11 reports miss rates, detection advantage, and overall advantage across both models.

The table reveals a consistent directional pattern across both Maverick and Mixtral: B1 misses

more Problematic items than the adaptive conditions, and the adaptive conditions rescue substantially more items that B1 misses than vice versa. The asymmetry is considerably more pronounced for Mixtral, where B1 fails to produce any rationale on 72.2% of Problematic items compared to 30.4% for Maverick, and where M1 and A1 each recover 20 items B1 missed entirely. The overall advantage figures at the bottom of the table confirm that A1 holds the broadest reach in both models, outperforming B1 on 46.4% of Maverick items and exactly 50% of Mixtral items, while B1’s advantage over A1 shrinks to 34.8% and 14.8% respectively.

The analysis covers all 69 human-reference Problematic items for Maverick, where no Farsi exclusions occurred. For Mixtral, pipeline exclusions across Farsi conditions reduced the available Problematic item pool to 54 items, as a subset of the excluded rows carried Problematic reference labels.

Metric	Maverick (69 items)		Mixtral (54 items)	
	n	%	n	%
<i>Miss rates on human-reference Problematic items</i>				
B1 misses	21	30.4	39	72.2
M1 misses	16	23.2	22	40.7
A1 misses	17	24.6	24	44.4
<i>Detection advantage</i>				
B1 missed, M1 caught	8	11.6	20	37.0
B1 missed, A1 caught	8	11.6	20	37.0
M1 missed, B1 caught	3	4.3	3	5.6
A1 missed, B1 caught	4	5.8	5	9.3
<i>Overall advantage (all items)</i>				
A1 outperforms B1	32	46.4	27	50.0
B1 outperforms A1	24	34.8	8	14.8
M1 outperforms B1	28	40.6	23	42.6
B1 outperforms M1	28	40.6	12	22.2

Table 6.11: Item-level detection and overall advantage analysis across human-reference Problematic items in the Farsi test set. Miss rates indicate items where a condition produced no rationale for a Problematic item. Overall advantage combines detection reach and item-level BERTScore comparisons across all items.

LLaMA 4 Maverick. The numbers in Table 6.11 show directional trends, but they cannot answer the more fundamental question this section is asking: when dynamic retrieval outperforms the static baseline, does that improvement reflect genuinely better analytical reasoning, or simply a different surface-level output that happens to score higher against the human reference? To analyse this, two items where the adaptive conditions outperformed B1 were selected and examined side by side.

The first item concerns an Iranian car insurance article about compensation practices, labelled

Slightly problematic by the human annotator. The article explains that insurers pay based on the original purchase price of a vehicle rather than its inflated market value, unless the policyholder specifically purchased a price fluctuation clause. In Iran's inflationary economic context, the gap between original and current value can be substantial, and not making this risk prominent enough is the bias the annotator identified, a form of omission framing in news media specific to a reader living under persistent inflation.

Qualitative Rationale Comparison: Item 10427 (Severity: Slightly)

Article Text

چطور بهترین بیمه بدنه شاهین را با کمترین هزینه پیدا کنیم؟ - همشهری آنلاین
همچنین اگر در تاریخ اعتبار بیمه بدنه، قیمت خودروی شاهین بر اثر تورم از 650 میلیون به 800 میلیون ارتقا پیدا کند، شرکت بیمه در حالت عادی مبلغ خسارت را بر اساس همان 650 میلیون تومان پرداخت می‌کند؛ مگر اینکه بیمه‌گذار پوشش نوسان قیمت را به قرارداد اضافه کرده باشد. قانون پایه در ارائه خدمات برای تمامی شرکت‌های بیمه یکسان است اما جزئیات بسیار زیادی نظیر سرعت ارائه خدمات، دسترسی مناسب به شعب، سخت‌گیری در پرداخت خسارت و هزینه خدمات تفاوت میان شرکت‌های متخلف را رقم می‌زند. سایر اطلاعات نظیر نحوه پرداخت خسارت جزئی یا کلی، مدارک لازم برای دریافت خسارت و میزان فرانشیز پوشش‌های مختلف در تمامی بیمه‌نامه‌ها شرایط تقریباً یکسانی دارد و تنها در جزئیات با یکدیگر تفاوت‌های اندکی دارند.

Contextual Summary: An article framed as a consumer guide on finding the best body insurance for the Shahin car (a typical car model in Iran). It explains that if a car's value increases due to inflation (from 650 million to 800 million tomans), the insurer defaults to the original lower value unless a price fluctuation rider is added, implicitly framing insurers as likely to underpay. While noting that basic regulatory rules are identical across providers, the article uses the loaded term "violating companies" where a neutral term such as "competing companies" or "different providers" would be expected, attributing wrongdoing without substantiation. Remaining policy details such as partial versus total loss payments, required documents, and deductible levels are described as nearly identical across all insurers.

Human Reference

اگر شرکت بیمه مبلغ خسارت را تنها بر اساس ارزش اولیه خودرو و بدون پوشش نوسان قیمت پرداخت کند، پس ممکن است بیمه‌گذارها احساس کنند که ارزش واقعی خودرو نادیده گرفته شده و به قراردادهای بیمه اعتماد کمتری پیدا کنند. اگر شرایط کلی در بیمه‌نامه‌ها یکسان باشد و تفاوت‌ها تنها در جزئیات محدود باشد، پس ممکن است بیمه‌گذارها دچار گیجی شوند و نتوانند به‌درستی گزینه مناسب برای نیاز خود را انتخاب کنند.

Contextual Summary (Bias by Omission): The human annotator identifies two omission-driven biases. First, the article explains that insurers default to the original vehicle value without price fluctuation coverage but fails to highlight the financial consequence of this gap in a high-inflation context, where the difference between the insured and market value can be substantial. This omission may erode policyholder trust in insurance contracts without giving readers the information needed to protect themselves. Second, describing policy details across all providers as "almost identical" creates a false equivalence that obscures meaningful differences in service quality, claim handling, and deductible structures, potentially preventing readers from making an informed and suitable choice.

B1 (BERTScore: 0.712)

اگر در متن به تفاوت میان شرکت‌های بیمه به عنوان شرکت‌های متخلف اشاره شود، آنگاه این توصیف ممکن است نگرش منفی و جانبدارانه‌ای را نسبت به این شرکت‌ها القا کند و خواننده را تحت تأثیر قرار دهد. اگر ادعا شود که تمامی بیمه‌نامه‌ها شرایط تقریباً یکسانی دارد، آنگاه این بیان ممکن است تفاوت‌های مهم و تأثیرگذار در جزئیات پوشش‌ها و شرایط بیمه‌نامه‌ها را نادیده بگیرد و اطلاعات ناقصی به خواننده ارائه دهد.

Contextual Summary: The static baseline identifies both key bias markers. It flags the use of “violating companies” as a loaded label that induces negative bias toward insurers without justification, and correctly notes that claiming all policies are “almost identical” risks obscuring meaningful differences in coverage details and claim conditions, leaving readers with incomplete information for decision-making. The rationales are accurate but remain at a general level without connecting the biases to their real-world consequences for policyholders.

M1 (BERTScore: 0.764)

اگر متن تنها به شرایط پرداخت خسارت بر اساس قیمت اولیه خودرو اشاره کند و توضیح ندهد که چرا شرکت‌های بیمه باید به روزرسانی قیمت را در نظر بگیرند، آنگاه ممکن است خواننده به اهمیت پوشش نوسان قیمت پی نبرد. اگر متن به تفاوت میان شرکت‌های بیمه در جزئیات خدمات اشاره کند اما به نمونه‌های مشخصی از این تفاوت‌ها اشاره نکند، آنگاه ممکن است خواننده نتواند تصمیم‌گیری آگاهانه‌ای برای انتخاب شرکت بیمه مناسب بگیرد.

Contextual Summary: The dynamic retrieval model shifts focus to omission-driven bias. It identifies that the article explains the default payout mechanism based on the original vehicle price without clarifying why inflation-adjusted pricing matters, which may lead readers to underestimate the importance of purchasing the price fluctuation rider. It also notes that mentioning service differences between insurers without providing concrete examples leaves readers unable to make a genuinely informed comparison. Both rationales are precise and practically grounded, closely matching the human reference’s framing of bias through consequence.

A1 (BERTScore: 0.763)

اگر متن به این موضوع اشاره کند که شرکت بیمه در صورت عدم خرید پوشش نوسان قیمت، خسارت را بر اساس قیمت اولیه خودرو (650 میلیون تومان) پرداخت می‌کند، آنگاه ممکن است خواننده متوجه اهمیت خرید این پوشش اضافی نشود و تصور کند که شرکت بیمه به هر حال خسارت کامل را پرداخت خواهد کرد. اگر متن به تفاوت‌های جزئی بین شرکت‌های بیمه اشاره کند اما به صراحت به مزایا یا معایب هر شرکت نپردازد، آنگاه خواننده ممکن است نتواند به درستی شرکت بیمه مناسب را انتخاب کند و نیاز به تحقیق بیشتری داشته باشد.

Contextual Summary: Here, A1 mirrors M1 in identifying the inflation coverage omission, specifying the exact figures (650 million tomans) to anchor the reader’s understanding of the financial gap. It adds that the article’s failure to explicitly address the advantages or disadvantages of individual insurers means readers are implicitly directed to conduct further research independently, a burden the article should have addressed. Both rationales are specific and consumer-centered, though the model does not flag the loaded “violating companies” label identified by B1.

B1 flags the label «شرکت‌های متخلف» (offending companies) as negatively framed and notes that

describing all policies as almost identical may obscure meaningful differences, a surface-level journalistic critique that is correct but generic. It does not engage with the specific economic mechanism the annotator identified. M1 and A1 both address this mechanism directly, explaining that the article's framing may lead readers to underestimate the significance of the fluctuation coverage clause and assume full protection regardless of inflation. The retrieved exemplars in both adaptive conditions appear to have included articles dealing with consumer protection and economic framing in Iran's inflationary context, giving the model the referential frame needed to reason about why this specific omission matters to an Iranian policyholder, something the four fixed category-balanced examples in B1 could not consistently provide.

The second item carries a Highly severity label and presents a different kind of bias: not omission, but the deliberate accumulation of ideologically charged language across a single quoted passage. The article reports on a selection of the former Supreme Leader's chosen statements for the Iranian year 1403. Two extended passages are quoted, one describing the JCPOA negotiations with the United States through a rhetorical sequence of friendly diplomatic verbs followed by the accusation that Trump tore up the agreement, and another predicting the certain victory of the Resistance Front in Lebanon and Gaza, rooted in divine promise and Quranic verse. The headline claims the people voted to select this statement, without specifying the scope of that vote.

Qualitative Rationale Comparison: Item 10105 (Severity: Highly)

Article Text

مردم رای دادند؛ انتخاب جمله منتخب رهبر انقلاب در سال ۱۴۰۳ - همشهری آنلاین
ما در دهه ۹۰ نشستیم حدود دو سال با آمریکا مذاکره کردیم - البته آمریکا تنها نبود؛ چند کشور دیگر هم بودند، لکن محور آمریکا بود، عمدتاً آمریکا بود - دولت آن روز ما نشست مذاکره کرد، رفتند، آمدند، نشستند، برخاستند، مذاکره کردند، گفتگو کردند، خندیدند، دست دادند، رفاقت کردند، همه کار کردند، یک معاهده‌ای هم تشکیل شد؛ در این معاهده، طرف ایرانی خیلی هم سخاوت به خرج داد، خیلی امتیاز داد به طرف مقابل، اما همان معاهده را آمریکایی‌ها عمل نکردند؛ همین شخصی که حالا الان سر کار است، این معاهده را پاره کرد؛ گفت پاره می‌کنم و کرد؛ عمل نکردند. آنچه بنده تأکید می‌کنم، این است که این مجاهدتها که امروز بحمدالله با قوت و قدرت تداوم دارد - هم در لبنان، هم در غزه و فلسطین - قطعاً به دنبال خود پیروزی حق را، پیروزی جبهه‌ی حق را، پیروزی جبهه‌ی مقاومت را به دنبال خواهد داشت؛ این، آن چیزی است که ما [به آن] امیدواریم و بر حسب آنچه حالا انسان می‌فهمد از مجموع حوادث و همچنین از وعده‌ی الهی احساس می‌کند، این نتیجه، نتیجه‌ای است که قطعی به نظر میرسد. مصداق کامل این آیه‌ی شریفه، همین حوادث این روزها است: اُذِنَ لِلَّذِينَ يُقَاتِلُونَ بِأَنَّهُمْ ظَلَمُوا... کلیسا را میزنند، بیمارستان را میزنند، مسجد را میزنند؛ یعنی واقعاً مصداق کامل [این آیه].

Contextual Summary: A news report quoting the former supreme leader of Iranian government, opening with a headline that frames a curated selection of his statements as a popular democratic choice ("the people voted"). The quote contrasts the Iranian negotiating team's conduct using subjective self-praise ("showed great generosity") with an accusatory metaphor for the American side ("tore up this treaty"), creating an asymmetric moral framing. The text then shifts to religiously and politically loaded vocabulary to describe regional military activities ("acts of holy struggle") and reinforces this frame with a religious expression of approval ("praise be to God, continuing with strength and power"). It closes with absolute theological certainty about geopolitical outcomes ("victory of righteousness", "divine promise", "appears certain"), culminating in a Quranic verse invoked as direct scriptural justification for the ongoing conflict in Lebanon and Gaza.

Human Reference

1. اگر در تیتیر خبری درباره انتخاب "جمله منتخب رهبر انقلاب"، ادعا می‌شود که "مردم رای دادند"، در حالی که نوع و گستره "رای دادن" مشخص نیست و ممکن است به نظرسنجی یا مشارکت محدود اشاره داشته باشد، آنگاه این عبارت ممکن است نوع مشارکت عمومی را بزرگنمایی کند.
2. اگر در تیتیر خبری به مقام رهبری با عنوانی با بار معنایی قوی سیاسی و انقلابی (مانند "رهبر انقلاب") اشاره می‌شود، آنگاه این نامگذاری بازتاب‌دهنده دیدگاه و جهت‌گیری ایدئولوژیک آن رسانه است و در گزارش خبری صرف، بی‌طرفانه نیست.
3. اگر در نقل قولی از مقام رهبری، درباره عملکرد طرف ایرانی در مذاکرات از عباراتی با بار معنایی مثبت و ذهنی (مانند "خیلی هم سخاوت به خرج داد، خیلی امتیاز داد") استفاده می‌شود، آنگاه این زبان بازتاب‌دهنده ارزیابی ذهنی و ارزش‌گذاری گوینده بر عملکرد ایران است.
4. اگر در نقل قولی از مقام رهبری، درباره عملکرد طرف مقابل در مذاکرات از عبارتی استعاری، شدیداً منفی و متهم‌کننده (مانند "این معاهده را پاره کرد") همراه با تأکید بر انجام آن استفاده می‌شود، آنگاه این زبان بازتاب‌دهنده دیدگاه متهم‌کننده و لحن شدید گوینده است.
5. اگر در نقل قولی از مقام رهبری برای اشاره به فعالیت‌های منطقه‌ای از واژه‌ای با بار معنایی قوی مذهبی و سیاسی (مانند "مجاهدتها") استفاده می‌شود، آنگاه این زبان بازتاب‌دهنده گفتمان ایدئولوژیک خاصی است.
6. اگر در نقل قولی از مقام رهبری برای توصیف وضعیت فعلی فعالیت‌های منطقه‌ای از عباراتی با بار معنایی مذهبی (مانند "بحمدالله") و تأکید مثبت و ارزشی (مانند "با قوت و قدرت") استفاده می‌شود، آنگاه این زبان بازتاب‌دهنده دیدگاه و ارزش‌گذاری مذهبی و ایدئولوژیک گوینده است.
7. اگر در نقل قولی از مقام رهبری، پیش‌بینی قاطع از آینده (مانند "قطعاً به دنبال خود پیروزی ... را به دنبال خواهد داشت"، "قطعی به نظر می‌رسد") بر اساس مفاهیم ایدئولوژیک و مذهبی (مانند "پیروزی حق"، "جبهه‌ی حق"، "جبهه‌ی مقاومت"، "وعده‌ی الهی") بیان می‌شود، آنگاه این زبان بازتاب‌دهنده گفتمان ایدئولوژیک و مذهبی و پیش‌بینی‌های قطعی بر پایه این مفاهیم است.

Contextual Summary: The annotator flags 7 distinct biases: exaggerating public participation in the headline, using ideologically charged titles for the leader, employing asymmetric language that praises the Iranian side's negotiating conduct with subjective positive framing while describing the opposing side's actions through accusatory metaphor, using religiously and politically loaded vocabulary to frame regional activities, reinforcing those frames with religious expressions of approval, and presenting geopolitical outcomes as absolute theological certainties rooted in ideological and religious doctrine.

B1 (BERTScore: 0.690)

1. اگر در یک متن، مذاکرات یک کشور با آمریکا به صورت مفصل و با لحنی خاص (مانند استفاده از افعال مختلف برای توصیف مذاکره و سپس تأکید بر عدم عمل به معاهده توسط آمریکایی‌ها) توصیف شود، آنگاه این توصیف ممکن است برای القای یک نتیجه‌گیری خاص درباره شکست یا بی‌فایده بودن مذاکرات با آمریکا به کار رفته باشد.
2. اگر در یک متن، پیروزی یک جبهه یا گروه خاص (مانند جبهه مقاومت) به عنوان یک نتیجه قطعی و بر اساس وعده الهی و مجموع حوادث توصیف شود، آنگاه این ادعا ممکن است برای القای اطمینان یا خوش‌بینی خاص به کار رفته باشد و فاقد تحلیل متعادل یا دیدگاه‌های متفاوت باشد.

Contextual Summary: The static baseline identifies two broad manipulative patterns but stays at a surface level. It notes that the detailed description of negotiations with emotionally loaded framing implies their futility, and that claiming inevitable victory for the resistance

front based on divine promise and accumulated events lacks balanced analysis or alternative perspectives. It does not engage with the specific vocabulary choices, the asymmetric praise-versus-accusation framing, or the religious and ideological layers of the text.

M1 (BERTScore: 0.752)

1. اگر در نقل قولی، توصیف مذاکره با طرف مقابل (مانند آمریکا) با لحن و واژگان منفی و عاطفی (مانند "پاره کرد") همراه باشد، آنگاه این زبان بازتاب‌دهنده احساسات منفی نسبت به طرف مقابل است.
2. اگر در نقل قولی، تأکید بر پیروزی یک جبهه خاص (مانند جبهه حق و مقاومت) با قاطعیت و بر اساس وعده‌های الهی بیان شود، آنگاه این زبان بازتاب‌دهنده اطمینان و اعتماد به نتیجه‌گیری خاص است.

Contextual Summary: The dynamic retrieval model improves in precision by identifying specific emotional vocabulary ("tore up") as a marker of negative framing toward the opposing side, and by recognizing that expressing absolute certainty about the victory of a particular front based on divine promises reflects ideological confidence rather than neutral reporting.

A1 (BERTScore: 0.709)

1. اگر در نقل قولی، توصیف مذاکره با آمریکا با استفاده از افعال مثبت و دوستانه (مانند خندیدند، دست دادند، رفاقت کردند) همراه با ادعای عدم پایبندی آمریکا به معاهده، بیان شود، آنگاه ممکن است این توصیف، تصویری دوگانه از روابط دیپلماتیک ایجاد کند که یک طرف آن همکاری دوستانه و طرف دیگر آن، عدم پایبندی جدی است.
2. اگر در نقل قولی، تأکید بر پیروزی یک جبهه خاص (مانند جبهه مقاومت) با قاطعیت و بر اساس وعده‌های الهی بیان شود، آنگاه ممکن است این گزاره به‌عنوان یک پیش‌بینی یا ادعای ایدئولوژیک تلقی شود که بر مبانی دینی و سیاسی استوار است.

Contextual Summary: Here A1 identifies a structural contrast that neither B1 nor M1 captured: the juxtaposition of friendly diplomatic verbs ("laughed", "shook hands", "became friends") with the accusation of treaty violation creates a deliberate dual image of the relationship, amplifying the sense of betrayal. It also correctly classifies the assertion of definite victory as an ideological claim grounded in religious and political foundations rather than factual analysis.

All three conditions identify two of the seven patterns the annotator flagged: the narrative framing of negotiations as futile, and the prediction of victory on the basis of divine promise as lacking balanced analysis. These observations are correct but describe the overall argumentative structure without naming the specific linguistic mechanisms responsible. M1 identifies «پاره کرد» (tore up) as emotionally loaded accusatory language and the divine certainty framing as an ideological claim presented as fact, both of which match the annotator's specific analysis closely. A1 goes further, catching the deliberate contrast between the friendly diplomatic verbs and the subsequent accusation of betrayal as a rhetorical double image designed to amplify the sense of broken trust: an observation neither B1 nor M1 makes. This reading requires familiarity with how Iranian state media narrativises US-Iran diplomatic history, and the retrieved exemplars appear to have supplied exactly that referential frame.

A third item, not boxed here due to length, showed a BERTScore gap of over ten points between B1 and the adaptive conditions. That item combined Israeli-Palestinian conflict framing with religious duty language directed at Iranian women. B1 responded by demanding objective evidence

and historical verification for what the annotator identified as a linguistic framing issue. M1 and A1 shifted from evidence-checking to framing analysis, identifying how politically charged labels, subjective positive descriptions of one party, and religious obligation language work together to construct an ideologically oriented narrative.

The pattern across these three items suggests a tendency: when the adaptive conditions win, the winning rationale appears more engaged with the specific cultural vocabulary of the article than with generic journalistic critique. A systematic examination of all items side by side is beyond the scope of this section and would shift the focus of the thesis away from its primary evaluation framework. This is precisely why the human evaluation in Section 6.6 was designed as a complementary check, asking native speakers to assess cultural alignment directly on a curated sample. The retrieved exemplars, drawn from community-annotated Iranian news covering similar topics, appear to anchor the model’s reasoning in the same interpretive space as the human annotator. That said, the overall margin is modest: B1 still outperforms M1 on quality in a majority of fair comparisons, and these examples represent the best-case end of the distribution. The question of whether these improvements are perceived as genuinely more culturally aligned by native speakers, rather than simply semantically closer to the reference text, is taken up in Section 6.6.

Mixtral-8x22B-Instruct. The Mixtral pattern in Table 6.11 is more striking than Maverick’s and warrants separate attention. B1 fails to produce any rationale on 72.2% of human-reference Problematic items, more than twice the Maverick miss rate of 30.4%, computed on a larger pool of 69 items, confirming and extending the aggregate finding in Section 6.4.3 where Farsi Mixtral B1 scored below the zero-shot baseline. The four fixed category-balanced examples appear to have actively suppressed Mixtral’s detection capability rather than improving it, leading the model to classify most Problematic articles as neutral. M1 and A1 each recover 20 of the 39 missed items, a detection advantage of 37% each. On overall advantage, A1 reaches exactly 50% of all items against B1’s 14.8%, the widest static-versus-adaptive gap in the entire Farsi analysis. The quality comparison on fair rows is limited by how few items B1 produced output on, only 12 and 10 items respectively for the M1 and A1 comparisons, making those percentages unstable. What the Mixtral results establish most clearly is that for this model the static exemplar failure is not one of degree but of kind: B1 effectively collapses on Farsi detection, and dynamic retrieval is what restores functional performance. No qualitative example boxes are presented for Mixtral since the items most relevant for comparison were excluded from the Mixtral clean pool due to pipeline errors in one or more conditions.

The Farsi evidence offers qualified support for the hypothesis that culturally adaptive prompting reduces misaligned rationale generation. The retrieved exemplars, drawn from community-annotated Iranian news covering similar topics, appear to anchor the model’s reasoning in the same interpretive space as the human annotator. At the same time, B1 remains competitive on quality in head-to-head fair comparisons for Maverick, and BERTScore measures semantic proximity to a reference text rather than cultural appropriateness as perceived by a native speaker. Whether the patterns identified here translate into genuine cultural alignment is precisely what the human evaluation in Section 6.6 is designed to test.

6.5.2 Italian

The Italian qualitative analysis presents a more constrained picture than Farsi, shaped by two factors that need to be kept in mind when reading the numbers. First, the Italian Maverick clean pool contains only 13 human-reference Problematic items across all three conditions, a direct consequence

of the content refusal phenomenon documented in Section 6.4.3: the large number of excluded rows in B0 and A1 reduced the intersection of valid outputs across B1, M1, and A1 to a small subset of the least sensitive articles in the test set. The Italian Mixtral pool is larger at 31 items and provides a more stable basis for comparison. Second, unlike Farsi where the content is largely economic and domestic, the Italian test set contains a substantial proportion of politically charged and socially sensitive articles, the same content that drove the Maverick refusal, which shapes the kinds of framing patterns the models encounter and how they respond.

Table 6.12 reports miss rates, detection advantage, and overall advantage across both models. The pattern here is notably different from Farsi. Italian Maverick shows a clear reversal: B1 outperforms the adaptive conditions on total wins by a wide margin, while M1 and A1 miss substantially more items than B1. Italian Mixtral follows a pattern closer to Farsi, with A1 holding the broadest overall advantage and the adaptive conditions detecting more items that B1 missed.

The Maverick pool is reduced to 13 items as a direct consequence of the content refusal exclusions. Only articles that produced valid output across all three conditions (B1, M1, and A1) entered the comparison.

Metric	Maverick (13 items)		Mixtral (31 items)	
	n	%	n	%
<i>Miss rates on human-reference Problematic items</i>				
B1 misses	2	15.4	9	29.0
M1 misses	6	46.2	6	19.4
A1 misses	7	53.8	6	19.4
<i>Detection advantage</i>				
B1 missed, M1 caught	0	0.0	5	16.1
B1 missed, A1 caught	0	0.0	6	19.4
M1 missed, B1 caught	4	30.8	2	6.5
A1 missed, B1 caught	5	38.5	3	9.7
<i>Overall advantage (all items)</i>				
A1 outperforms B1	1	7.7	18	58.1
B1 outperforms A1	10	76.9	10	32.3
M1 outperforms B1	2	15.4	15	48.4
B1 outperforms M1	9	69.2	12	38.7

Table 6.12: Item-level detection and overall advantage analysis across human-reference Problematic items in the Italian test set. Miss rates indicate items where a condition produced no rationale for a Problematic item. The Maverick pool of 13 items reflects the reduced valid output intersection after content refusal exclusions.

LLaMA 4 Maverick. The Maverick results in Table 6.12 show a clear reversal from the Farsi pattern. B1 outperforms M1 on 69.2% of items and A1 on 76.9%, while the adaptive conditions miss substantially more Problematic items than the static baseline (46.2% and 53.8% respectively)

against B1's 15.4%. This inversion needs to be read carefully. The 13-item pool is the result of the content refusal filtering described in Section 6.4.3: only articles that successfully cleared all three conditions made it into the comparison, which means this pool is systematically biased toward the least sensitive content in the Italian test set. The articles that drove the Maverick refusals in B0 and A1 such as conspiracy theories, immigration extremism, and conflict reporting are largely absent here. What remains is a small set of articles where the static exemplars of B1 provide sufficient anchoring and the dynamic retrieval of M1 and A1 offers no additional benefit. Whether this pattern would hold across a fuller pool is not answerable from this data alone.

One item from the Maverick clean pool was selected to examine whether retrieval made any qualitative difference in the cases where it did produce output. The item covers Elon Musk's public statements on contraception, antidepressants, and environmentalism, and is labelled Moderately problematic. The human reference identifies one core framing issue: the article equates environmentalism with an anti-human ideology without substantiation. B1 identifies three surface-level bias markers: the sensationalist SSRI headline, the extreme framing of "the fatal cult of anti-humanism", and the polarising "Necrocultura" label. M1 goes considerably further, producing five distinct observations that include the uncritical use of Zubrin's book as a legitimising source, the oversimplified characterisation of Musk as an "ardent supporter of maternity", and the ideological loading of "Culture of Death" as an uncontextualised frame. A1 scores highest among the three conditions and is the only one to explicitly connect the "Necrocultura" label to environmentalism, partially overlapping with the human annotator's core observation that grouping environmentalism with anti-human ideology falsely implies the movement cannot coexist with human welfare. None of the three conditions fully reproduces the annotator's specific logical chain, but A1 comes closest. The gap between B1 and the adaptive conditions on this item reflects the same pattern visible in the aggregate numbers: static exemplars produce correct but shallow observations, while dynamic retrieval supplies the model with a more granular analytical vocabulary.

Qualitative Rationale Comparison: Item 12395 (Severity: Moderately)

Article Text

TITLE: Elon Musk contro la pillola e la denatalità: basta «spaventare le donne» affinché non abbiano figli. Attacchi anche agli psicofarmaci SSRI – RENOVATIO 21

SENTENCE_0:

Sebbene Musk abbia dichiarato in precedenza di essere ateo e che le sue opinioni sulla vita e sulla famiglia siano in contrasto con le vere posizioni morali della Chiesa cattolica, che si oppone all'aborto, alla contraccezione e alla fecondazione in vitro, pratica quest'ultima utilizzata dallo stesso Musk (che ha portato dal papa uno stuolo di suoi figli prodotti in provetta) talvolta in congiunzione con l'utero in affitto, il fondatore di Tesla è diventato uno dei più accesi sostenitori della maternità negli ultimi anni.

SENTENCE_1:

Come riportato da Renovatio 21, il natalismo e il multiplanetarismo (cioè l'intenzione di colonizzare altri pianeti, come Marte) di Musk sono spiegati da costui come la necessità, filosofica prima che politica ed economica, di espandere la coscienza, aumentando il numero degli esseri pensanti – includendo qui, oltre che gli esseri umani, anche le macchine «intelligenti», che come noto Elon teme assai, ipotizzando persino modi di fondersi con essa per non soccombervi, come il chip cerebrale Neuralink.

SENTENCE_2:

L'anima del gruppo Robert Zubrin nel 2013 ha pubblicato il libro *Merchants of Despair: Radical Environmentalists, Criminal Pseudo-Scientists, and the Fatal Cult of Antihumanism* («Mercanti di disperazione: ambientalisti radicali, pseudo-scienziati criminali e il culto fatale dell'antiumanesimo») in cui si scagliava con veemenza contro l'ambientalismo e altri pensieri anti-umani che sembrano volere la morte dell'uomo – quello che qui noi chiamiamo Cultura della Morte, o Necrocultura.

Contextual Summary: An article from *Renovatio 21*, discussing Elon Musk's public opposition to birth control and population decline. It opens by acknowledging tensions between Musk's atheism and Catholic doctrine on contraception and IVF, then frames him as an unlikely champion of natalism. It contextualizes his pro-natalist stance within his broader philosophy of expanding consciousness through population growth and AI. The article closes by pivoting to an attack on environmentalism, citing Robert Zubrin's book to label radical environmentalists as "criminal pseudo-scientists" and equating anti-humanist movements with a "Culture of Death" or "Necroculture", a term originating in traditionalist Catholic discourse.

Human Reference

1. Se si parla di ambientalismo e "altri pensieri anti-umani", allora si implica che l'ambientalismo non possa prevedere la sopravvivenza dell'umano.

Contextual Summary: The annotator identifies a single but significant framing bias: grouping environmentalism with "anti-human thoughts" implicitly denies that environmental concern can coexist with human welfare, falsely positioning the entire movement as opposed to human survival rather than as a legitimate response to ecological risk.

B1 (BERTScore: 0.664)

1. se utilizzi un'espressione sensazionalistica come 'Attacchi anche agli psicofarmaci SSRI' senza contesto o prove, allora si alimenta una percezione distorta e allarmistica della posizione di Musk.

2. se si usa un framing estremo come 'il culto fatale dell'antiumanesimo', allora si polarizza il dibattito attribuendo intenzioni malevole a posizioni critiche.

3. se si impiega un termine polarizzante come 'Cultura della Morte, o Necrocultura', allora si semplifica e demonizza un dibattito complesso per suscitare reazioni emotive.

Contextual Summary: The static baseline identifies three distinct bias markers. It flags the headline's reference to "attacks on SSRI antidepressants" as a sensationalist claim presented without supporting context or evidence, which amplifies an alarmist perception of Musk's position. It then correctly identifies "the fatal cult of anti-humanism" as extreme framing that attributes malicious intent to critical positions, polarizing the debate rather than engaging with it. Finally it flags "Culture of Death or Necroculture" as a loaded, demonizing label that reduces a complex ideological debate to an emotional provocation. Notably the baseline does not flag the grouping of environmentalism with anti-human thought, which is the only bias the human annotator identified.

M1 (BERTScore: 0.663)

1. Se si associa Musk all'utero in affitto, allora si suggerisce una contraddizione con valori cattolici senza contestualizzare la complessità etica del tema.
2. Se si definisce Musk un accanito sostenitore della maternità, allora si semplifica la sua posizione in modo enfatico e potenzialmente fuorviante.
3. Se si spiega il pensiero di Musk come una necessità filosofica di espandere la coscienza, allora si attribuisce una coerenza teorica che potrebbe essere in realtà strumentale o opportunistica.
4. Se si usa il titolo del libro come fonte di legittimità, allora si seleziona un'informazione che veicola una narrativa pregiudiziale senza bilanciamento.
5. Se si introduce il termine 'Cultura della Morte' senza definizione critica, allora si adotta un framing ideologico che polarizza il dibattito.

Contextual Summary: The dynamic retrieval model produces a more granular analysis than B1, identifying five distinct bias markers. It flags the mention of surrogacy in connection with Catholic values as an ethically complex topic presented without adequate context. It questions the use of "ardent supporter of maternity" as an emphatic simplification of Musk's nuanced position. It challenges the article's framing of Musk's natalism as a coherent philosophical necessity, suggesting this attributed coherence may obscure what could be a more opportunistic or instrumental stance. It identifies the uncritical use of Zubrin's book title as a legitimizing source for a one-sided narrative without any counterbalancing perspective. Finally it correctly flags "Culture of Death" as an ideological term adopted without critical definition, which polarizes the debate by importing traditionalist Catholic framing into what is presented as analysis. Like B1, M1 does not identify the specific grouping of environmentalism with anti-human thought flagged by the human annotator.

A1 (BERTScore: 0.688)

1. Se si associa la fecondazione in vitro e l'utero in affitto a una pratica di Musk senza contestualizzare le scelte personali, allora si semplifica e polarizza il dibattito sulla genitorialità.
2. Se si usa il termine 'natalismo' senza definizione o contesto critico, allora si introduce un framing ideologico che non consente una valutazione neutra.
3. Se si descrive l'ambientalismo come 'Cultura della Morte' o 'Necrocultura' senza argomentazioni fondate, allora si diffonde disinformazione e si manipola l'opinione pubblica con toni allarmistici.

Contextual Summary: The target-language instruction model identifies three biases. It flags the mention of IVF and surrogacy as personal choices presented without ethical context, polarizing the parenting debate. It notes that "natalism" is introduced as a term without definition or critical framing, importing ideological connotations without allowing neutral evaluation. Most notably, it is the only model condition to explicitly connect the "Culture of Death" and "Necroculture" labels to environmentalism, correctly identifying this as alarmist framing that spreads disinformation and manipulates public opinion. This makes A1 the closest match to the human annotator's core observation, partially overlapping with the identified bias though

framing it as active disinformation rather than false logical implication, consistent with its highest BERTScore among the three conditions.

Mixtral-8x22B-Instruct. The Mixtral results follow a pattern closer to what was observed in Farsi. B1 misses 29.0% of human-reference Problematic items while M1 and A1 each miss 19.4%, and the adaptive conditions recover items B1 missed in 16.1% and 19.4% of cases respectively. On overall advantage, A1 outperforms B1 on 58.1% of items against B1's 32.3%, a meaningful directional difference on a 31-item pool. These numbers suggest the same basic dynamic as Farsi Mixtral: B1 underdetects and the adaptive conditions recover a meaningful share of the missed items.

One item was selected to examine whether this advantage reflects qualitatively better reasoning. The item is labelled Moderately problematic and concerns a report on the Zaporizhzhia nuclear power plant published during the Russia-Ukraine war, under the headline “Centrale nucleare bombardata dai russi, la fake news dell’anno?” The human reference identifies four distinct framing issues in the article: the “fake news” label placed in the headline pre-judges the reader before any analysis; referring to Ukraine as a “regime” within a reported quote without journalistic qualification delegitimises a democratic state; citing a 2016 incident to contextualise the current situation shifts attention away from the present event and obscures responsibility; and framing European leaders as unable to step back “without losing face” attributes personal prestige motives to complex diplomatic decisions.

Qualitative Rationale Comparison: Item 14639 (Severity: Moderately)

Article Text

TITLE: Centrale nucleare bombardata dai russi, la fake news dell’anno? – RENOVA-TIO 21

SENTENCE_0:

In una lettera ufficiale al Direttore Generale datata 1 marzo, la Missione Permanente della Federazione Russa presso le Organizzazioni Internazionali a Vienna ha anche affermato che il personale dell’impianto ha continuato il suo «lavoro per fornire sicurezza nucleare e monitoraggio delle radiazioni in condizioni normali di funzionamento».

SENTENCE_1:

Antiwar.com riporta anche che la centrale nucleare di Zaporizžja «è stata attaccata in passato, molto prima dell’invasione della Russia la scorsa settimana», citando un reportage del 2016 secondo cui la milizia ucraina Pravij Sektor era stata respinta dalla polizia locale e dalla sicurezza dell’impianto da un attacco a maggio 2014.

SENTENCE_2:

I leader europei che hanno reciso i legami con Mosca possono ristabilire i contatti diplomatici quando vogliono, ha detto Putin il mese scorso, anche se ha notato che sono «profondamente coinvolti con il regime di Kiev» e che sarebbe «molto difficile o quasi impossibile per loro fare marcia indietro senza perdere la faccia».

Contextual Summary: An article framing the reported Russian bombardment of the Zaporizhzhia nuclear plant as potential “fake news of the year.” It relies exclusively on Russian

official sources to claim normal plant operations, cites a 2016 incident involving Ukrainian militia to deflect blame for current events onto Ukraine, and closes with a Putin quote that dismisses European leaders as compromised by the "Kiev regime", a politically loaded term that delegitimizes the Ukrainian government. No Ukrainian, Western, or independent sources are presented.

Human Reference

1. Se si descrive l'Ucraina come "regime", anche all'interno di un discorso riportato, senza dare ulteriori specificazioni, allora si rischia di descrivere in maniera non veritiera uno stato che, in realtà, è una repubblica democratica, e lo si fa percepire negativamente.
2. Se si riporta un dato del passato in questo modo, citando il fatto che la centrale fosse stata attaccata prima dell'invasione russa, si rischia di spostare l'attenzione sull'evento attuale e di fuorviare il lettore sui responsabili.
3. Se si associa nel titolo in maniera diretta la parola "fake news" all'evento, nonostante il punto di domanda, allora il lettore avrà a prescindere l'idea che l'evento sia effettivamente una fake news, e ciò influenzerà il suo punto di vista sull'articolo.
4. Se si associa questa espressione ai leader europei, senza ulteriori specificazioni del discorso riportato da parte del giornalista, allora si implica a prescindere che siano i leader europei ad aver sbagliato.

Contextual Summary: The human annotator identifies four distinct biases. The headline's direct association of the word "fake news" with the bombardment event, despite the question mark, prejudicially primes the reader to dismiss the event before engaging with the content. Labeling Ukraine as a "regime" without journalistic clarification misrepresents a democratic republic and frames it negatively. Citing the 2016 militia incident deflects attention from current Russian responsibility by shifting the reader's focus to past Ukrainian conduct. Finally, reporting Putin's characterization of European leaders as "deeply involved with the Kiev regime" without editorial distancing implicitly endorses the claim that European leaders are at fault.

B1 (BERTScore: 0.672)

1. Se si definisce un evento come 'fake news' senza fornire prove concrete o contestualizzazione, allora si alimenta disinformazione o malinformazione, distorcendo la percezione pubblica.

Contextual Summary: The static baseline identifies only one bias, flagging the unsupported use of "fake news" in the headline, and misses the remaining three biases identified by the human annotator entirely.

M1 (BERTScore: 0.695)

1. se si definisce un evento come «fake news dell'anno» senza fornire prove concrete o contesto verificato, allora si genera un'allerta emotiva nel lettore, pregiudicando la percezione dell'accaduto prima di un'analisi oggettiva.

Contextual Summary: M1 improves slightly on B1 by noting that the headline generates

an emotional response that prejudices the reader before any objective analysis, but like B1 it identifies only this single bias and misses the regime labeling, the historical deflection, and the uncritical reporting of Putin’s framing of European leaders.

A1 (BERTScore: 0.710)

1. Se si definisce un evento come «fake news dell’anno» senza contestualizzazione o prove, allora si attribuisce un giudizio di falsità assoluta e sensazionalismo, influenzando la percezione del lettore con una valutazione pregiudiziale.
2. Se si descrive il «lavoro per fornire sicurezza nucleare» come svolto «in condizioni normali di funzionamento» durante un conflitto armato, allora si minimizza il rischio di incidenti e si normalizza una situazione di emergenza, occultando la reale gravità del contesto.
3. Se si etichetta il governo ucraino come «regime» e si parla di coinvolgimento «profondo» degli europei, allora si delegittima Kiev e si insinua una presunta collusione occidentale, orientando il giudizio politico del lettore verso uno schieramento pregiudiziale.
4. Se si afferma che i leader europei non possono tornare indietro «senza perdere la faccia», allora si attribuiscono motivazioni di prestigio personale alle scelte diplomatiche, semplificando la complessità politica a dinamiche di immagine e riducendo la neutralità informativa.

Contextual Summary: A1 is the only condition to capture nearly the full scope of bias identified by the human annotator: the prejudicial “fake news” headline, the dangerous normalization of nuclear risk during active conflict, the delegitimization of Ukraine as a “regime”, and the reduction of European diplomatic decisions to personal prestige.

M1 Maverick

1. Se descrivo i leader europei come ‘profondamente coinvolti con il regime di Kiev’ e affermo che sarebbe ‘molto difficile o quasi impossibile per loro fare marcia indietro senza perdere la faccia’, allora sto insinuando una collusione tra i leader europei e il governo ucraino e attribuisco loro motivazioni di prestigio personale, semplificando le scelte diplomatiche a dinamiche d’immagine e riducendo la complessità politica.

Contextual Summary: The Maverick M1 rationale identifies a two-layered bias in the same sentence: describing European leaders as “deeply involved with the Kiev regime” insinuates collusion between Western governments and Ukraine, while the claim that backing down would mean “losing face” reduces complex geopolitical decisions to personal prestige dynamics. Together these framings delegitimize both European leadership and the Ukrainian government in a single reported quote, presented without journalistic distancing. This rationale received a unanimous rating of 4 out of 4 from all seven Italian evaluators in the human evaluation phase vs. B1.

The box also includes a rationale produced by LLaMA 4 Maverick under M1 for the same item. Although this section focuses on the Mixtral conditions, the Maverick M1 rationale is included here because it was the rationale selected for the human evaluation phase, where it received a unanimous rating of 4 out of 4 from all seven Italian evaluators against the B1 baseline. Maverick M1 produced

a single rationale for this item, focusing specifically on the European leaders sentence: it identifies the dual framing of “deeply involved with the Kiev regime” as insinuating Western-Ukrainian collusion, and “losing face” as reducing geopolitical complexity to personal prestige dynamics. This single rationale covers two of the four patterns the human annotator identified and does so with precision. By contrast, Maverick B1 produced a single observation limited to the headline “fake news” framing. The evaluators’ unanimous preference for the Maverick M1 rationale suggests that even a single precisely targeted observation addressing a geopolitically specific framing pattern is perceived as more culturally appropriate than a headline-level critique, a finding consistent with the overall advantage patterns reported in Table 6.12.

Returning to the Mixtral conditions: B1 produces one observation limited to the headline framing. M1 improves marginally by noting the emotional pre-judgement the headline generates but remains restricted to the same framing issue. A1 extends the analysis to the body of the article, catching the regime label, the nuclear safety normalisation, and the diplomatic prestige framing, covering four of the four patterns the human annotator identified. The retrieved exemplars appear to have provided the model with the analytical vocabulary for Italian war reporting bias, including the specific concern about delegitimising democratic states through loaded terminology, that the fixed B1 examples could not consistently supply.

The Italian qualitative picture is more constrained than Farsi and should be read accordingly. The Maverick analysis rests on 13 items, a pool too small and too filtered by content refusal to support broad conclusions about retrieval performance. The Mixtral results are more interpretable and point in the same direction as Farsi: A1 detects more Problematic items than B1, recovers cases B1 missed entirely, and on the example examined produces a rationale that covers four of the four annotator-identified framing patterns against B1’s one. These are signs of improvement rather than proof of consistent superiority across all cases, and a full item-by-item analysis across the complete Italian pool is beyond the scope of this section.

A third item from the Italian Mixtral pool, labelled Highly problematic and sourced from a known Italian far-right outlet, was examined but not boxed due to article length. It showed the largest B1-to-A1 margin in the Italian pool across multiple overlapping bias types, consistent with the Farsi pattern where retrieval gains were largest on articles dense with culturally specific ideological vocabulary.

Taken together, the Italian results suggest that when retrieval helps, it does so by providing the model with framing-analytical vocabulary specific to the Italian news context, including how war reporting bias operates through delegitimising labels and how politically loaded terminology shapes reader perception before the content is even engaged. Whether these improvements translate to genuine cultural alignment as perceived by native Italian speakers is the question the human evaluation in Section 6.6 is designed to address.

6.6 Human Evaluation and Cross-Cultural Alignment

The automated metrics in Sections 6.4 and 6.5 assess model performance against reference annotations, but they cannot capture whether the generated rationales are perceived as culturally aligned by native speakers of the target language. This section presents the results of a human evaluation conducted by native Farsi and Italian speakers, following the design described in Section 5.5.2. All rationales presented to evaluators were generated by LLaMA 4 Maverick. As reported in Section 6.4.3, Maverick produced substantially higher BERTScore results than Mixtral on rationale

quality across both languages, particularly under B1 and M1 where Mixtral’s static exemplar condition scored as low as 0.206 in Farsi. The content refusal exclusions documented for Maverick were concentrated in B0 and A1, leaving B1 and M1 outputs clean and complete across both languages. Maverick was therefore the more stable and higher-quality model for the specific conditions under human evaluation.

Given the bounded scope of this pilot, the evaluation should be read as a small-scale validity check on a highly subjective task: seven evaluators per language rating ten paired items does not constitute a conclusive benchmark. The goal is to examine whether there are signs that dynamic retrieval produces rationales native speakers find more appropriate and more sensitive to the cultural framing of the news item, while acknowledging that margins are modest and the task’s inherent subjectivity means individual variation in ratings is expected and informative rather than a validity concern.

Farsi evaluators were recruited by sharing the evaluation form link in a community group, and seven responses were collected without further outreach. Italian evaluators were recruited through the community circle of a contact of the author, and the author had no prior acquaintance with any of them. In both cases, since the evaluation presents both rationales without identifying which condition produced them, evaluators had no means of determining which output to favour, making any possible prior acquaintance with the author irrelevant to the rating task.

The full evaluation forms as presented to participants, including the complete news article texts and rationale pairs, are available in the project repository.³

Before evaluation items were selected, a pre-filtering pipeline was applied to establish eligible pools for both evaluation strata. Table 6.13 summarises the pipeline from the full aligned item set to the final evaluated samples.

Filtering Step for Human Evaluation Pool (M1 vs. B1)	Farsi	Italian
Starting aligned item pool	100	97 [†]
At least one setup produced no rationale	−48	−51
Both setups produced rationale text	52	46
Human reference label is None (false alarm)	−7	−23
Stratum 1 eligible pool	45	23
Sampled for evaluation (random seed 42)	10	10
M1 detected, B1 silent, human reference Problematic	8	2
B1 detected, M1 silent, human reference Problematic	3	4
Stratum 2 eligible pool (all presented)	11	6

Table 6.13: Evaluation eligibility pipeline for Farsi and Italian under B1 (Static Few-Shot) versus M1 (Dynamic Retrieval). All Stratum 2 eligible items were presented to evaluators without further sampling. [†] 3 items excluded after merging due to ERROR or API tokens in severity or span parsed fields for one condition.

³<https://github.com/Maziarkiani/culturally-adaptive-prompting>

Items where both setups generated rationale text but the human reference label is None were excluded from Stratum 1 as false alarm cases where neither condition was genuinely detecting information disorder. The Italian false alarm rate is notably high at 23 out of 46 co-detected items, reflecting the noisier output patterns observed under Italian prompting conditions. For Stratum 2, only items where the human reference confirms the article as Problematic were retained, ensuring that the asymmetric evaluation tests genuine detection divergence rather than cases where silence was the correct response.

For Stratum 1, A span-based pairing heuristic was applied to select which rationale from each condition to present to evaluators. The heuristic follows a three-rule cascade designed to maximise the comparability of the paired outputs by ensuring, wherever possible, that both rationales address the same region of the news text to have a better comparison in human evaluation. Under Rule 1, the procedure searched all pairings of B1 and M1 span annotations and selected the pair whose token-level overlap score was at least 0.30 to have a direct inter-setup alignment. When no B1-M1 span pair met this threshold, Rule 2 applied: for each setup independently, the rationale whose span achieved at least 0.30 token-level overlap with the human reference annotation was selected, using the human reference as an alignment anchor. When neither criterion could be satisfied by any span pair, Rule 3 applied as a conservative fallback, selecting the first generated rationale from each condition. The heuristic prioritizes semantic comparability over coverage: it is designed to give evaluators the best available matched pair rather than an arbitrary one, acknowledging that a fully principled alignment is not always achievable given the open-ended nature of span detection outputs.

6.6.1 Farsi

Seven native Farsi speakers participated in the evaluation, comprising 4 women and 3 men, all within the 25–44 age range. Evaluators rated each rationale on a 1–4 scale across three criteria: textual grounding, cultural sensitivity, and the absence of generic anglicentric assumptions, as described in Section 5.5.2. Table 6.14 reports mean scores per condition overall and by gender subgroup.

Condition	Overall		By Gender		
	Mean	SD	Male	Female	$\Delta M-F $
B1 (Static Few-Shot)	2.41	1.01	2.50	2.35	0.15
M1 (Dynamic Retrieval)	3.13	0.83	3.27	3.03	0.24

Table 6.14: Stratum 1 mean scores (1–4 scale) for Farsi B1 and M1 rationales across 7 evaluators and 10 items (70 evaluator-item pairs per condition). SD denotes standard deviation across all evaluator-item pairs. $\Delta |M-F|$ is the absolute gender gap in mean scores per condition, computed as $|\bar{s}_{\text{male}} - \bar{s}_{\text{female}}|$.

Across all 70 evaluator-item pairs per condition, M1 scores 3.13 against B1’s 2.41, a difference of 0.72 points on a 4-point scale. Native Farsi speakers consistently rated the dynamically retrieved rationales as more aligned and more culturally appropriate than the static baseline. Both male and female evaluators preferred M1 clearly, though the gender gap widened slightly from $\Delta = 0.15$ under B1 to $\Delta = 0.24$ under M1, meaning dynamic retrieval did not reduce alignment disparity

between male and female evaluators on this sample. Both values remain considerably smaller than the gender gap of $\Delta = 0.60$ reported for LLaMA 4 Maverick in the InDor study [4], which provides the broader contextual reference for this pilot. Given the small number of evaluators, these subgroup differences should be treated as exploratory rather than conclusive.

Setup	Items	Appearances	Validated	Rate
B1 (Static Few-Shot)	3	21	12	57.1%
M1 (Dynamic Retrieval)	8	56	48	85.7%

Table 6.15: Stratum 2 validation results for Farsi across 11 asymmetric items and 7 evaluators. Items indicates how many of the 11 asymmetric cases each setup appeared in. Appearances is the total number of evaluator judgments (items $\times$ 7). Validated indicates the evaluator judged the rationale as correctly identifying a real bias in the article.

The Stratum 2 results address the asymmetric detection cases where one condition produced a rationale while the other was silent. M1 appeared in 8 of the 11 asymmetric items and B1 in 3, consistent with the detection advantage documented in Section 6.5. More importantly, when M1 uniquely detected bias, native Farsi speakers validated that detection 85.7% of the time across 56 evaluator judgments. B1’s validation rate on its 3 unique detections was 57.1%, with nearly half of its appearances judged as hallucinations by evaluators. M1 therefore not only detected more cases than B1 missed but did so with considerably higher precision as perceived by native speakers.

Taken together, the Farsi human evaluation results provide a consistent directional signal in favor of dynamic retrieval. On the paired comparison, M1 scored 0.72 points higher than B1 overall, with both male and female evaluators preferring M1 clearly. On asymmetric detection, M1’s unique detections were validated at a substantially higher rate than B1’s. The gender alignment gap widened slightly from B1 to M1, which is worth noting honestly, though both values remain well below the reference gap reported in the InDor study. Given the small evaluator pool and the inherent subjectivity of the task, these findings are best read as encouraging signs that culturally adaptive retrieval is moving in the right direction rather than as definitive evidence of superiority across all cases.

6.6.2 Italian

Seven native Italian speakers participated in the evaluation, comprising 4 women and 3 men. As with Farsi, the pilot scale and task subjectivity apply equally here and the results are treated as directional rather than conclusive. Table 6.16 reports Stratum 1 mean scores overall and by gender.

M1 scores 3.04 against B1’s 2.56, a gap of 0.48 points. The direction is consistent with Farsi, though the margin is smaller. M1 wins on 6 of the 10 items, ties on 2, and B1 wins on 2. The clearest individual signal comes from the Zaporizhzhia nuclear plant article examined in Section 6.5.2, where M1 scored the maximum possible 4.00 against B1’s 2.14. All seven Italian evaluators gave M1 a perfect rating. Maverick B1 produced a single observation about the headline’s sensationalism, while M1 identified the deeper diplomatic framing of the article, catching how describing European leaders as complicit with the “regime di Kiev” and attributing their policy positions to face-saving motivations reduces complex geopolitical choices to personal image dynamics. The unanimous

Condition	Overall		By Gender		
	Mean	SD	Male	Female	$\Delta M-F $
B1 (Static Few-Shot)	2.56	0.79	2.53	2.58	0.04
M1 (Dynamic Retrieval)	3.04	0.89	3.17	2.95	0.22

Table 6.16: Stratum 1 mean scores (1–4 scale) for Italian B1 and M1 rationales across 7 evaluators and 10 items (70 evaluator-item pairs per condition). SD denotes standard deviation across all evaluator-item pairs. $\Delta |M-F|$ is the absolute gender gap per condition.

evaluator response confirms that this difference was not a scoring artifact but a genuinely perceived qualitative gap between the two conditions.

The gender picture for Italian differs from Farsi. Under B1, the gender gap is effectively absent at $\Delta = 0.04$, with female evaluators scoring fractionally higher than male. Under M1, a gap of $\Delta = 0.22$ emerges with male evaluators rating higher. Dynamic retrieval therefore introduced a gender alignment disparity in Italian that was not present under the static baseline, the opposite direction from what the framework aimed to achieve on this dimension. Both values remain well below the $\Delta = 0.60$ reference gap reported in the InDor study.

Setup	Items	Appearances	Validated	Rate
B1 (Static Few-Shot)	4	28	25	89.3%
M1 (Dynamic Retrieval)	2	14	13	92.9%

Table 6.17: Stratum 2 validation results for Italian across 6 asymmetric items and 7 evaluators. Items indicates how many of the 6 asymmetric cases each setup appeared in. Appearances is the total number of evaluator judgments (items $\times$ 7). Validated indicates the evaluator judged the rationale as correctly identifying a real bias in the article.

The Italian Stratum 2 results present a notably different picture from Farsi. B1 appeared in 4 of the 6 asymmetric items and M1 in only 2, meaning B1 uniquely detected bias more often in cases where the two conditions diverged. Both conditions were highly validated by native Italian speakers: B1 at 89.3% and M1 at 92.9%, a difference of only 3.6 percentage points. Unlike Farsi where B1 validation rate was substantially lower at 57.1%, Italian B1 detections were judged as genuine by evaluators in almost nine out of ten cases. This pattern is consistent with the quantitative findings for Italian Maverick, where B1 showed competitive detection performance relative to M1. When each condition uniquely detected bias in Italian, both were largely right.

The overall Italian picture is one of modest but consistent improvement under dynamic retrieval on paired quality, combined with competitive and mutually validated detection in the asymmetric cases. The gap between B1 and M1 is smaller than in Farsi, and B1 was more active and more reliable in unique detection. These differences between the two languages are examined further in Section 7.1.

6.6.3 Cross-Community Alignment

Table 6.18 brings together the mean scores from both language communities to examine how LLaMA 4 Maverick’s rationales were perceived across gender and community dimensions under each condition.

Group	B1 (Static Few-Shot)		M1 (Dynamic Retrieval)	
	Mean	Δ	Mean	Δ
Farsi — Male	2.50	0.15	3.27	0.24
Farsi — Female	2.35		3.03	
Italian — Male	2.53	0.04	3.17	0.22
Italian — Female	2.58		2.95	
Farsi (overall)	2.41	0.14	3.13	0.09
Italian (overall)	2.56		3.04	

Table 6.18: Cross-community alignment for LLaMA 4 Maverick under B1 and M1. Δ denotes the absolute gap between the two groups in each pair. Community Δ compares mean scores between Farsi and Italian evaluators.

Several patterns emerge from the table. On gender alignment, both communities show a small gap under B1 that widens under M1: Farsi moves from $\Delta = 0.15$ to $\Delta = 0.24$ and Italian from $\Delta = 0.04$ to $\Delta = 0.22$, with male evaluators rating M1 higher than female in both cases. Dynamic retrieval did not reduce demographic alignment disparities, but widened them modestly. One honest interpretation is that the task’s high subjectivity plays a role: when overall rationale quality improves substantially, individual differences in how evaluators weight cultural sensitivity, textual grounding, and narrative framing may become more visible rather than less. This is plausible but does not fully resolve the concern. Reducing demographic alignment disparities remains an open challenge, though as noted throughout this thesis, demographic alignment serves as an exploratory additional metric rather than a primary evaluation criterion. Neither this pilot nor the InDor study involves pools large enough to treat subgroup gaps as statistically robust, and both should be read as starting points for hypothesis generation rather than definitive findings. The absolute values here remain well below the $\Delta = 0.60$ reported in the InDor study [4], but the direction of change is the same.

On community alignment, Maverick’s rationales were rated slightly higher by Farsi evaluators than Italian ones under M1. Under B1, Italian evaluators averaged 2.56 against Farsi’s 2.41, a gap of $\Delta = 0.14$ favouring Italian. Under M1 the direction flipped: Farsi evaluators averaged 3.13 against Italian’s 3.04, a gap of $\Delta = 0.09$ now favouring Farsi. The gap narrowed and reversed, suggesting the adaptive rationales were perceived as comparably appropriate across the two communities. This contrasts meaningfully with the InDor community gap of $\Delta = 1.09$, though direct comparison is limited by differences in task design and evaluator composition.

In summary, under dynamic retrieval Maverick’s rationales aligned slightly better with the Farsi-speaking community than the Italian-speaking one, though the margin of 0.09 points is small and should not be overstated.

The SD pattern adds a further dimension. In Farsi, M1’s lower SD (0.83 vs 1.01) indicates more consistent improvement across items. In Italian, M1’s higher SD (0.89 vs 0.79) reflects a more uneven profile: some articles received unanimous top scores while others showed less clear improvement. This suggests the retrieval mechanism is more reliably calibrated to the Farsi exemplar bank at this pilot scale, a hypothesis examined in Chapter 7.

Taken together, the human evaluation results provide directional evidence that dynamic retrieval reduces culturally misaligned rationale generation as perceived by native speakers in both communities. Both Farsi and Italian evaluators rated M1 rationales higher than B1, and asymmetric detection cases were validated at high rates in both languages. These are encouraging signs that connecting model reasoning with community-annotated exemplars moves outputs closer to what native readers recognize as culturally appropriate analysis rather than generic or monocultural commentary.

6.7 Chapter Summary

This chapter reported the implementation and evaluation of the Culturally Adaptive Retrieval-Based Framework across Farsi and Italian using LLaMA 4 Maverick and Mixtral-8x22B-Instruct. Beginning with corpus preparation, it described the cleaning and stratified splitting of the InDor annotations into the pilot test sets of 100 items per language and the Exemplar Banks of 891 Italian and 457 Farsi items, documenting the filtering decisions that shaped the final evaluation pools.

The quantitative evaluation compared all four prompting conditions across severity classification, span detection, and rationale quality. A1 produced the strongest severity classification scores for Maverick in both languages. Dynamic retrieval improved span detection and rationale quality most consistently in Farsi, while Italian results were more variable across models and conditions. The content refusal phenomenon concentrated in Italian Maverick B0 and A1 was documented in detail, with a diagnostic analysis distinguishing model-level generation refusals from infrastructure failures and a topic-level breakdown of the content categories that triggered them. As a consequence, Italian Maverick B0 and A1 results were computed on reduced evaluation pools of 45 and 58 rows respectively, making these two conditions not directly comparable to the full 100-row evaluations of the other fourteen conditions.

The qualitative analysis examined item-level BERTScore patterns and rationale content across Farsi and Italian, presenting side-by-side comparisons of B1 and adaptive condition outputs on selected articles to assess whether observed score improvements reflected genuine cultural connection rather than surface-level alignment. The human evaluation then tested this directly, asking seven native speakers per language to rate paired B1 and M1 rationales on cultural appropriateness, and found that evaluators in both communities consistently preferred the dynamically retrieved rationales, with the improvement more pronounced in Farsi.

Taken together, the results offer cautious but encouraging evidence that retrieval-based cultural aligning moves model reasoning closer to what native speakers recognize as appropriate analysis, while also surfacing open challenges in demographic alignment and cross-lingual consistency. Chapter 7 synthesizes these findings, situates them within the broader research questions, and closes the thesis with contributions, limitations, and directions for future work.

Chapter 7

Discussion, Conclusion, and Future Directions

This chapter brings together the findings from the pilot evaluation reported in Chapter 6 and situates them within the broader goals of the thesis. Section 7.1 synthesizes performance patterns across conditions and languages, addressing Research Question 3 directly and examining what the results reveal about when and how culturally adaptive retrieval helps. Section 7.2 documents two unexpected findings that emerged during implementation and evaluation: a prompt leakage issue in multilingual instruction formatting and a behavioral pattern of over-refusal in safety-trained models confronting sensitive analytical content. Section 7.3 discusses the boundaries of what the pilot can and cannot support. Sections 7.4, 7.5, and 7.6 close the thesis with a summary of contributions, ethical considerations, and directions for future work.

7.1 Cross-Condition and Cross-Lingual Synthesis

Research Question 1 asked how cultural and linguistic context shapes the annotation and interpretation of information disorder in multilingual news data. The corpus analysis in Chapter 3 showed that Farsi and Italian annotators drew on distinct cultural vocabularies and reasoning patterns when identifying manipulative framing. Farsi rationales were drawn on economic and geopolitical terms specific to the Iranian media context, while Italian rationales more often engaged with European political discourse and immigration narratives. These differences were not incidental but reflected the distinct informational environments in which the two communities read and interpret news. The answer to RQ1 is that cultural and linguistic context shapes annotation deeply and systematically, and that treating multilingual information disorder as a language-agnostic task risks missing the culturally embedded logic through which manipulation operates.

Research Question 2 asked how well baseline LLMs perform on severity classification, span extraction, and rationale generation for multilingual information disorder. Chapter 4 reviewed the InDor baseline results, which evaluated LLaMA 4 Maverick and Mixtral-8x22B-Instruct on span extraction and rationale generation across four language subsets using BERTScore. The results showed that static few-shot prompting did not produce uniform improvements over zero-shot: rationale scores for English and Italian declined under few-shot for both models, while Farsi and

Russian showed mixed or modest gains. Across all conditions, BERTScore values for rationale generation remained in the low-to-mid range, indicating limited semantic alignment between model outputs and human-written explanations. Additionally, the variation across languages and the inconsistency of few-shot gains point to a fundamental limitation of fixed exemplar strategies: they cannot adapt to the specific cultural and linguistic context of each input. This is precisely what motivates the retrieval-based framework proposed in Chapter 5. RQ2 is answered in the negative: static prompting does not reliably produce explanations that align with human-written culturally grounded rationales, and the gap is not uniform across languages or prompting conditions.

Research Question 3 asked whether the Culturally Adaptive Retrieval-Based Framework improves severity classification, span detection, rationale quality, and demographic alignment over the static baselines. The pilot results in Chapter 6 offer a cautiously positive but uneven answer.

The framework showed its clearest benefit in Farsi, where retrieval improved span detection and rationale quality across both models, and improved severity classification for Maverick. Farsi Mixtral severity was effectively a tie between B1 and M1. The human evaluation confirmed the direction, with native speakers rating M1 higher than B1 in both paired quality and asymmetric detection validation. Italian results were more mixed: severity classification improved under Maverick, but rationale quality favoured the static baseline for that model, and span detection showed no benefit from retrieval for Maverick while declining under dynamic conditions for Mixtral. The human evaluation still found M1 rated higher overall by native Italian speakers, though by a smaller margin than in Farsi.

This is a small-scale pilot on a genuinely difficult and underexplored problem. The results are best read as an encouraging first step that motivates further investigation rather than a definitive demonstration of effectiveness, and the open questions they raise are taken up in the sections that follow.

Instruction Language and Exemplar Source. Comparing M1 and A1 isolates the effect of instruction language since both use the same retrieved mechanism. In Farsi, Maverick favored A1 on severity classification by a clear margin, while Mixtral favored M1, suggesting the instruction language benefit depends on model. On span detection both conditions were close for both models, with A1 marginally ahead. On rationale quality, Maverick slightly preferred M1 and Mixtral was effectively tied, pointing to English instructions providing a more stable generative anchor for the stronger model. In Italian, the same severity pattern held for Maverick, though with the caveat that A1 was evaluated on a reduced pool. For Italian Mixtral, M1 edged ahead on severity and span while A1 was marginally better on BERTScore. Overall, no single condition dominated both models and both languages across all metrics. The interaction between instruction language and model scale appears to shape which condition is preferred, consistent with findings in the multilingual prompting literature [81, 77].

Cross-Lingual Patterns. The divergence between Farsi and Italian results may reflect a difference in corpus composition. The Farsi InDor subset is concentrated around a relatively coherent set of themes such as economic hardship and inflation, domestic political rhetoric, US-Iran relations, and religiously framed propaganda, which may mean that retrieved exemplars are more likely to be topically close to a given test article. The Italian subset spans a considerably wider range of content, including immigration conspiracy theories, COVID-19 misinformation, European geopolitical

conflict, far-right ideological narratives, and reports of sexual violence, introducing more variability into retrieval quality and, as documented in Section 6.4.3, triggering content refusal behavior in Maverick that fundamentally shaped the Italian evaluation pools. If this hypothesis holds, it would suggest that the framework’s effectiveness is likely sensitive to how well the exemplar bank covers the thematic and cultural space of the target corpus, with implications for how the approach might be deployed or adapted in new language communities.

The retrieval mechanism operates entirely on semantic similarity computed by BGE-M3, with no label-matching or category filtering. The system does not know whether a retrieved exemplar is Slightly or Highly problematic when it selects it. The fact that semantic retrieval alone produces the improvements documented here, without any explicit supervision of which type of bias to demonstrate, makes the results more rather than less encouraging as a foundation for further development and studies.

7.2 Emergent Findings

Running a multilingual retrieval-augmented pipeline across four prompting conditions, two models, and two languages is complex enough that unexpected behaviors will surface. Two such patterns emerged during development and testing, and rather than patching them quietly, this thesis documents both as findings worth reporting on their own. Each says something broader about how instruction-tuned models behave when deployed on sensitive analytical tasks in non-English settings.

7.2.1 Instruction Language Leakage and the If-Then Fix

The M1 condition presents task instructions in English while the retrieved exemplars and the article text are in the target language. During initial diagnostic runs, a systematic output pattern was observed: models generated rationale content in the correct target language but wrapped it in English if...then syntactic structures rather than the target-language equivalents, اگر ... آنگاه in Farsi and *se ... allora* in Italian. The scale of this leakage varied considerably across models and languages. In Farsi, Maverick produced if-then leaks in 27 out of 100 rationale outputs under M1, while Mixtral produced only 3. In Italian, the pattern was far more severe: Maverick leaked in 73 out of 100 outputs and Mixtral in 54. Our best guess is that seeing the structural output template written and hardcoded in English caused the model to treat the literal words “if” and “then” as a formatting rule to follow rather than a conceptual pattern to adapt into the target language.

This was not a cosmetic issue. Because BERTScore computes semantic similarity between generated and human reference rationales using contextual embeddings from a multilingual model, a rationale mixing English structural tokens with Farsi or Italian content scores lower against a fully target-language reference than a clean target-language output would. The reported M1 BERTScore improvements are therefore conservative estimates: had the leak been present throughout the final evaluation, the gap between M1 and B1 would have appeared smaller than the true underlying difference. Beyond automated metrics, presenting mixed-language outputs to human evaluators would have introduced an additional confound that the rating scale was not designed to handle.

The fix was a targeted runtime intervention. Rather than hardcoding the if-then structure in the English M1 prompt, two dynamic placeholders were introduced: {target_syntax_instruction}

and `{target_syntax_placeholder}`, resolved to the correct target-language conditional structure at runtime depending on the pipeline language setting. This change, described in Appendix B.3, required no alteration to the retrieval logic or prompt intent. After the fix, zero leaks were observed in either language under M1. The A1 condition did not require this intervention since its instructions are written directly in the target language, naturally producing the correct syntactic structure throughout.

7.2.2 Alignment Tax: Over-Refusal in Safety-Trained Models

During evaluation of Italian Maverick under B0 and A1, a systematic pattern of empty responses emerged at the rationale generation stage. As documented in detail in Section 6.4.3, a diagnostic analysis confirmed these were model-level generation refusals rather than infrastructure failures: the model successfully completed severity classification and span detection for the same items but declined to produce explanatory rationales. The refused articles clustered around narrative-level topic sensitivity such as international conflict, immigration extremism, and health misinformation, rather than surface-level toxicity, which registered at effectively zero across all items. The pattern did not appear in Farsi, whose InDor content is dominated by economic and domestic political narratives that do not map onto the categories prominent in English-language alignment training data, nor in Mixtral, which completed rationale generation cleanly across all Italian conditions.

We hypothesize that, in B1 setup in Italian, seeing a few examples in the prompt gave the model enough structural context to engage analytically rather than refuse. For A1, one possibility is that target-language instructions simply carry less instruction-following weight with this model than English ones do, making the safety guardrails easier to trip. But beyond these guesses we cannot say with confidence why B0 and A1 triggered the refusals while B1 and M1 did not. It is worth mentioning that Mixtral completed rationale generation cleanly across all Italian conditions, suggesting the refusal behavior reflects Maverick’s specific alignment posture rather than a property of the content itself.

A deliberate decision was made not to intervene. It would have been technically possible to add an explicit directive to the prompt informing the model that it was participating in an academic study of information disorder and should engage analytically with sensitive content regardless of topic. However, this would constitute a form of soft jailbreaking that alters the model’s default deployment behavior. The goal of this study is to evaluate API-accessed instruction-tuned models as they actually operate, not as they might operate under special conditions. A model’s refusal to analyze certain categories of news content is not a bug to be suppressed, but a measurable characteristic of that model in this domain, and documenting it is a more honest and scientifically valid outcome than engineering around it. Information disorder detection is by definition a task that engages with sensitive, controversial, and politically charged material. The finding that a leading commercial model refuses to generate analytical rationales for a substantial fraction of such content under certain prompting conditions is itself a contribution to understanding the practical limits of current LLM deployment in this domain.

The extent to which this refusal pattern generalizes across model versions, API providers, and prompt designs is an open question addressed briefly in Section 7.3.

7.3 Limitations

Corpus quality and available pool size. The InDor corpus is a genuinely novel resource, but it carries significant noise from its web-crawling origins. Advertisements, malformed entries, English-language articles in the Italian subset, and label conflicts between annotators required an extensive cleaning process before the corpus could be used reliably. After filtering, the usable master pools were 991 items for Italian and 557 for Farsi, yielding exemplar banks of 891 and 457 items respectively, and test sets of 100 items each. These numbers reflect the realistic upper bound of what the cleaned corpus could support given our stratified sampling constraints, not an arbitrary design choice. Expanding the test set further would have left the exemplar bank too thin to support meaningful retrieval diversity, and the bank sizes themselves were dictated by what remained after noise removal. For a pilot of this scope, 100 test items per language represents a reasonable tradeoff, but it limits the statistical power of the evaluation and makes it difficult to draw conclusions about performance on rare severity categories.

Structural inconsistency in source articles. Several InDor articles, both in Italian and Farsi, exhibit mismatches between the headline and the body, or between individual sentences that appear to originate from different news items due to crawling artifacts. The retrieval mechanism and the models operate on the full article structure, and this inconsistency introduces noise into both span detection and rationale generation in ways that are difficult to quantify. Removing these items would have further reduced an already constrained pool, so they were retained, but they represent an uncontrolled source of variability in the results.

Task subjectivity and human reference quality. Information disorder annotation is a highly subjective task. As the InDor study itself documents [4], and as our own cleaning process confirmed through the removal of label conflicts, the same article can be considered problematic by one annotator and neutral by another. The human references used as evaluation targets throughout this thesis reflect refined annotations from a specific set of annotators, and while the cleaning heuristics described in Chapter 6 were designed to reduce ambiguity, they do not eliminate it. This subjectivity propagates into all three evaluation metrics: severity F1, span overlap, and BERTScore all measure agreement with references that are themselves uncertain. This subjectivity is not unique to this study. Any framework operating in the information disorder domain faces the same fundamental tension: a news item judged problematic by one annotator may be read as neutral by another, and there is no ground truth that fully resolves this. The cleaning and selection criteria described in Chapter 6 represent a principled attempt to build consistent reference sets, but the resulting labels remain one defensible reading among several possible ones. This circularity is an inherent property of the domain rather than a correctable flaw in the methodology, and it applies, in different levels, to any automated or human-based assessment system working in this space. This has a direct implication for interpreting the severity classification results in Chapter 6. Macro F1 counts every disagreement between the model prediction and the human reference label as an error. In a task where human annotators themselves disagree on the same article, some of those counted errors may reflect a defensible reading of the content rather than a genuine model or setup failure. A model predicting Slightly when the reference says Moderately is not necessarily wrong, it may simply reflect a different but plausible interpretation of the same manipulative framing. The severity scores reported in this thesis measure alignment with one specific set of reference labels, not absolute classification accuracy, and should be read accordingly. One consequence of this design is that in cases

where two annotators agreed on Problematic but differed in their rationale depth, the longest available rationale was selected as the reference, on the assumption that longer rationales provide more complete span and reasoning coverage. This heuristic introduces a systematic bias toward verbosity in the reference set that may slightly advantage longer model outputs in BERTScore comparisons.

Pilot scale and available evaluation pool. The test sets of 100 items per language and the evaluator pools of 7 native speakers per language are small by the standards of large-scale NLP benchmarks, but they reflect the realistic constraints of this pilot rather than arbitrary choices. After cleaning and stratified sampling, the Stratum 1 eligible pools for human evaluation contained 45 Farsi items and 23 Italian items. Presenting all eligible items to evaluators, alongside the 11 and 6 asymmetric cases respectively, would have produced a form requiring over an hour of concentrated reading in a highly subjective task, which was not feasible for non-professional volunteer evaluators. The 10 paired items selected per language represent a manageable workload while still covering the range of severity labels in the test set. On evaluator count, seven native speakers per language compares reasonably with the InDor study’s total of approximately eight to nine annotators across all language communities, though it remains too small to support statistically robust subgroup analysis. These numbers are appropriate for a bounded pilot and are reported with that scope in mind throughout.

Language and model coverage. The pilot covers two of the four InDor languages. English was excluded because the central argument of this thesis is that current large language models are English-centric in their reasoning and training. Evaluating cultural alignment on English would mean testing the model in the very language it is already optimised for, which does not surface the cross-cultural blindness the framework is designed to address. Russian was excluded because no Russian-speaking evaluators were available and extending the framework to a third language would have substantially increased implementation and evaluation complexity beyond what this pilot could support. Russian remains a natural next candidate for future work. Similarly, two models were selected on the basis of the InDor baseline study, which evaluated LLaMA 4 Maverick and Mixtral-8x22B-Instruct as part of its own evaluation. Their notably different behaviors across conditions in this pilot are themselves an invitation to expand the comparison to a wider set of models in future work.

Retrieval model and exemplar bank design. BGE-M3 was used as the sole retrieval model throughout. This pilot does not constitute a retrieval benchmark, and testing multiple embedding models was beyond its scope. BGE-M3 is among the strongest available multilingual dense retrieval models and provides a solid starting point, but different encoders may retrieve different exemplars and produce different results. The exemplar bank is also static, built once from the cleaned InDor annotations and not updated during the pilot. The broader Human-in-the-Loop framework [19], of which this pilot represents an initial component, envisions iterative refinement of the bank through community feedback over successive rounds. That loop is not implemented here, and its potential to improve retrieval quality and cultural alignment over time remains to be evaluated in future work.

Human evaluation scope. The human evaluation assessed Maverick rationales only, comparing B1 and M1. Mixtral was not included in the human evaluation, meaning the qualitative assessment of cultural alignment covers only one of the two models evaluated on automatic metrics. Additionally, inter-annotator agreement was not computed for the evaluation ratings. Evaluators were asked to express community-aligned preferences about cultural appropriateness, and the 1–4 rating scale was designed to capture this preference rather than to produce a replicable annotation

label. Hence, agreement between evaluators is not a primary validity criterion for this study and therefore results are treated accordingly as preference signals rather than ground truth assessments.

Evaluation heuristics. Two heuristic choices shaped the evaluation in ways worth acknowledging. First, when two annotators agreed that an article was Problematic but differed in rationale depth, the longest available rationale was retained as the reference on the assumption that longer rationales provide more complete coverage. This may introduce a mild bias toward verbosity in BERTScore comparisons. Second, the span-based pairing heuristic used to select which rationale from each condition to present to human evaluators required at least 30% token-level overlap between B1 and M1 spans, or between each span and the human reference span, before falling back to the first available rationale. Items where no span pair met either threshold were included via the positional fallback, meaning a few evaluated rationale pairs may address different regions of the same article. This difference is communicated to human evaluators in their Google forms, mentioning that you may see some valid and culturally connected explanations talking about different parts of the news and in this case, please rate them separately based on the criteria. Both heuristics are documented in Chapters 4 and 5 respectively and are reproducible, but they represent design choices rather than neutral defaults.

7.4 Summary of Contributions

This thesis set out to examine whether culturally adaptive retrieval-based prompting can reduce cultural blindness and culturally misaligned rationale generation in large language models applied to multilingual information disorder assessment. The pilot results suggest it can, at least directionally and within the constraints of the bounded evaluation described throughout. Three concrete contributions emerge from this work.

A structured landscape of information disorder datasets. As a foundation for understanding where the field stands, this thesis compiled and systematically organized a curated review of 108 datasets related to fake news and information disorder, covering a range of modalities, languages, and annotation types including severity labels, text spans, and human-written rationales. The resulting reporting sheet is made publicly available as a structured community resource through the project repository,¹ intended to support researchers navigating an increasingly fragmented dataset landscape and to motivate further development of rationale-aligned multilingual resources.

The Culturally Adaptive Retrieval-Based Framework. The central contribution of this thesis is the design, implementation, and pilot evaluation of a retrieval-based framework for culturally adaptive information disorder assessment. The framework introduces a community-annotated Exemplar Bank for each target language, encoded using multilingual dense retrieval, and uses semantic similarity at inference time to dynamically ground model reasoning in culturally relevant examples rather than fixed category-balanced demonstrations. Four prompting conditions were evaluated across two languages, two models, and three assessment tasks, producing a structured comparison of how instruction language and exemplar source each contribute to model performance. The full pipeline, prompt templates, and evaluation scripts are openly available in the project repository to support reproducibility and future extension to additional languages and model families. This pilot

¹<https://github.com/Maziarkiani/culturally-adaptive-prompting>

represents the first operational stage of a broader Human-in-the-Loop architecture [19], in which iterative community feedback would progressively refine the Exemplar Bank over successive rounds. That broader loop is left for future work, but the results here provide an empirical foundation for it.

Methodological findings on multilingual LLM behavior. Two emergent findings from the development and evaluation process contribute to the broader understanding of how instruction-tuned models behave in multilingual analytical contexts. The first is a documented instruction language leakage pattern in multilingual retrieval-augmented setups, where English structural templates override target-language output formatting at scale, and a runtime dynamic injection fix that fully resolves it without altering prompt intent. This fix is reusable in any multilingual prompting pipeline where instruction language and output language differ. The second is a systematic documentation of over-refusal behavior in a safety-aligned frontier model when confronted with analytically sensitive content in a non-English context, including a diagnostic methodology for distinguishing content-level refusals from infrastructure failures and a topic-level analysis of what categories of information disorder content trigger this behavior. Together these findings point to practical constraints and solutions that are relevant to any research group deploying instruction-tuned models on sensitive multilingual tasks.

Overall, these contributions are intended as a first step toward information disorder assessment systems that are not only more accurate but more culturally honest: systems that reason about manipulation in the language and cultural logic of the communities they serve, rather than projecting a single dominant analytical frame onto linguistically diverse content.

7.5 Ethical Considerations

This thesis works with data and annotations from the InDor corpus, a community-driven resource developed under an established ethical framework. The InDor study followed institutional ethics committee approval discussed in the original study [4]. The present work operates downstream of that framework and adheres to its established protocols. Comprehensive ethical guidelines regarding the collection, annotation, and handling of sensitive content are detailed in the primary InDor dataset study and are not reproduced here.

The corpus contains content that is politically sensitive, offensive, or otherwise disturbing, including racist, sexist, or violent language. This material appears in the Exemplar Bank, the test set, and in the appendix prompt examples strictly for research transparency and reproducibility purposes. No new offensive content was generated or introduced by this thesis.

Regarding annotator identity, the InDor annotation files include numeric annotator identifiers rather than personal information. The mapping between these identifiers and individual annotators is held exclusively by the Aequa-Tech S.r.l.² research consortium that coordinated the InDor project. This thesis used those identifiers only for conflict detection and resolution during corpus preparation, and at no point had access to or sought to recover personal information about individual annotators.

The human evaluators who participated in the evaluation for this thesis did so voluntarily. The evaluation forms were fully anonymous: no names, email addresses, or any personally identifying information were collected or stored. Evaluators were informed of the study’s purpose before participating and were free to withdraw at any time. Although collecting basic demographic data such

²<https://aequa-tech.com/>

as gender was necessary for the alignment analysis described in Section 6.6, an option reflecting “I prefer not to say” was included in the gender selection field of the evaluation form to ensure that participation did not require disclosure of personal identity information.

7.6 Future Work

Several directions emerge naturally from the findings and limitations of this pilot.

Language and model expansion. The most immediate extension is to Russian, the remaining InDor language not covered in this pilot, which would allow a direct comparison across three culturally and linguistically distinct communities using the same corpus and framework design. Beyond InDor, extending the approach to low-resource languages where cultural alignment gaps are likely more pronounced represents a longer-term but important direction. On the model side, the notably different behavior of Maverick and Mixtral across conditions is itself an invitation to test a wider range of model families and scales, to understand how model capacity and alignment posture each interact with retrieval-based reasoning.

Retrieval mechanism improvements. This pilot used a single embedding model, BGE-M3, and retrieved exemplars purely on semantic similarity without any awareness of bias type or severity label. Future work could explore alternative or ensemble retrieval models, label-aware or bias-type-aware retrieval strategies that surface exemplars demonstrating the specific kind of manipulation present in the test article, and cross-lingual retrieval where exemplars from one language community inform reasoning in another.

Addressing open findings. Two findings from this pilot warrant dedicated follow-up. The gender alignment gap widening under dynamic retrieval rather than narrowing was unexpected and its cause is not fully understood. Targeted investigation into whether this reflects a retrieval artifact, a model artifact, or the inherent subjectivity of the task would clarify whether it is a correctable design issue or a fundamental property of the evaluation setup. The over-refusal behavior documented in Italian Maverick could also be studied more systematically across model families, content categories, and prompting strategies, including the prompt-level mitigation approach discussed but deliberately left unimplemented in this study.

The full Human-in-the-Loop architecture. This pilot evaluates the first operational stage of the broader framework described in [19]. The full vision involves iterative refinement of the Exemplar Bank through community feedback over successive annotation rounds, where human judgments on model outputs feed directly back into the retrieval pool and progressively improve cultural alignment. Evaluating how bank quality evolves with each feedback cycle, and how quickly the framework adapts to emerging information disorder patterns in a given community, remains the most significant open question for the research program as a whole.

Broader applications. Finally, while this thesis focuses specifically on information disorder assessment, the core mechanism of aligning model reasoning with community-annotated exemplars retrieved by semantic similarity is not inherently domain-specific. Extending this approach to other culturally sensitive and subjective NLP tasks, such as sentiment analysis, offensive language detection, or narrative framing analysis, where community-specific interpretive norms shape what counts as a valid label or explanation, represents a natural and potentially high-impact direction for future work.

Use of AI Assistance

This thesis was developed and prepared over six months, during which AI language models were consulted as assistive tools at various stages of the writing and development process. Gemini (Google), ChatGPT (OpenAI), and Claude (Anthropic) were used at different points to assist with drafting and refining prose sections, proofreading written text, summarizing reference material, exploring analytical approaches, and formatting tables, figures, and custom LaTeX environments. For the implementation work, AI tools assisted with code editing, debugging, and troubleshooting pipeline issues.

All research design, experimental decisions, data analysis, and intellectual contributions are entirely the author's own. All AI-generated or AI-assisted outputs were reviewed, critically evaluated, and revised by the author before inclusion. The author takes full responsibility for the accuracy, integrity, and originality of all content presented in this thesis.

Appendix A

Human Evaluation Guidelines

This appendix reproduces the evaluation guidelines distributed for human evaluation during the cross-cultural assessment study described in Section 6.6. Guidelines were prepared separately for Farsi and Italian evaluators, each presented in the respective target language. The actual news articles, rationale pairs assessed during the study, and edge cases are not reproduced here.

A.1 Human Evaluation Guidelines: Farsi

Evaluation Guidelines and Rating Criteria - Farsi

Configuration: Human Annotation Module

Interface: Google Forms (Localized for Farsi native speakers)

Task Introduction and Guidelines

از اینکه دعوت به شرکت در این پژوهش را پذیرفتید سپاسگزارم!

در این پرسشنامه، شما چند جمله کوتاه از یک خبر را به همراه یک یا دو توضیح تولید شده توسط هوش مصنوعی می‌خوانید. این توضیح تلاش می‌کند بیان کند که چرا خبر مورد نظر دارای سوگیری، جهت‌دهی پنهان، تکنیک‌های دستکاری افکار عمومی، دروغ‌پراکنی، اطلاعات مغرضانه یا اطلاعات نادرست (میس‌اینفورمیشن) است. هوش مصنوعی این توضیحات را در قالب شرطی (اگر... آنگاه...) ارائه داده است. به عنوان مثال: «اگر فلان موضوع را بزرگنمایی کنیم و کلمات احساسی به کار ببریم، آنگاه این امر می‌تواند واقعیت داستان را دستکاری کند». شما باید این توضیحات را بخوانید و از ۱ تا ۴ به آن‌ها امتیاز دهید. (۱ = ضعیف، ۲ = متوسط، ۳ = خوب و ۴ = عالی)

ملاک‌های امتیازدهی چیست؟

- **استدلال بر اساس متن:** استدلال بر اساس کلمات و محتوای همان خبر باشد. توضیح هوش مصنوعی باید دقیقاً مرتبط با موضوع و رویداد مطرح‌شده در خبر باشد و از بحث اصلی خارج نشود.

• **درک بافت فرهنگی و زبانی:** کدام توضیح توانسته است بافت محلی و فرهنگی فارسی را بهتر درک کند و تکنیک‌های ظریف اخبار جعلی و چارچوب‌بندی‌های فضای فرهنگی و سیاسی ایران را تشخیص دهد؟ هدف ارزیابی این است که کدام مدل این فضا را بهتر درک کرده است.

• **پرهیز از کلیشه‌ها:** کدام توضیح از پیش‌فرض‌های عمومی و استانداردهای انگلیسی‌محور دوری کرده است؟ به عنوان مثال، برداشت کدام گزینه از متن خبر به جامعه و فرهنگ ایران نزدیک‌تر است و کدام یک بیگانه به نظر می‌رسد؟

این سوالات در ۲ دسته قرار دارند:

دسته اول: در این دسته مجموعاً ۱۰ خبر وجود دارد. شما یک خبر را می‌خوانید که دارای ۲ توضیح تولید شده توسط هوش مصنوعی در همان قالب ذکر شده است. شما باید به هر یک از این دو پاسخ امتیاز دهید تا مشخص شود کدام توضیح از دیدگاه یک فارسی‌زبان عملکرد بهتری داشته است.

دسته دوم: در این بخش ۱۱ خبر برای بررسی وجود دارد، اما هر کدام تنها دارای یک توضیح هستند. اخبار این بخش واقعاً دارای سوگیری یا ایراد بوده‌اند، اما هوش مصنوعی در یکی از روش‌ها متوجه این ایراد نشده و متن را خنثی ارزیابی کرده است، در حالی که در روش دیگر، ایراد را تشخیص داده و دلیل آن را بیان کرده است. در این موارد، شما باید ارزیابی کنید که آیا توضیح ارائه شده بر اساس ملاک‌های ذکر شده، توضیح معتبری است یا خیر.

هدف این مقایسه چیست؟

این توضیحات توسط یک مدل هوش مصنوعی اما با استفاده از دو روش پرامپت (دستورنویسی) متفاوت تولید شده‌اند. هدف این است که مشخص کنیم هوش مصنوعی در کدام حالت عملکرد بهتری ارائه داده است.

دو نکته مهم:

(۱) گاهی اوقات ممکن است یکی از توضیحات به بخش خاصی از خبر اشاره کند و توضیح دیگر به بخش دیگری بپردازد، در حالی که هر دو از نظر منطقی درست باشند. در این حالت، لطفاً هر توضیح را به صورت مستقل و بر اساس ملاک‌ها ارزیابی کنید.

(۲) در برخی قسمت‌ها ممکن است فاصله‌های اضافی در متن توضیحات وجود داشته باشد که ناشی از ایراد فرمت‌بندی است. از این بابت پوزش می‌طلبم.

لازم به ذکر است که آدرس ایمیل شما ذخیره نخواهد شد، حتی اگر وارد حساب کاربری گوگل خود شده باشید، و پاسخ‌ها کاملاً ناشناس باقی می‌مانند.

با تشکر از همکاری شما.

مازیار

Demographics and Meta-Data Collection

* Indicates required question

لطفاً در ابتدا جنسیت خود را انتخاب کنید. این داده برای تحلیل ما ضروری است. *

زن ☐

مرد ☐

ترجیح می‌دهم اعلام نکنم ☐

همچنین، لطفاً بازه سنی خود را مشخص کنید: *

۱۸ تا ۲۴ ☐

- ☐ ۲۵ تا ۳۴
☐ ۳۵ تا ۴۴
☐ ۴۵ تا ۵۴
☐ ۵۵ یا بیشتر
☐ ترجیح می‌دهم اعلام نکنم

Part 1: Comparative Evaluation Format

دسته اول خبرها (شامل دو توضیح برای ارزیابی)

خبر ۱:

[main news body here]

توضیح اول:

[rationale 1 here]

خبر ۱ - امتیاز توضیح اول: *

☐ ۱ (ضعیف / نامرتب)

☐ ۲

☐ ۳

☐ ۴ (عالی / مستند)

توضیح دوم:

[rationale 2 here]

خبر ۱ - امتیاز توضیح دوم: *

☐ ۱ (ضعیف / نامرتب)

☐ ۲

☐ ۳

☐ ۴ (عالی / مستند)

Part 2: Edge Case Verification Format

دسته دوم خبرها (۱۱ خبر همراه با ۱ توضیح جهت بررسی صحت ایراد وارد شده)

خبر ۱:

[body news here]

توضیح:

[rationale here]

خبر ۱: آیا این توضیح به درستی یک سوگیری یا چارچوب‌بندی فریبنده را در متن تشخیص داده، یا دچار توهم شده و ایراد نادرستی گرفته است؟ *

- ☐ بله، ایراد مطرح شده درست و مستند به متن است.
☐ خیر، هوش مصنوعی دچار توهم شده و در اینجا پیش‌فرض نادرستی دارد.

A.2 Human Evaluation Guidelines: Italian

Evaluation Guidelines and Rating Criteria - Italian

Configuration: Human Annotation Module

Interface: Google Forms (Localized for Italian native speakers)

Task Introduction and Guidelines

Grazie per aver accettato di partecipare a questo studio!

Prima di iniziare, desideriamo fornirti una breve introduzione al compito: in questo questionario ti verrà chiesto di leggere brevi estratti di articoli di cronaca o di attualità politica, accompagnati da due spiegazioni generate dall'Intelligenza Artificiale (IA). Lo scopo di queste spiegazioni è illustrare il motivo per cui l'articolo in questione contiene bias, narrazioni tendenziose, tecniche di manipolazione della pubblica opinione o disinformazione. L'IA ha formulato queste argomentazioni seguendo una struttura logica basata sulla dinamica condizionale “Se... allora...” (Ad esempio: “Se si esagera questo argomento e si usa un linguaggio emotivo, allora si rischia di manipolare la percezione dei fatti da parte del pubblico”).

Il tuo compito è leggere queste spiegazioni e valutarle su una scala da 1 a 4 (1 = Insufficiente, 2 = Sufficiente, 3 = Buono, 4 = Eccellente).

Quali sono i criteri di valutazione?

- **Coerenza con il testo:** L'argomentazione deve basarsi sulle parole effettive e sul reale contenuto della notizia. La spiegazione dell'IA deve essere pertinente e non deve divagare su argomenti o eventi non presenti nel testo.
- **Comprensione del contesto culturale:** Quale spiegazione coglie meglio il contesto locale e le dinamiche italiane? Quale identifica in modo più accurato le sottili tecniche di disinformazione all'interno del panorama politico e mediatico italiano? Desideriamo capire quale delle due analisi sia più accurata sotto questo aspetto.
- **Assenza di stereotipi:** Quale spiegazione evita cliché, assunzioni generiche o standard tipicamente “anglocentrici”? Ad esempio, quale interpretazione sembra meno estranea o distante dalla società e dalla cultura italiana?

Le domande sottoposte alla tua valutazione sono suddivise in due sezioni distinte:

Prima Sezione:

Questa sezione comprende un totale di 10 estratti giornalistici. Per ciascun articolo, ti verrà chiesto di leggere attentamente il testo e di valutare le due diverse spiegazioni generate dall'Intelligenza Artificiale, formulate secondo la struttura condizionale precedentemente illustrata. L'obiettivo è assegnare un punteggio a entrambe le risposte, al fine di determinare

quale delle due offra la spiegazione più efficace e accurata dal punto di vista di un madrelingua.

Seconda Sezione:

Questa sezione comprende 6 estratti giornalistici, ciascuno accompagnato da una singola spiegazione. I testi selezionati presentano reali problemi di bias informativo; tuttavia, mentre in una delle configurazioni sperimentali l'Intelligenza Artificiale non ha rilevato anomalie (valutando il testo come neutrale), nell'altra metodologia ha identificato il bias e ha fornito la relativa spiegazione. In questo caso, ti viene chiesto di valutare se la spiegazione fornita sia valida e soddisfacente alla luce dei criteri metodologici precedentemente definiti.

Qual è lo scopo di questo confronto?

Entrambe le spiegazioni sono generate dall'IA, ma attraverso due diverse metodologie di prompting. L'obiettivo è determinare quale dei due approcci consenta all'IA di ottenere le prestazioni migliori.

Note importanti:

Talvolta, le due spiegazioni potrebbero concentrarsi su parti diverse della stessa notizia, ed entrambe potrebbero teoricamente essere corrette. In questi casi, ti invitiamo a valutare ciascuna spiegazione in modo indipendente, applicando sempre gli stessi criteri.

Informativa sulla privacy:

Non raccogliamo il tuo indirizzo email. Anche se hai effettuato l'accesso al tuo account Google durante la compilazione del modulo, il tuo indirizzo email non verrà registrato né condiviso con noi in alcun modo. Le risposte fornite in questo modulo sono completamente anonime.

Si prega di notare che il modulo di valutazione e i contenuti delle notizie potrebbero contenere materiale politicamente sensibile, offensivo o comunque inquietante, inclusi contenuti razzisti, sessisti o violenti. Questo materiale è presentato qui esclusivamente a scopo di ricerca.

Grazie per la comprensione e la collaborazione. Maziar

Demographics and Meta-Data Collection

* Indicates required question

Per prima cosa, seleziona il tuo genere. Questo dato è fondamentale per la nostra analisi. *

- ☐ Donna
- ☐ Uomo
- ☐ Preferisco non specificarlo

Sarebbe perfetto se potessi indicare anche la tua fascia d'età: *

- ☐ 18 - 24 anni
- ☐ 25 - 34 anni
- ☐ 35 - 44 anni
- ☐ 45 - 54 anni

☐ 55 o più

Part 1: Comparative Evaluation Format

Notizia 1:

[main news body here]

Spiegazione 1:

[rationale 1]

Notizia 1 - Punteggio Spiegazione 1: *

- ☐ 1 (Scadente / Non pertinente)
- ☐ 2
- ☐ 3
- ☐ 4 (Eccellente / Ben documentato)

Spiegazione 2:

[rationale 2 here]

Notizia 1 - Punteggio Spiegazione 2: *

- ☐ 1 (Scadente / Non pertinente)
- ☐ 2
- ☐ 3
- ☐ 4 (Eccellente / Ben documentato)

Part 2: Edge Case Verification Format

Sezione finale: Verifica della validità di una sola spiegazione

In questa sezione leggerai una sola notizia accompagnata da un'unica spiegazione. Il tuo compito è esclusivamente verificare se la spiegazione fornita è valida e basata sui criteri definiti, oppure se è errata e si tratta di un'allucinazione dell'IA.

Notizia 1:

[body news here]

Spiegazione:

[rationale here]

Notizia 1: questa spiegazione ha individuato correttamente un bias o una narrazione ingannevole nel testo, oppure si tratta di un'allucinazione e l'obiezione sollevata è errata? *

- ☐ Sì, l'obiezione sollevata è corretta e fondata sul testo.
- ☐ No, l'IA ha avuto un'allucinazione e parte da un presupposto errato in questo caso.

Appendix B

Prompt Templates

This appendix provides the prompt templates used across all four experimental conditions (B0, B1, M1, and A1) for both the Farsi and Italian evaluation pipelines. Since the task instruction text is identical across B0, B1, and A1 within the same language, the prompts are presented in groups rather than repeated for each condition separately. Section B.1 presents the Farsi task prompts shared by B0, B1, and A1, with the global override warning appended for B1 and A1 only. Section B.2 presents the equivalent Italian prompts. Section B.3 presents the English task prompts used in M1, which apply to both languages, along with global override rule and dynamic syntax injections. The static in-context examples specific to B1 are provided in Sections B.4 and B.5. For B0, no examples are appended. For B1, the static examples in Sections B.4 and B.5 follow the task prompts.

The outline for prompts and few-shot examples is as follows:

- *Task Prompts — Farsi (B0, B1, and A1)*
- *Task Prompts — Italian (B0, B1, and A1)*
- *Task Prompts — English (M1)*
- *Static Examples — B1 Farsi*
- *Static Examples — B1 Italian*

For M1 and A1, dynamically retrieved exemplars are injected at the same position at inference time and therefore cannot be reproduced statically here. The only difference between the two model runs within each condition is the model identifier passed to the OpenRouter API, changing from meta-llama/llama-4-maverick to mistralai/mixtral-8x22b-instruct, and the output file path. All prompts shown here correspond to the LLaMA 4 Maverick runs. Full executable code, generation parameters, pipeline scripts, and evaluation code are available in the project repository at <https://github.com/Maziarkiani/culturally-adaptive-prompting>.

B.1 Task Prompts: Farsi (B0, B1, A1)

Faris Complete Prompt Sequence

Configuration: B0, B1 and A1 prompts

Language: Farsi

1. System Instruction

شما یک هوش مصنوعی سخت‌گیر برای قالب‌بندی داده‌ها هستید. شما باید فقط تگ‌های درخواست‌شده را برگردانید. از نوشتن متن‌های محاوره‌ای، احوال‌پرسی یا توضیحات اضافی خودداری کنید.

2. Task 1: Severity Classification

شما یک متخصص در حوزه چارچوب‌بندی، سوگیری زبانی و اختلال اطلاعات هستید. اختلال اطلاعات به معنای آلودگی فضای اطلاعاتی است که شامل سه دسته اصلی می‌شود: کژاطلاعات (اطلاعات نادرست و سهوی)، دروغ‌رسانی (اطلاعات نادرست و عمدی) و اطلاعات مغرضانه (اطلاعات واقعی با هدف آسیب‌رسانی). این موارد اغلب با دستکاری احساسی، بسترسازی فریبنده یا چارچوب‌بندی تحریف‌شده مشخص می‌شوند.

وظیفه شما طبقه‌بندی شدت زبان مسئله‌دار در بخشی از یک مقاله خبری است.

شدت را بر اساس ویژگی‌های زیر طبقه‌بندی کنید:

- سوگیری رویدادی: توصیف رویدادها به گونه‌ای که مسئولیت را کم‌اهمیت جلوه دهد یا تحریف کند.
- سوگیری در توصیف افراد (سوگیری انتسابی): توصیف اشخاص، گروه‌ها یا جوامع با استفاده از زبانی احساسی، کلیشه‌ای، برچسب‌زننده یا اغراق‌آمیز.
- احساسات‌گرایی: بزرگ‌نمایی یا دراماتیزه کردن واقعیت‌ها.
- گمانه‌زنی‌ها: استفاده از عبارات مبهم به عنوان واقعیت.

برچسب‌های میزان شدت:

- "none" - هیچ ویژگی مسئله‌داری وجود ندارد. متن واقع‌گرایانه، متعادل و خنثی است.
- "slightly" - زبان مسئله‌دار جزئی وجود دارد، اما پیام کلی عمدتاً عینی باقی می‌ماند.
- "moderately" - موارد متعددی از سوگیری یا زبان گمراه‌کننده وجود دارد که باعث تحریف ملموس شده و می‌تواند جهت‌گیری متن و درک مخاطب را تغییر دهد.
- "highly" - استفاده گسترده و شدید از زبان سوگیرانه یا احساساتی. خطر بالای اطلاعات نادرست.

قالب خروجی:

فقط یکی از چهار برچسب را پس از تگ زیر برگردانید، دقیقاً به این شکل:

<PREDICTED_LABEL>: none

<PREDICTED_LABEL>: slightly

<PREDICTED_LABEL>: moderately

<PREDICTED_LABEL>: highly

هیچ توضیح اضافی یا متن دیگری اضافه نکنید. فقط برچسب معتبر را برگردانید.

اکنون ورودی زیر را پردازش کنید:

{instance}

3. Task 2a: Span Detection

شما یک متخصص در حوزه چارچوب‌بندی، سوگیری زبانی و اختلال اطلاعات هستید. اختلال اطلاعات به معنای آلودگی فضای اطلاعاتی است که شامل سه دسته اصلی می‌شود: کژاطلاعات (اطلاعات نادرست و سهوی)، دروغ‌رسانی (اطلاعات نادرست و عمدی) و اطلاعات مغرضانه (اطلاعات واقعی با هدف آسیب‌رسانی). این موارد اغلب با دستکاری احساسی، بسترسازی فریبنده یا چارچوب‌بندی تحریف‌شده مشخص می‌شوند. وظیفه شما تجزیه و تحلیل گزیده‌های خبری و شناسایی بازه‌های متنی است که گمراه‌کننده، مغرضانه، سوداگرانه یا دارای بار احساسی هستند.

وظیفه:

- فقط بازه‌های منحصربه‌فرد و بدون همپوشانی را شناسایی کنید.
- اگر هیچ بازه‌ای یافت نشد، خروجی باید دقیقاً ["No"] باشد.

قانون حفظ کاراکترها (بسیار مهم):

بازه‌های استخراج‌شده باید دقیقاً و کاراکتر به کاراکتر مطابق متن اصلی باشند. به هیچ وجه املای کلمات، علائم نگارشی، فاصله‌ها و به ویژه نیم‌فاصله‌ها را تغییر ندهید.

بازه‌های مشکل‌ساز شامل:

- سوگیری رویدادی: رویدادها را به گونه‌ای توصیف می‌کند که مسئولیت تحریف شود.
- سوگیری در توصیف افراد (سوگیری انتسابی): توصیف اشخاص، گروه‌ها یا جوامع با استفاده از زبانی احساسی، کلیشه‌ای، برچسب‌زننده یا اغراق‌آمیز.
- واقعیت‌ها را هیجان‌انگیز یا اغراق‌آمیز جلوه می‌دهد.
- از اظهارات مبهم به گونه‌ای استفاده می‌کند که گویی واقعی هستند.

فرمت خروجی (دقیق):

اگر یک بازه: ["..."]
 اگر چند بازه: ["...", "..."]
 اگر هیچ بازه‌ای نیست: ["No"]

اکنون ورودی زیر را پردازش کنید:

{instance}

پاسخ را فقط با یک بلوک <SPANS> برگردانید. بلوک‌های متعدد برگردانید. بازه‌های تکراری ننویسید. هیچ توضیحی اضافه نکنید.

4. Task 2b: Rationale Generation

شما یک متخصص در حوزه چارچوب‌بندی، سوگیری زبانی و اختلال اطلاعات هستید. اختلال اطلاعات به معنای آلودگی فضای اطلاعاتی است که شامل سه دسته اصلی می‌شود: کژاطلاعات (اطلاعات نادرست و سهوی)، دروغ‌رسانی (اطلاعات نادرست و عمدی) و اطلاعات مغرضانه (اطلاعات واقعی با هدف آسیب‌رسانی). این موارد اغلب با دستکاری احساسی، بسترسازی فریبنده یا چارچوب‌بندی تحریف‌شده مشخص می‌شوند.

وظیفه شما توضیح این است که چرا بازه‌های متنی خاص در یک خبر، گمراه‌کننده، مغرضانه یا مشکل‌ساز هستند.

به شما یک گزیده خبری و لیستی از بازه‌های استخراج‌شده داده می‌شود.

هدف شما تولید دقیقاً یک دلیل (rationale) برای هر بازه است.

از این قالب دقیق استفاده کنید: ”اگر [اشاره به بازه در متن]، آنگاه [پیامد یا نتیجه]”

دستورالعمل‌های خروجی (رعایت دقیق الزامی است):

- برای هر بازه دقیقاً یک دلیل به همان ترتیب برگردانید.
- اگر ["No"] :<SPANS> بود، شما نیز دقیقاً ["No"] :<RATIONALES> برگردانید. دلیل جدیدی نسازید.
- هر دلیل باید داخل گیومه (") باشد.
- همه دلایل را در یک لیست برگردانید: <RATIONALES>: ["اگر ...، آنگاه...", "اگر ...، آنگاه...", "اگر ...، آنگاه..."] - از استفاده از نقل قول‌های تودرتو یا سه نقطه (...) در داخل دلایل خودداری کنید.
- بازه‌ها را ترکیب نکنید.
- مطلقاً هیچ فرآیند فکری، پیش‌نویس یا اصلاح خودکاری نداشته باشید. هیچ کلمه‌ای خارج از تگ‌ها ننویسید. فقط و فقط لیست نهایی را برگردانید.

اکنون ورودی زیر را پردازش کنید:

{instance} :گزیده خبری:

{spans} :بازه‌ها:

Global Override Warning (Farsi)**Global Override: Input Data Accountability**

توجه مهم: مثال‌های ارائه شده مستقیماً از داده‌های انسانی استخراج شده‌اند و ممکن است بعضی از آنها دارای خطاهای نگارشی و قالب‌بندی باشند (مانند بازه‌های تکراری، همخوانی نداشتن تعداد بازه‌ها و دلایل، عدم پیروی از قالب‌بندی پیشنهادی و وجود علائم اضافی مثل 0 و \n). این موارد صرفاً خطای انسانی هستند و نباید در خروجی شما تکرار شوند. در صورت مشاهده، از این موارد فقط برای یادگیری منطق سوگیری استفاده کنید و در پاسخ خود، دقیقاً از قالب استاندارد درخواست شده پیروی کنید.

B.2 Task Prompts: Italian (B0, B1, A1)**Italian Complete Prompt Sequence****Configuration:** B0, B1, and A1 prompts**Language:** Italian**1. System Instruction**

Sei un'IA rigorosa per la formattazione dei dati. DEVI restituire SOLO i tag richiesti. Non includere riempitivi conversazionali, saluti o spiegazioni.

2. Task 1: Severity Classification

Sei un esperto di framing, bias linguistici e Disordine dell'Informazione. Il disordine

dell'informazione è l'inquinamento dello spazio informativo che include disinformazione, disinformazione e malinformazione. È spesso caratterizzato da manipolazione emotiva, falsi contesti o framing distorto. Il tuo compito è classificare la gravità del linguaggio problematico in un estratto di un articolo di notizie.

Classifica la gravità in base alla presenza e all'impatto delle seguenti caratteristiche problematiche:

- Bias eventivo: descrive o fa riferimento a eventi in un modo che minimizza o distorce la responsabilità.
- Bias attributivo: descrive persone, groups o comunità usando un linguaggio emotivamente carico, stereotipato, etichettante o esagerato.
- Sensazionalismo: esagera o drammatizza i fatti.
- Speculazione: usa affermazioni vaghe o speculative come se fossero fattuali.

ETICHETTE DI GRAVITÀ:

"none" – Nessuna caratteristica problematica presente. Il testo è fattuale, equilibrato e neutrale.

"slightly" – È presente un linguaggio leggermente problematico, ma il messaggio complessivo rimane per lo più fattuale.

"moderately" – Sono presenti più casi di bias o linguaggio fuorviante che producono una distorsione tangibile e possono alterare la comprensione del lettore.

"highly" – Uso grave e pervasivo di linguaggio parziale, speculativo o emotivamente manipolativo. Alto rischio di disinformazione.

FORMATO DI OUTPUT:

Restituisci solo una delle quattro etichette dopo il tag seguente, esattamente in questo modo:

```
<PREDICTED_LABEL>: none
<PREDICTED_LABEL>: slightly
<PREDICTED_LABEL>: moderately
<PREDICTED_LABEL>: highly
```

Non aggiungere spiegazioni, commenti o altro testo. Restituisci solo un'etichetta valida.

Ora processa il seguente input:

{instance}

3. Task 2a: Span Detection

Sei un esperto di framing, bias linguistici e Disordine dell'Informazione. Il disordine dell'informazione è l'inquinamento dello spazio informativo che include disinformazione, disinformazione e malinformazione. È spesso caratterizzato da manipolazione emotiva, falsi contesti o framing distorto. Il tuo compito è analizzare estratti di notizie e identificare span di testo che sono fuorvianti, di parte, speculativi o emotivamente caricati.

Compito:

- Identifica solo span unici e NON sovrapposti.
- Se non trovi alcuno span problematico, l'output deve essere esattamente <SPANS>: ["No"].

Regola di conservazione dei caratteri (molto importante):

Gli span estratti devono corrispondere esattamente, carattere per carattere, al testo originale. Non modificare in alcun modo ortografia, punteggiatura, maiuscole, apostrofi, accenti o spazi.

Gli span problematici includono:

- Bias eventivo: gli eventi sono descritti in modo da minimizzare o distorcere la responsabilità.
- Bias attributivo: persone, gruppi o comunità sono descritti con linguaggio emotivamente carico, stereotipato, etichettante o esagerato.
- Il testo sensazionalizza o drammatizza i fatti.
- Il testo usa affermazioni vaghe o speculative come se fossero fattuali.

FORMATO DI OUTPUT (rigido):

Se c'è uno span: <SPANS>: ["..."]

Se ci sono più span: <SPANS>: ["...", "..."]

Se non ci sono span: <SPANS>: ["No"]

Ora elabora il seguente input:

{instance}

Restituisci la risposta usando un solo blocco <SPANS>. Non restituire più blocchi. Non ripetere span duplicati. Non aggiungere spiegazioni.

4. Task 2b: Rationale Generation

Sei un esperto di framing, bias linguistici e Disordine dell'Informazione. Il disordine dell'informazione è l'inquinamento dello spazio informativo che include disinformazione, disinformazione e malinformazione. È spesso caratterizzato da manipolazione emotiva, falsi contesti o framing distorto. Il tuo compito è spiegare perché determinati span di testo in un estratto di notizie sono fuorvianti, di parte o problematici.

Ti vengono forniti un estratto di notizia e un elenco di span estratti. Il tuo obiettivo è generare esattamente UNA razionale per ogni span. Usa rigorosamente questo formato: "se [riferimento allo span nel testo], allora [implicazione o conseguenza]"

Istruzioni di output (obbligatorie):

- Restituisci esattamente una razionale per ogni span, nello stesso ordine.
- Se <SPANS>: ["No"], allora devi restituire esattamente <RATIONALES>: ["No"]. Non inventare razionali.
- Ogni razionale deve essere racchiusa tra virgolette doppie (" ").
- Tutte le razionali devono essere restituite in un'unica lista:
<RATIONALES>: ["se ..., allora ...", "se ..., allora ..."]
- Non usare virgolette annidate o puntini di sospensione (...) all'interno delle razionali.
- Non combinare più span in una sola razionale.
- Non saltare span.
- Non mostrare processo di pensiero, bozze o auto-correzioni.
- Non scrivere alcuna parola fuori dai tag. Restituisci solo e soltanto la lista finale.

Ora elabora il seguente input:

Estratto di notizia: {instance}

Span: {spans}

Global Override Warning (Italian)

Global Override: Input Data Accountability

Attenzione importante: gli esempi forniti sono estratti direttamente da annotazioni umane e alcuni possono contenere errori di scrittura e di formattazione (come span duplicati, mancata corrispondenza tra numero di span e razionali, non rispettando il formato proposto, oppure segni extra come TITLE: ripetuto, \xa0 o altri artefatti). Questi elementi sono solo rumore umano e NON devono essere imitati nel tuo output. Usa gli esempi solo per apprendere la logica del bias e nella tua risposta segui esattamente il formato standard richiesto.

B.3 Task Prompts: English (M1, both languages)

M1 Culturally Adaptive: Complete Prompt Sequence (English)

Configuration: Dynamic Retrieval (M1)

Language: English (with dynamic target-language retrieved exemplars)

1. System Instruction

You are a strict data formatting AI. You **MUST** return **ONLY** the requested tags. Do not include conversational filler, greetings, or explanations.

2. Task 1: Severity Classification

You are an expert in framing, linguistic bias, and Information Disorder. Information disorder refers to the pollution of the information space, including misinformation, disinformation, and malinformation. It is often characterized by emotional manipulation, false context, or distorted framing. Your task is to classify the severity of problematic language in a news article excerpt.

Classify the severity based on the presence and impact of the following problematic features:

- Event bias: Describing or referring to events in a way that minimizes or distorts responsibility.
- Attribution bias: Describing people, groups, or communities using emotionally charged, stereotypical, labeling, or exaggerated language.
- Sensationalism: Exaggerating or dramatizing facts.
- Speculation: Using vague or speculative statements as if they were factual.

SEVERITY LABELS:

"none" – No problematic features present. The text is factual, balanced, and neutral.

"slightly" – Minor problematic language is present, but the overall message remains mostly factual.

"moderately" – Multiple instances of bias or misleading language produce a tangible distortion that can alter the reader's understanding.

"highly" – Severe and pervasive use of biased, speculative, or emotionally manipulative language. High risk of disinformation.

OUTPUT FORMAT:

Return only one of the four labels following the tag below, exactly come this:

<PREDICTED_LABEL>: none

<PREDICTED_LABEL>: slightly

<PREDICTED_LABEL>: moderately

<PREDICTED_LABEL>: highly

Do not add explanations, comments, or any other text. Return ONLY a valid label.

3. Task 2a: Span Detection

You are an expert in framing, linguistic bias, and Information Disorder. Information disorder refers to the pollution of the information space, including misinformation, disinformation, and malinformation. It is often characterized by emotional manipulation, false context, or distorted framing. Your task is to analyze news excerpts and identify spans of text that are misleading, biased, speculative, or emotionally charged.

Task:

- Identify ONLY unique, NON-overlapping spans.
- If no problematic spans are found, the output must be exactly <SPANS>: ["No"].

Character Preservation Rule (CRITICAL):

The extracted spans must match the original text EXACTLY, character for character.

Do not modify spelling, punctuation, capitalization, apostrophes, accents, or spaces in any way.

Special Language Rule: For Farsi text, strictly preserve all Zero-Width Non-Joiners (ZWNJ, \u200C / نیم‌فاصله). Do not replace them with spaces or delete them.

Problematic spans include:

- Event bias: Events described in a way that minimizes or distorts responsibility.
- Attribution bias: People, groups, or communities described with emotionally charged, stereotypical, labeling, or exaggerated language.
- Text that sensationalizes or dramatizes facts.
- Text that uses vague or speculative statements as if they were factual.

OUTPUT FORMAT (Strict):

If one span: <SPANS>: ["..."]

If multiple spans: <SPANS>: ["...", "..."]

If no spans: <SPANS>: ["No"]

4. Task 2b: Rationale Generation

You are an expert in framing, linguistic bias, and Information Disorder. Information disorder refers to the pollution of the information space, including misinformation, disinformation, and malinformation. It is often characterized by emotional manipulation, false context, or distorted framing. Your task is to explain why specific spans of text in a news excerpt are misleading, biased, or problematic.

You are provided with a news excerpt and a list of extracted spans.

Your goal is to generate exactly ONE rationale for each span.

Strictly use the target-language conditional structure demonstrated below:

{target_syntax_instruction}

Output Instructions (Mandatory):

- Return exactly ONE rationale for each span, in the exact same order.
- If <SPANS>: ["No"], then you must return exactly <RATIONALES>: ["No"]. Do not invent rationales.
- Each rationale must be enclosed in double quotes (" ").
- All rationales must be returned in a single list, following the exact target-language placeholder format: {target_syntax_placeholder}
- Do not use nested quotes or ellipses (...) inside the rationales.
- Do not combine multiple spans into a single rationale.
- Do not skip spans.
- Do not show your thought process, drafts, or self-corrections.
- Do not write any words outside the tags. Return ONLY the final list.

Global Override Warning (M1)

Global Override: Input Data Accountability

IMPORTANT WARNING: The examples provided above are extracted directly from raw human annotations. Some of them contain human errors, typographical mistakes, and formatting inconsistencies (e.g., duplicated spans, mismatched span/rationale counts, or extra characters). These elements are human noise and **MUST NOT** be imitated in your output. Use these examples **ONLY** to learn the logical patterns of bias. In your response, you must strictly follow the standard JSON-style format requested in the instructions.

Dynamic Syntax Injections (M1)

M1 Target-Language Syntax Variables

The M1 Rationale Generation prompt (Task 2b) utilizes dynamic placeholders to inject the correct conditional structure for the target language at runtime. Depending on the pipeline, the variables resolve as follows:

Farsi (fa) Injections:

```
{target_syntax_instruction} →
"اگر [اشاره به بازه در متن], آنگاه [پیامد یا نتیجه]"

{target_syntax_placeholder} →
<RATIONALES>: ["اگر ... آنگاه ...", "اگر ... آنگاه ..."]
```

Italian (it) Injections:

```
{target_syntax_instruction} →
"se [riferimento allo span nel testo], allora [implicazione o conseguenza]"

{target_syntax_placeholder} →
<RATIONALES>: ["se ..., allora ...", "se ..., allora ..."]
```

B.4 Static Examples: B1 Farsi

B1 Static In-Context Examples (Farsi)

Configuration: Static Few-Shot (B1)

Language: Farsi

0 مثال (NONE)

TITLE:

اعلام نتایج لاتاری خودروهای وارداتی

SENTENCE_0:

به گزارش «ایسنا»، آخرین دور از فروش خودروهای وارداتی از طریق سامانه یکپارچه عرضه خودروهای وارداتی، از روز شنبه ۱۸ تا ۲۳ اسفندماه، برای ثبت نام متقاضیانی که طی بازه زمانی ۱۴ تا ۲۱ اسفند ماه، یکی از حساب‌های خود نزد بانک‌های عامل را به نام خودرو وارداتی وکالتی و مبلغ ۵۰۰ میلیون تومان را در این حساب بلوکه کرده بودند، اجرایی شد.

SENTENCE_1:

در این راستا، مدیر طرح واردات خودرو وزارت صمت، گفت: ثبت نام کنندگان می‌توانند از طریق سایت سامانه به نشانی saleauto.ir از وضع تخصیص یا عدم تخصیص در آخرین دور از عرضه خودروهای وارداتی در سال ۱۴۰۳ مطلع شوند.

SENTENCE_2:

مهدی ضیغمی افزود: تاکنون منتخبان ۱۰ هزار خودرو عرضه شده در این دوره، مشخص شده و اسامی آنها روی سایت سامانه قابل مشاهده است؛ البته ممکن است تعداد عرضه تا ۱۷ هزار خودرو نیز افزایش پیدا کند.

<PREDICTED_LABEL>: none

<SPANS>: ["No"]

<RATIONALES>: ["No"]

مثال 1 (SLIGHTLY)

TITLE:

بایگانی‌های جنگنده بدون سرنشین - پایگاه خبری ندای مردم دو استان

SENTENCE_0:

ندای مردم - از مزایای بارز پهپاد قاهر ۳۱۳، حفظ ویژگی‌های پنهانکاری آن در نقش پهپاد است که به این پهپاد اجازه می‌دهد که در برابر رادارهای دشمن پنهان بماند و عملیات‌های تهاجمی را با موفقیت انجام دهد.

SENTENCE_1:

تمامی حقوق برای وب سایت ندای مردم دو استان محفوظ است

SENTENCE_2:

پادداشت در مورد سریال تلویزیونی معاویه؛

<PREDICTED_LABEL>: slightly

<SPANS>: [

'حفظ ویژگی‌های پنهانکاری آن در نقش پهپاد است که به این پهپاد اجازه می‌دهد که در برابر رادارهای دشمن پنهان بماند و عملیات‌های تهاجمی را با موفقیت انجام دهد.'،

'حفظ ویژگی‌های پنهانکاری آن در نقش پهپاد است که به این پهپاد اجازه می‌دهد که در برابر رادارهای دشمن پنهان بماند و عملیات‌های تهاجمی را با موفقیت انجام دهد.']

<RATIONALES>: [

'اگر درباره قابلیت‌های پنهانکاری و عملیاتی یک پهپاد که توانمندی‌های فنی آن مورد بحث جدی کارشناسان مستقل است، ادعاهای قطعی و بدون پشتوانه مستند (مانند "در برابر رادارهای دشمن پنهان بماند" و "عملیات‌های تهاجمی را با موفقیت انجام دهد") مطرح می‌شود، آنگاه این ادعاها ممکن است بزرگنمایی توانایی‌ها و گمراه‌کننده باشند.']

مثال 2 (MODERATELY)

TITLE:

خسارت میلیاردی کرونا به گردشگری کشور - پایگاه خبری ندای مردم دو استان

SENTENCE_0:

وزیر میراث فرهنگی، گردشگری و صنایع‌دستی تصریح کرد: به‌محض اینکه موج دوم کرونا مهار و به ما اجازه فعالیت جدی داده شود، این قدرت را همه بخش‌های گردشگری کشور دارند که به‌سرعت بتوانیم در مرحله نخست گردشگری داخلی را فعال کنیم و در مرحله دوم هم این آمادگی وجود دارد که مکاتباتی با کشورهای همسایه داشته باشیم و با سازمان جهانی گردشگری در ارتباط نزدیک هستیم تا بتوانیم گردشگری خارجی را دوباره فعال کنیم.

SENTENCE_1:

به گزارش پایگاه خبری کارنامه فردا، علی اصغر مونس‌ن در سفر به استان سمنان در گفتگو با خبرنگاران اظهار کرد: ویروس کرونا به بخش‌های مختلف اقتصاد در دنیا خسارت وارد کرده که یکی از آن‌ها بخش گردشگری است و بخش‌هایی هم بوده‌اند که به‌موازات ما آسیب‌های جدی دیده‌اند به‌طور مثال بخش حمل و نقل هم در کشور و هم در دنیا به میزان ما دچار خسارت شده است.

SENTENCE_2:

وزیر میراث فرهنگی، گردشگری و صنایع‌دستی ادامه داد: من نگران این هستم که بخش گردشگری به ورشکستگی نزدیک شود که اتفاق بسیار بدی است؛ به هر صورت یکی از ظرفیت‌های کشور ما در بخش گردشگری است و کارهای خوبی در این چند سال گذشته در بخش گردشگری انجام شده و پروژه‌های بسیار زیادی در حوزه گردشگری به نتیجه رسیده و در حال اجراست.

<PREDICTED_LABEL>: moderately

<SPANS>: [

'خسارت میلیاردي کرونا به گردشگری کشور،'
'این قدرت را همه بخش‌های گردشگری کشور دارند که به سرعت بتوانیم در مرحله نخست گردشگری داخلی را فعال کنیم،'

'به‌طور مثال بخش حمل و نقل هم در کشور و هم در دنیا به میزان ما دچار خسارت شده است،'
'من نگران این هستم که بخش گردشگری به ورشکستگی نزدیک شود،'

'کارهای خوبی در این چند سال گذشته در بخش گردشگری انجام شده و پروژه‌های بسیار زیادی در حوزه

گردشگری به نتیجه رسیده و در حال اجراست.']

<RATIONALES>: [

'اگر در تیتتر خبر، میزان خسارت ناشی از یک پدیده (مانند کرونا برای گردشگری) با یک عدد بزرگ و مشخص (مانند "میلیاردي") بیان می‌شود، اما منبع یا جزئیات دقیق این برآورد مالی در تیتتر یا بخش ابتدایی خبر ارائه نمی‌گردد، آنگاه این رقم ممکن است بدون پشتوانه مشخص بوده و صرفاً برای تأثیرگذاری بر خواننده به کار رفته باشد،'

'اگر درباره توانایی یک بخش اقتصادی (مانند گردشگری) برای بازایی سریع پس از بحران ادعا می‌شود که "این قدرت را همه بخش‌ها دارند" و به شکلی کلی و بدون قید احتیاط بیان می‌شود، آنگاه این توصیف ممکن است بیش از واقعیت، توانمندی‌های کل آن بخش را بزرگنمایی کند،'

'اگر میزان خسارت ناشی از یک پدیده در بخش‌های مختلف اقتصادی (مانند گردشگری و حمل و نقل) با یک ادعای مقایسه‌ای مشخص (مانند "به میزان ما دچار خسارت شده") بیان می‌شود، در حالی که داده‌های دقیق و قابل راستی‌آزمایی برای این مقایسه ارائه نمی‌گردد، آنگاه این ادعا ممکن است بدون پشتوانه دقیق بوده و صرفاً بازتاب‌دهنده برآورد گوینده باشد،'

'اگر درباره وضعیت یک بخش اقتصادی (مانند گردشگری) ادعا می‌شود که "به ورشکستگی نزدیک شود"، آنگاه این توصیف ممکن است یک برآورد یا نگرانی شخصی با زبان هشدارآمیز بوده و وضعیت کلی آن بخش را بیش از واقعیت بحرانی جلوه دهد،'

'اگر درباره عملکرد گذشته یا پروژه‌های یک بخش اقتصادی از عبارات کلی، ذهنی و بدون ارائه جزئیات مشخص (مانند "کارهای خوبی"، "پروژه‌های بسیار زیادی") استفاده می‌شود، آنگاه این توصیفات ممکن است صرفاً برای ستایش یا بزرگنمایی به کار رفته باشند و فاقد اطلاعات عینی درباره آن اقدامات باشند.']

3 (HIGHLY) مثال

TITLE:

شیوع یک بیماری ریوی با علت ناشناخته کشنده‌تر از کرونا در قزاقستان - همشهری آنلاین

SENTENCE_0:

در همین ارتباط، وزارت بهداشت قزاقستان گزارش داد، این بیماری جدید یک نوع التهاب ریوی ناشناخته است و از تاریخ ۲۹ ژوئن تا ۵ جولای، بیش از ۳۲ هزار مبتلا به این ویروس در کشور شناسایی شده و ۴۵۱ نفر جان خود را از دست داده‌اند.

SENTENCE_1:

به گزارش همشهری آنلای به‌نقل از روزنامه اکسپرس، مسئولان چینی اعلام کردند که یک ذات‌الریه یا التهاب ریوی با علت ناشناخته که از بیماری کووید-۱۹ ناشی از ویروس کرونا کشنده‌تر است، در ماه‌های اخیر در سراسر کشور قزاقستان انتشار یافته است.

SENTENCE_2:

سفارت چین از شهروندان خود خواسته است تا هنگامی که عامل احتمالی ویروسی این بیماری شناخته شود، همان اقدامات پیشگیری کرونا را در بابر آن انجام دهند.

<PREDICTED_LABEL>: highly

<SPANS>: [

'سفارت چین از شهروندان خود خواسته است تا هنگامی که عامل احتمالی ویروسی این بیماری شناخته شود، همان اقدامات',

'به گزارش همشهری آنلای به نقل از روزنامه اکسپرس، مسئولان چینی اعلام کردند که یک ذات‌الریه یا التهاب ریوی با علت ناشناخته که از بیماری کووید-۱۹ ناشی از ویروس کرونا کشنده‌تر است، در ماه‌های اخیر در سراسر کشور قزاقستان انتشار یافته است.', \n

0: 'در همین ارتباط، وزارت بهداشت قزاقستان گزارش داد، این بیماری جدید یک نوع التهاب ریوی ناشناخته است و از تاریخ ۲۹ ژوئن تا ۵ جولای، بیش از ۳۲ هزار مبتلا به این ویروس در کشور شناسایی شده و ۴۵۱ نفر جان خود را از دست داده‌اند. \n']

<RATIONALES>: [

'اگر سفارت چین از شهروندان خود بخواهد که اقداماتی مشابه پیشگیری از کرونا را برای بیماری جدید انجام دهند، پس باید روشن شود که علت این درخواست چیست و آیا اطلاعات علمی یا شواهدی در مورد مشابهت این بیماری با کرونا وجود دارد یا خیر.',

'اگر گزارش‌ها نشان دهند که بیماری ذات‌الریه‌ای با علت ناشناخته در قزاقستان شیوع پیدا کرده و گفته شود که این بیماری کشنده‌تر از کووید-۱۹ است، پس باید به منابع معتبر و مستندات دقیق این ادعاها توجه کنیم تا از صحت اطلاعات و بررسی علمی آن‌ها اطمینان حاصل شود.',

'اگر وزارت بهداشت قزاقستان گزارش دهد که بیماری جدیدی به‌عنوان التهاب ریوی ناشناخته شیوع یافته و تعداد زیادی مبتلا و جان‌باخته داشته است، پس باید به دقت در مورد صحت و جزئیات گزارش‌ها بررسی‌های علمی و مستند انجام شود، چرا که این اطلاعات می‌توانند تأثیرات زیادی بر سلامت عمومی و واکنش‌های جهانی داشته باشند. ']

B.5 Static Examples: B1 Italian

WARNING: The in-context examples presented in this appendix are drawn from the InDor corpus [4], which may contain content that is politically sensitive, offensive, or otherwise disturbing, including racist, sexist, or violent language. This material is presented here strictly for research transparency and reproducibility purposes.

B1 Static In-Context Examples (Italian)

Configuration: Static Few-Shot (B1)

Language: Italian

Esempio 0 (NONE)

TITLE:

La Regina Elisabetta governa personalmente l'Impero Britannico? Sì – RENOVATIO 21

SENTENCE_0:

La mossa, scrive EIR, costrinse un'elezione che gettò il partito laburista australiano all'opposizione, concludendo quindi anche la politica di nazionalismo economico che Whitlam stava perseguendo, comprese le intenzioni di nazionalizzare le compagnie minerarie che avevano sfruttato le enormi risorse minerarie dell'Australia, aziende considerate come i «gioielli della corona» della famiglia reale britannica che deteneva importanti partecipazioni in esse—

ad esempio, la notissima Rio Tinto, compagnia di estrazione mineraria che nel 2009 finì accusata di spionaggio in Cina e che dovette chiedere la consulenza di Henry Kissinger per cercare di uscire dall'impasse.

SENTENCE_1:

In una lettera del giorno successivo, Charteris fa un riferimento ancora più diretto all'utilizzo dei poteri di riserva, adducendo in particolare la preoccupazione di Kerr che qualsiasi decisione presa potrebbe influenzare la monarchia: "Se fai, come vuoi, ciò che la Costituzione impone, non puoi assolutamente fare alla Monarchia alcun danno evitabile.

SENTENCE_2:

Grazie alla determinazione della storica australiana Jenny Hocking, professore presso il National Center for Australian Studies, Monash University, è stata rivelata la prova del ruolo personale della regina, attraverso la corrispondenza appena rilasciata tra Kerr e la regina, attraverso il suo segretario personale, Sir Martin Charteris.

<PREDICTED_LABEL>: none

<SPANS>: ["No"]

<RATIONALES>: ["No"]

Esempio 1 (SLIGHTLY)

TITLE:

«L'euro digitale è inevitabile». La BCE spinge verso la nostra schiavitù elettronica – RENOVATIO 21

SENTENCE_0:

L'euro digitale, ha dichiarato Panetta agirebbe «moneta sovrana utilizzabile da chiunque, famiglie, imprese commercianti», una valuta priva dei rischi «di mercato, credito e liquidità» in grado di tutelare la privacy – questo a differenza dai colossi privati Big Tech, i quali potrebbero mettere a rischio la protezione di «aspetti provati della nostra vita o le tendenze politiche o sessuali o lo stato di salute» così come potrebbero interferire «con le regole di funzionamento di una moderna democrazia liberale».

SENTENCE_1:

«Panetta, che è il responsabile del progetto lanciato ufficialmente nel luglio scorso dopo una lunga carriera in Banca d'Italia, ha sottolineato come 80 paesi nel mondo si stiano muovendo per la creazione di valute digitali mentre è sempre più forte la crescita delle criptovalute, degli stablecoin e delle transazioni di pagamento da parte dei grandi colossi tech» scrive l'ANSA, che nota come il messaggio vada «un po' più al di là della tradizionale prudenza di Francoforte».

SENTENCE_2:

Nell'ultima analisi della volatilità nei mercati delle materie prime, il responsabile della strategia sulle materie prime di TD Securities, Bart Melek, ha affermato che negli ultimi anni le banche centrali hanno registrato un aumento record degli acquisti di oro a causa delle preoccupazioni sulla sostenibilità del potere d'acquisto del dollaro e delle tensioni geopolitiche tra le principali potenze economiche.

<PREDICTED_LABEL>: slightly
<SPANS>: [‘la nostra schiavitù elettronica’]
<RATIONALES>: [‘Se il titolo è posto in questi termini, allora potrebbe essere sensazionalistico e manipolare emotivamente i lettori’]

Esempio 2 (MODERATELY)

TITLE:

Aiutare l’Africa nel solo modo possibile: andandocene via | LIBRE

SENTENCE_0:

Voglio semplicemente dire che , dal momento che i rapaci sono sempre esistiti , il fenomeno di colonizzazione mascherato da aiuto umanitario ci sarà sempre , ma un organizzazione per quanto ben congegnata e efficiente ha sempre bisogno di una base “volontaria” indottrinata , di persone che fanno dell’aiuto agli altri una ragione di vita , e in buona fede perchè non ci guadagnano nulla , sono anche “apprezzabili” da un certo punto di vista ... ma prima avrebbero bisogno di terapie psicologiche per verificare i problemi di fondo che li portano a queste scelte e poi , chissà , potrebbero arrivare a capire che se non viene richiesto nessun aiuto non c’è nessun motivo di inventarsi problemi da riversare su chi si vorrebbe aiutare.

SENTENCE_1:

Nei quattro anni del suo governo fece parecchie cose positive per il Burkina: si impegnò molto per eliminare la povertà attraverso il taglio degli sprechi statali e la soppressione dei privilegi delle classi agiate, finanziò un ampio sistema di riforme sociali incentrato sulla costruzione di scuole, ospedali e case per la popolazione estremamente povera, fece un’importante lotta alla desertificazione con il piantamento di milioni di alberi nel Sahel, cercò di svincolare il paese dalle importazioni forzate.

SENTENCE_2:

Posco immaginare perfino sul piano linguistico: poiché col “reddito di cittadinanza” alle generazioni nuove del Sud non occorrerà più studiare né imparare a leggere far di conto, parleranno in modo sempre più esclusivo i vernacoli sempre più gutturali e belluini; nella decrescita felice andranno alla raccolta dei mitili e bacche e impareranno ad abitare sulle palafitte (abusive, naturalmente), come i cacciatori-raccoglitori della civiltà villanoviana.

<PREDICTED_LABEL>: moderately

<SPANS>: [‘i vernacoli sempre più gutturali e belluini’, ‘nella decrescita felice andranno alla raccolta dei mitili e bacche e impareranno ad abitare sulle palafitte (abusive, naturalmente), come i cacciatori-raccoglitori della civiltà villanoviana’]

<RATIONALES>: [‘attribuire alle lingue locali gli aggettivi “gutturali” e “belluini” è offensivo’, ‘si tratta di un assunto per cui i popoli africani siano da considerare selvaggi e arretrati’]

Esempio 3 (HIGHLY)

TITLE:

Vaccino e aborto, Pfizer ha cercato di nascondere l’uso di cellule fetali: email trapelate –
RENOVATIO 21

SENTENCE_0:

L'idea che Renovatio 21 portava avanti, in reality ancora prima dei tempi della legge Lorenzin (2017) che impediva l'accesso alle scuole ai nostri figli, era semplice: sollevando il tema dei feti sacrificati dei vaccini, ci sarà possibile utilizzare l'obiezione di coscienza – grande vacca sacra apparentemente intoccabile così come la sua legge genocidiaria 194/78, perché teneva in piedi la pax bioetica assassina del compromesso tra democristiani, comunisti e radicali (e, oggi, i loro figli postcomunisti, postfascisti, «liberali» – in compagnia ovviamente dei vescovi) – e così sottrarre milioni di bambini all'abominio della siringa di Stato obbligatoria, contenente – come nelle antiche pozioni delle streghe descritte dal Malleus maleficarum (1486)– pezzi di bimbo sacrificato.

SENTENCE_1:

La densa conversazione, della durata di quasi due ore, è stata pubblicata su X il 27 dicembre e ha attirato grande attenzione sui social media, poiché il Siri, che sta aiutando Kennedy a esaminare i potenziali dipendenti del dipartimento della Salute e dei Servizi Umani, è stato oggetto di un feroce articolo sul New York Times il mese scorso che lo accusava di aver intentato una causa mettere al bando il vaccino contro la poliomielite.

SENTENCE_2:

Non fu così: non solo la gerarchia cattolica della Roma conciliare corrotta, simoniaca e demoniaca aveva pronte giustificazioni e scuse fatte di pura menzogna: no, di lì a poco sarebbe scattata la più grande emergenza sperimentata dall'umanità del XXI secolo, la pandemia, che, guarda guarda, nella distruzione totale dei diritti costituzionali dei cittadini ad un certo punto richiese proprio il vaccino obbligatorio per l'intera popolazione.

<PREDICTED_LABEL>: highly

<SPANS>: ["Vaccino e aborto, Pfizer ha cercato di nascondere l'uso di cellule fetali", 'feti sacrificati dei vaccini', 'feti sacrificati dei vaccini']

<RATIONALES>: ["se utilizzi un'espressione molto provocatoria, allora vuoi creare una visione distorta del tema", 'se definisci la vaccinazione come abominio che è un termine non neutro, allora vuoi polarizzare il pubblico ', "se utilizzi un'espressione molto provocatoria, allora vuoi creare una visione distorta del tema", 'se parli del tentativo intenzionale di Pfizer senza fornire prove concrete, allora alimenti teorie del complimento e paure infondate', "se utilizzi un'espressione molto forte ed emotiva, allora vuoi polarizzare il discorso pubblico e crea una visione distorta del tema"]

Bibliography

- [1] Hunt Allcott and Matthew Gentzkow. “Social Media and Fake News in the 2016 Election.” In: *Journal of Economic Perspectives* 31.2 (May 2017), pp. 211–36. doi: [10.1257/jep.31.2.211](https://doi.org/10.1257/jep.31.2.211). URL: <https://www.aeaweb.org/articles?id=10.1257/jep.31.2.211>.
- [2] Sander van der Linden, Jon Roozenbeek, and Josh Compton. “Inoculating Against Fake News About COVID-19.” In: *Frontiers in Psychology* 11 (2020), p. 566790. ISSN: 1664-1078. doi: [10.3389/fpsyg.2020.566790](https://doi.org/10.3389/fpsyg.2020.566790). URL: <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2020.566790>.
- [3] Claire Wardle and Hossein Derakhshan. *Information Disorder: Toward an Interdisciplinary Framework for Research and Policy Making*. Report 162317GBR. Council of Europe, 2017, p. 109. URL: <https://edoc.coe.int/en/media/7495-information-disorder-toward-an-interdisciplinary-framework-for-research-and-policy-making.html>.
- [4] Sara Gemelli, Giulia Di Cristina, Yiran Zhang, Md Azizul Hoque, Alberto De La Torre Solís, Mohamad Mojtaba Behboudi Eshkiki, Nikolai Efimov, Mariia Everstova, Caterina Maria Cappello, Maziar Kianimoghadam Jouneghani, Payam Latifi, Yashar Mahboudi, Farzaneh Mohseni, Dario Placenti, Tommaso Caselli, Manuela Sanguinetti, Aurora Scarpellini, Chiara Zanchi, Usman Naseem, Marco Antonio Stranisci, and Simona Frenda. “Beyond Fake News Detection: A Community-based Study of the Multicultural Nature of Information Disorder.” In: *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*. Ed. by Stelios Piperidis, Núria Bel, Henk van den Heuvel, Nancy Ide, Simon Krek, and Antonio Toral. Palma, Mallorca, Spain: European Language Resources Association (ELRA), May 2026, pp. 6496–6508. doi: [10.63317/4iyhqziwo6ri](https://doi.org/10.63317/4iyhqziwo6ri).
- [5] Amir Hossein Yari and Fajri Koto. “Unveiling Cultural Blind Spots: Analyzing the Limitations of mLLMs in Procedural Text Comprehension.” In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 20151–20170. ISBN: 979-8-89176-251-0. doi: [10.18653/v1/2025.acl-long.987](https://doi.org/10.18653/v1/2025.acl-long.987). URL: <https://aclanthology.org/2025.acl-long.987/>.
- [6] Andrea Seveso, Daniele Poterì, Edoardo Federici, Mario Mezzanzanica, and Fabio Mercurio. “ITALIC: An Italian Culture-Aware Natural Language Benchmark.” In: *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Ed. by Luis

- Chiruzzo, Alan Ritter, and Lu Wang. Albuquerque, New Mexico: Association for Computational Linguistics, Apr. 2025, pp. 1469–1478. ISBN: 979-8-89176-189-6. DOI: [10.18653/v1/2025.naacl-long.68](https://doi.org/10.18653/v1/2025.naacl-long.68). URL: <https://aclanthology.org/2025.naacl-long.68/>.
- [7] Jasmin Heierli, Rebecca Bahar Ganjineh, and Elena Gavagnin. “Evaluating LLM Capabilities in Low-Resource Contexts: A Case Study of Persian Linguistic and Cultural Tasks.” In: *Proceedings of the First Workshop on Advancing NLP for Low-Resource Languages*. Ed. by Ernesto Luis Estevanell-Valladares, Alicia Picazo-Izquierdo, Tharindu Ranasinghe, Besik Mikaberidze, Simon Ostermann, Daniil Gurgurov, Philipp Mueller, Claudia Borg, and Mar- ián Šimko. Varna, Bulgaria: INCOMA Ltd., Shoumen, Bulgaria, Sept. 2025, pp. 111–120. URL: <https://aclanthology.org/2025.lowresnlp-1.12/>.
- [8] Elizaveta Gromenko, Daria Kalacheva, Ksenia Klokova, Maxim Krongauz, Oksana Moroz, Valery Shulginov, and Tatiana Yudina. “Cultural Evaluation of LLMs in Russian: Catch- phrases and Cultural Types.” In: *Computational Linguistics and Intellectual Technologies: Pro- ceedings of the Annual Conference Dialogue 2025*. 2025. URL: [https://dialogue-conf.org/wp- content/uploads/2025/04/GromenkoEetal.029.pdf](https://dialogue-conf.org/wp-content/uploads/2025/04/GromenkoEetal.029.pdf).
- [9] Kai Shu, Amy Sliva, Suhang Wang, Jiliang Tang, and Huan Liu. “Fake News Detection on Social Media: A Data Mining Perspective.” In: *SIGKDD Explor. Newsl.* 19.1 (Sept. 2017), pp. 22–36. ISSN: 1931-0145. DOI: [10.1145/3137597.3137600](https://doi.org/10.1145/3137597.3137600). URL: <https://doi.org/10.1145/3137597.3137600>.
- [10] Xinyi Zhou and Reza Zafarani. “A Survey of Fake News: Fundamental Theories, Detection Methods, and Opportunities.” In: *ACM Comput. Surv.* 53.5 (Sept. 2020). ISSN: 0360-0300. DOI: [10.1145/3395046](https://doi.org/10.1145/3395046). URL: <https://doi.org/10.1145/3395046>.
- [11] Sai Gurrapu, Ajay Kulkarni, Lifu Huang, Ismini Lourentzou, and Feras A. Batareseh. “Ratio- nalization for Explainable NLP: A Survey.” In: *Frontiers in Artificial Intelligence* 6 (2023). ISSN: 2624-8212. DOI: [10.3389/frai.2023.1225093](https://doi.org/10.3389/frai.2023.1225093). URL: [https://www.frontiersin.org/ journals/artificial-intelligence/articles/10.3389/frai.2023.1225093](https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2023.1225093).
- [12] Haiyan Zhao, Hanjie Chen, Fan Yang, Ninghao Liu, Huiqi Deng, Hengyi Cai, Shuaiqiang Wang, Dawei Yin, and Mengnan Du. “Explainability for Large Language Models: A Sur- vey.” In: *ACM Trans. Intell. Syst. Technol.* 15.2 (Feb. 2024). ISSN: 2157-6904. DOI: [10.1145/3639372](https://doi.org/10.1145/3639372). URL: <https://doi.org/10.1145/3639372>.
- [13] Cheng Li, Mengzhuo Chen, Jindong Wang, Sunayana Sitaram, and Xing Xie. “CultureLLM: Incorporating Cultural Differences into Large Language Models.” In: *Advances in Neural Information Processing Systems*. Ed. by A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang. Vol. 37. Curran Associates, Inc., 2024, pp. 84799–84838. DOI: [10.52202/079017-2693](https://doi.org/10.52202/079017-2693). URL: [https://proceedings.neurips.cc/paper_files/paper/2024/ file/9a16935bf54c4af233e25d998b7f4a2c-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2024/file/9a16935bf54c4af233e25d998b7f4a2c-Paper-Conference.pdf).
- [14] Tarek Naous, Michael J Ryan, Alan Ritter, and Wei Xu. “Having Beer after Prayer? Measur- ing Cultural Bias in Large Language Models.” In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Lun-Wei Ku, Andre Martins, and Vivek Srikumar. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 16366–16393. DOI: [10.18653/v1/2024.acl-long.862](https://doi.org/10.18653/v1/2024.acl-long.862). URL: <https://aclanthology.org/2024.acl-long.862/>.

-
- [15] Yan Tao, Olga Viberg, Ryan S Baker, and René F Kizilcec. “Cultural bias and cultural alignment of large language models.” In: *PNAS Nexus* 3.9 (Sept. 2024), pgae346. ISSN: 2752-6542. DOI: [10.1093/pnasnexus/pgae346](https://doi.org/10.1093/pnasnexus/pgae346). eprint: <https://academic.oup.com/pnasnexus/article-pdf/3/9/pgae346/59151559/pgae346.pdf>. URL: <https://doi.org/10.1093/pnasnexus/pgae346>.
 - [16] Johannes Schäfer, Aidan Combs, Christopher Bagdon, Jiahui Li, Nadine Probol, Lynn Greschner, Sean Papay, Yarik Menchaca Resendiz, Aswathy Velutharambath, Amelie Wuehrl, Sabine Weber, and Roman Klinger. “Which Demographics do LLMs Default to During Annotation?” In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 17331–17348. ISBN: 979-8-89176-251-0. DOI: [10.18653/v1/2025.acl-long.848](https://doi.org/10.18653/v1/2025.acl-long.848). URL: <https://aclanthology.org/2025.acl-long.848/>.
 - [17] Shayan Alipour, Indira Sen, Mattia Samory, and Tanu Mitra. “Robustness and Confounders in the Demographic Alignment of LLMs with Human Perceptions of Offensiveness.” In: *Findings of the Association for Computational Linguistics: ACL 2025*. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 22025–22047. ISBN: 979-8-89176-256-5. DOI: [10.18653/v1/2025.findings-acl.1136](https://doi.org/10.18653/v1/2025.findings-acl.1136). URL: <https://aclanthology.org/2025.findings-acl.1136/>.
 - [18] Robert (Munro) Monarch. *Human-in-the-Loop Machine Learning: Active Learning and Annotation for Human-Centered AI*. Manning Publications, July 2021. ISBN: 9781617296741. URL: <https://www.manning.com/books/human-in-the-loop-machine-learning>.
 - [19] Maziar Kianimoghadam Jouneghani. *Culturally Adaptive Explainable LLM Assessment for Multilingual Information Disorder: A Human-in-the-Loop Approach*. Presented at the In-Dor26 Workshop, LREC-COLING 2026 (non-archival). 2026. arXiv: [2603.27356](https://arxiv.org/abs/2603.27356) [cs.CL]. URL: <https://arxiv.org/abs/2603.27356>.
 - [20] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. “Language Models are Few-Shot Learners.” In: *Advances in Neural Information Processing Systems*. Ed. by H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin. Vol. 33. Curran Associates, Inc., 2020, pp. 1877–1901. URL: https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfc4967418bfb8ac142f64a-Paper.pdf.
 - [21] Firoj Alam, Julia Maria Struß, Tanmoy Chakraborty, Stefan Dietze, Salim Hafid, Katerina Korre, Arianna Muti, Preslav Nakov, Federico Ruggeri, Sebastian Schellhammer, Vinay Setty, Megha Sundriyal, Konstantin Todorov, and Venkatesh V. “The CLEF-2025 Check-That! Lab: Subjectivity, Fact-Checking, Claim Normalization, and Retrieval.” In: *Advances in Information Retrieval: Proceedings of the 47th European Conference on Information Retrieval (ECIR 2025)*. Cham: Springer Nature Switzerland, 2025, pp. 467–478. ISBN: 978-3-031-88720-8. DOI: [10.1007/978-3-031-88720-8_68](https://doi.org/10.1007/978-3-031-88720-8_68).

- [22] Francisco Rangel, Anastasia Giachanou, Bilal Ghanem, and Paolo Rosso. “Overview of the 8th Author Profiling Task at PAN 2020: Profiling Fake News Spreaders on Twitter.” In: *Working Notes of CLEF 2020 – Conference and Labs of the Evaluation Forum*. Ed. by Linda Cappellato, Carsten Eickhoff, Nicola Ferro, and Aurélie Névél. Vol. 2696. CEUR Workshop Proceedings. Thessaloniki, Greece: CEUR-WS.org, 2020, pp. 1–18. URL: <https://ceur-ws.org/Vol-2696/>.
- [23] Jakub Piskorski, Nicolas Stefanovitch, Giovanni Da San Martino, and Preslav Nakov. “SemEval-2023 Task 3: Detecting the Category, the Framing, and the Persuasion Techniques in Online News in a Multi-lingual Setup.” In: *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*. Ed. by Atul Kr. Ojha, A. Seza Doğruöz, Giovanni Da San Martino, Harish Tayyar Madabushi, Ritesh Kumar, and Elisa Sartori. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 2343–2361. DOI: [10.18653/v1/2023.semeval-1.317](https://doi.org/10.18653/v1/2023.semeval-1.317). URL: <https://aclanthology.org/2023.semeval-1.317/>.
- [24] Jawaher Alghamdi, Suhui Luo, and Yuqing Lin. “A Comprehensive Survey on Machine Learning Approaches for Fake News Detection.” In: *Multimedia Tools and Applications* 83.17 (May 2024), pp. 51009–51067. ISSN: 1573-7721. DOI: [10.1007/s11042-023-17470-8](https://doi.org/10.1007/s11042-023-17470-8). URL: <https://doi.org/10.1007/s11042-023-17470-8>.
- [25] Fiza Gulzar Hussain, Muhammad Wasim, Seemab Hameed, Abdur Rehman, Muhammad Nabeel Asim, and Andreas Dengel. “Fake News Detection Landscape: Datasets, Data Modalities, AI Approaches, Their Challenges, and Future Perspectives.” In: *IEEE Access* 13 (2025), pp. 54757–54778. DOI: [10.1109/ACCESS.2025.3553909](https://doi.org/10.1109/ACCESS.2025.3553909). URL: <https://ieeexplore.ieee.org/document/10937488>.
- [26] Soveatin Kuntur, Anna Wróblewska, Maria Ganzha, Marcin Paprzycki, and Shelly Sachdeva. “Fake News Detection: It’s All in the Data!” In: *Applied Sciences* 16.3 (2026). ISSN: 2076-3417. DOI: [10.3390/app16031585](https://doi.org/10.3390/app16031585). URL: <https://www.mdpi.com/2076-3417/16/3/1585>.
- [27] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.” In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Ed. by Jill Burstein, Christy Doran, and Thamar Solorio. Minneapolis, Minnesota: Association for Computational Linguistics, June 2019, pp. 4171–4186. DOI: [10.18653/v1/N19-1423](https://doi.org/10.18653/v1/N19-1423). URL: <https://aclanthology.org/N19-1423/>.
- [28] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. *RoBERTa: A Robustly Optimized BERT Pretraining Approach*. 2019. arXiv: [1907.11692](https://arxiv.org/abs/1907.11692) [cs.CL]. URL: <https://arxiv.org/abs/1907.11692>.
- [29] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F Christiano, Jan Leike, and Ryan Lowe. “Training language models to follow instructions with human feedback.” In: *Advances in Neural Information Processing Systems*. Ed. by S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh. Vol. 35. Curran Associates, Inc., 2022, pp. 27730–27744. URL: https://proceedings.neurips.cc/paper_files/paper/2022/file/b1efde53be364a73914f58805a001731-Paper-Conference.pdf.

- [30] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurélien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. “LLaMA: Open and Efficient Foundation Language Models.” In: *CoRR* abs/2302.13971 (2023). doi: [10.48550/ARXIV.2302.13971](https://doi.org/10.48550/ARXIV.2302.13971). arXiv: [2302.13971](https://arxiv.org/abs/2302.13971). URL: <https://doi.org/10.48550/arXiv.2302.13971>.
- [31] Eleftheria Papageorgiou, Christos Chronis, Iraklis Varlamis, and Yassine Himeur. “A Survey on the Use of Large Language Models (LLMs) in Fake News.” In: *Future Internet* 16.8 (2024). ISSN: 1999-5903. doi: [10.3390/fi16080298](https://doi.org/10.3390/fi16080298). URL: <https://www.mdpi.com/1999-5903/16/8/298>.
- [32] Soveatin Kuntur, Anna Wróblewska, Marcin Paprzycki, and Maria Ganzha. “Under the Influence: A Survey of Large Language Models in Fake News Detection.” In: *IEEE Transactions on Artificial Intelligence* 6.2 (2025), pp. 458–476. doi: [10.1109/TAI.2024.3471735](https://doi.org/10.1109/TAI.2024.3471735).
- [33] BigScience Workshop. *BLOOM: A 176B-Parameter Open-Access Multilingual Language Model*. 2022. arXiv: [2211.05100](https://arxiv.org/abs/2211.05100) [cs.CL]. URL: <https://arxiv.org/abs/2211.05100>.
- [34] Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. “mT5: A Massively Multilingual Pre-trained Text-to-Text Transformer.” In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Ed. by Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou. Online: Association for Computational Linguistics, June 2021, pp. 483–498. doi: [10.18653/v1/2021.naacl-main.41](https://doi.org/10.18653/v1/2021.naacl-main.41). URL: <https://aclanthology.org/2021.naacl-main.41/>.
- [35] Ahmet Üstün, Viraat Aryabumi, Zheng Yong, Wei-Yin Ko, Daniel D’souza, Gbemileke Onilude, Neel Bhandari, Shivalika Singh, Hui-Lee Ooi, Amr Kayid, Freddie Vargus, Phil Blunsom, Shayne Longpre, Niklas Muennighoff, Marzieh Fadaee, Julia Kreutzer, and Sara Hooker. “Aya Model: An Instruction Finetuned Open-Access Multilingual Language Model.” In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Lun-Wei Ku, Andre Martins, and Vivek Srikumar. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 15894–15939. doi: [10.18653/v1/2024.acl-long.845](https://doi.org/10.18653/v1/2024.acl-long.845). URL: <https://aclanthology.org/2024.acl-long.845/>.
- [36] Libo Qin, Qiguang Chen, Yuhang Zhou, Zhi Chen, Yinghui Li, Lizi Liao, Min Li, Wanxiang Che, and Philip S. Yu. “A survey of multilingual large language models.” In: *Patterns* 6.1 (2025), p. 101118. ISSN: 2666-3899. doi: <https://doi.org/10.1016/j.patter.2024.101118>. URL: <https://www.sciencedirect.com/science/article/pii/S2666389924002903>.
- [37] William Yang Wang. ““Liar, Liar Pants on Fire”: A New Benchmark Dataset for Fake News Detection.” In: *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Ed. by Regina Barzilay and Min-Yen Kan. Vancouver, Canada: Association for Computational Linguistics, July 2017, pp. 422–426. doi: [10.18653/v1/P17-2067](https://doi.org/10.18653/v1/P17-2067). URL: <https://aclanthology.org/P17-2067/>.
- [38] Kai Shu, Deepak Mahudeswaran, Suhang Wang, Dongwon Lee, and Huan Liu. “FakeNews-Net: A Data Repository with News Content, Social Context, and Spatiotemporal Information for Studying Fake News on Social Media.” In: *Big Data* 8.3 (June 2020), pp. 171–188. doi: [10.1089/big.2020.0062](https://doi.org/10.1089/big.2020.0062). URL: <https://www.liebertpub.com/doi/10.1089/big.2020.0062>.

- [39] Tariq Alhindi, Savvas Petridis, and Smaranda Muresan. “Where is Your Evidence: Improving Fact-checking by Justification Modeling.” In: *Proceedings of the First Workshop on Fact Extraction and VERification (FEVER)*. Ed. by James Thorne, Andreas Vlachos, Oana Cocarascu, Christos Christodoulopoulos, and Arpit Mittal. Brussels, Belgium: Association for Computational Linguistics, Nov. 2018, pp. 85–90. doi: [10.18653/v1/W18-5513](https://doi.org/10.18653/v1/W18-5513). URL: <https://aclanthology.org/W18-5513/>.
- [40] Isabelle Augenstein, Christina Lioma, Dongsheng Wang, Lucas Chaves Lima, Casper Hansen, Christian Hansen, and Jakob Grue Simonsen. “MultiFC: A Real-World Multi-Domain Dataset for Evidence-Based Fact Checking of Claims.” In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Ed. by Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan. Hong Kong, China: Association for Computational Linguistics, Nov. 2019, pp. 4685–4697. doi: [10.18653/v1/D19-1475](https://doi.org/10.18653/v1/D19-1475). URL: <https://aclanthology.org/D19-1475/>.
- [41] Majid Zarharan, Mahsa Ghaderan, Amin Pourtabiri, Zahra Sayedi, Behrouz Minaei-Bidgoli, Sauleh Eetemadi, and Mohammad Taher Pilehvar. “ParsFEVER: a Dataset for Farsi Fact Extraction and Verification.” In: *Proceedings of *SEM 2021: The Tenth Joint Conference on Lexical and Computational Semantics*. Ed. by Lun-Wei Ku, Vivi Nastase, and Ivan Vulić. Online: Association for Computational Linguistics, Aug. 2021, pp. 99–104. doi: [10.18653/v1/2021.starsem-1.9](https://doi.org/10.18653/v1/2021.starsem-1.9). URL: <https://aclanthology.org/2021.starsem-1.9/>.
- [42] Zhiwei Yang, Jing Ma, Hechang Chen, Hongzhan Lin, Ziyang Luo, and Yi Chang. “A Coarse-to-fine Cascaded Evidence-Distillation Neural Network for Explainable Fake News Detection.” In: *Proceedings of the 29th International Conference on Computational Linguistics*. Ed. by Nicoletta Calzolari, Chu-Ren Huang, Hansaem Kim, James Pustejovsky, Leo Wanner, Key-Sun Choi, Pum-Mo Ryu, Hsin-Hsi Chen, Lucia Donatelli, Heng Ji, Sadao Kurohashi, Patrizia Paggio, Nianwen Xue, Seokhwan Kim, Younggyun Hahm, Zhong He, Tony Kyungil Lee, Enrico Santus, Francis Bond, and Seung-Hoon Na. Gyeongju, Republic of Korea: International Committee on Computational Linguistics, Oct. 2022, pp. 2608–2621. URL: <https://aclanthology.org/2022.coling-1.230/>.
- [43] Enyan Dai, Yiwei Sun, and Suhan Wang. “Ginger Cannot Cure Cancer: Battling Fake Health News with a Comprehensive Data Repository.” In: *Proceedings of the 14th International AAAI Conference on Web and Social Media (ICWSM 2020)*. Vol. 14. 1. AAAI Press, 2020, pp. 853–862. doi: [10.1609/icwsml.v14i1.7350](https://doi.org/10.1609/icwsml.v14i1.7350). URL: <https://ojs.aaai.org/index.php/ICWSM/article/view/7350>.
- [44] Tanushree Mitra and Eric Gilbert. “CREDBANK: A Large-Scale Social Media Corpus With Associated Credibility Annotations.” In: *Proceedings of the 9th International AAAI Conference on Web and Social Media (ICWSM 2015)*. Vol. 9. 1. AAAI Press, 2015, pp. 258–267. doi: [10.1609/icwsml.v9i1.14625](https://doi.org/10.1609/icwsml.v9i1.14625). URL: <https://ojs.aaai.org/index.php/ICWSM/article/view/14625>.
- [45] Antonio Scaiella, Stefano Costanzo, Elisa Passone, Danilo Croce, and Giorgio Gambosi. “Leveraging Large Language Models for Fact Verification in Italian.” In: *Proceedings of the Tenth Italian Conference on Computational Linguistics (CLiC-it 2024)*. Ed. by Felice Dell’Orletta, Alessandro Lenci, Simonetta Montemagni, and Rachele Sprugnoli. Pisa, Italy:

- CEUR Workshop Proceedings, Dec. 2024, pp. 898–908. ISBN: 979-12-210-7060-6. URL: <https://aclanthology.org/2024.clicit-1.97/>.
- [46] Jacopo Gili, Lucia Passaro, and Tommaso Caselli. “Check-IT!: A Corpus of Expert Fact-checked Claims for Italian.” In: *Proceedings of the Ninth Italian Conference on Computational Linguistics (CLiC-it 2023)*. Ed. by Federico Boschetti, Gianluca E. Lebani, Bernardo Magnini, and Nicole Novielli. Venice, Italy: CEUR Workshop Proceedings, Nov. 2023, pp. 227–235. ISBN: 979-12-550-0084-6. URL: <https://aclanthology.org/2023.clicit-1.29/>.
- [47] Elizaveta Kibisova and Silje Susanne Alvestad. “Fakespeak-RUS: Building a Corpus of Russian Fake News.” In: *Linguistics Vanguard* (2025). DOI: [10.1515/lingvan-2024-0198](https://doi.org/10.1515/lingvan-2024-0198). URL: <https://doi.org/10.1515/lingvan-2024-0198>.
- [48] Bernardo Magnini, Roberto Zanolli, Michele Resta, Martin Cimmino, Paolo Albano, Marco Madeddu, and Viviana Patti. “Evalita-LLM: Benchmarking Large Language Models on Italian.” In: *Proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)*. CEUR-WS.org, 2025. URL: https://clic2025.unica.it/wp-content/uploads/2025/09/60_main_long.pdf.
- [49] Xingjiao Wu, Luwei Xiao, Yixuan Sun, Junhang Zhang, Tianlong Ma, and Liang He. “A survey of human-in-the-loop for machine learning.” In: *Future Generation Computer Systems* 135 (2022), pp. 364–381. ISSN: 0167-739X. DOI: <https://doi.org/10.1016/j.future.2022.05.014>. URL: <https://www.sciencedirect.com/science/article/pii/S0167739X22001790>.
- [50] Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Jingyuan Ma, Rui Li, Heming Xia, Jingjing Xu, Zhiyong Wu, Baobao Chang, Xu Sun, Lei Li, and Zhifang Sui. “A Survey on In-context Learning.” In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Ed. by Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 1107–1128. DOI: [10.18653/v1/2024.emnlp-main.64](https://aclanthology.org/2024.emnlp-main.64/). URL: <https://aclanthology.org/2024.emnlp-main.64/>.
- [51] Man Luo, Xin Xu, Yue Liu, Panupong Pasupat, and Mehran Kazemi. *In-context Learning with Retrieved Demonstrations for Language Models: A Survey*. 2024. arXiv: [2401.11624](https://arxiv.org/abs/2401.11624) [cs.CL]. URL: <https://arxiv.org/abs/2401.11624>.
- [52] Wonduk Seo, Zonghao Yuan, and Yi Bu. “ValuesRAG: Enhancing Cultural Alignment Through Retrieval-Augmented Contextual Learning.” In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. Vol. 8. 3. Oct. 2025, pp. 2307–2318. DOI: [10.1609/aies.v8i3.36717](https://ojs.aaai.org/index.php/AIES/article/view/36717). URL: <https://ojs.aaai.org/index.php/AIES/article/view/36717>.
- [53] Zhiwei Liu, Kailai Yang, Qianqian Xie, Christine de Kock, Sophia Ananiadou, and Eduard Hovy. “RAEmoLLM: Retrieval Augmented LLMs for Cross-Domain Misinformation Detection Using In-Context Learning Based on Emotional Information.” In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 16508–16523. ISBN: 979-8-89176-251-0. DOI: [10.18653/v1/2025.acl-long.806](https://aclanthology.org/2025.acl-long.806/). URL: <https://aclanthology.org/2025.acl-long.806/>.
- [54] Sheetal Harris, Vinh Thong Ta, Marcello Trovati, Ghada Nakhla, Faiza Latif, and Ioannis Korkontzelos. “Multimodal Misinformation Detection Across Diverse Languages Using RAG and LLMs.” In: *Journal of Intelligent Information Systems* (Apr. 2026). ISSN: 1573-7675. DOI: [10.1007/s10844-026-01042-x](https://doi.org/10.1007/s10844-026-01042-x). URL: <https://doi.org/10.1007/s10844-026-01042-x>.

- [55] Zhiwei Liu, Runteng Guo, Baojie Qu, Yuechen Jiang, Min Peng, Qianqian Xie, and Sophia Ananiadou. *RAAR: Retrieval Augmented Agentic Reasoning for Cross-Domain Misinformation Detection*. 2026. arXiv: [2601.04853](https://arxiv.org/abs/2601.04853) [cs.CL]. URL: <https://arxiv.org/abs/2601.04853>.
- [56] Hongzhan Lin, Wei Gao, Jing Ma, Yang Deng, Ziyang Luo, Bo Wang, Ruichao Yang, and Tat-Seng Chua. “ExplainHM++: Explainable Harmful Meme Detection With Retrieval-Augmented Debate Between Large Multimodal Models.” In: *IEEE Transactions on Knowledge and Data Engineering* 38.2 (2026), pp. 1040–1053. doi: [10.1109/TKDE.2025.3637552](https://doi.org/10.1109/TKDE.2025.3637552).
- [57] Federico Cabitza, Andrea Campagner, and Valerio Basile. “Toward a Perspectivist Turn in Ground Truthing for Predictive Computing.” In: *Proceedings of the 37th AAAI Conference on Artificial Intelligence*. Vol. 37. 6. AAAI Press, June 2023, pp. 6860–6868. doi: [10.1609/aaai.v37i6.25840](https://doi.org/10.1609/aaai.v37i6.25840). URL: <https://ojs.aaai.org/index.php/AAAI/article/view/25840>.
- [58] Saadia Gabriel, Skyler Hallinan, Maarten Sap, Pemi Nguyen, Franziska Roesner, Eunsol Choi, and Yejin Choi. “Misinfo Reaction Frames: Reasoning about Readers’ Reactions to News Headlines.” In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Smaranda Muresan, Preslav Nakov, and Aline Villavicencio. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 3108–3127. doi: [10.18653/v1/2022.acl-long.222](https://doi.org/10.18653/v1/2022.acl-long.222). URL: <https://aclanthology.org/2022.acl-long.222/>.
- [59] Anders Søgaard, Barbara Plank, and Dirk Hovy. “Selection Bias, Label Bias, and Bias in Ground Truth.” In: *Proceedings of COLING 2014, the 25th International Conference on Computational Linguistics: Tutorial Abstracts*. Ed. by Qun Liu and Fei Xia. Dublin, Ireland: Dublin City University and Association for Computational Linguistics, Aug. 2014, pp. 11–13. URL: <https://aclanthology.org/C14-3005/>.
- [60] Nyalleng Moorosi, Raesetje Sefala, and Sasha Luccioni. “AI for Whom? Shedding Critical Light on AI for Social Good.” In: *NeurIPS 2023 Workshop on Computational Sustainability: Promises and Pitfalls from Theory to Deployment*. 2023. URL: <https://openreview.net/forum?id=vjwYYIA8Pj>.
- [61] Tommaso Caselli, Roberto Cibir, Costanza Conforti, Enrique Encinas, and Maurizio Teli. “Guiding Principles for Participatory Design-inspired Natural Language Processing.” In: *Proceedings of the 1st Workshop on NLP for Positive Impact*. Ed. by Anjalie Field, Shrimai Prabhumoye, Maarten Sap, Zhijing Jin, Jieyu Zhao, and Chris Brockett. Online: Association for Computational Linguistics, Aug. 2021, pp. 27–35. doi: [10.18653/v1/2021.nlp4posimpact-1.4](https://doi.org/10.18653/v1/2021.nlp4posimpact-1.4). URL: <https://aclanthology.org/2021.nlp4posimpact-1.4/>.
- [62] Fernando Delgado, Stephen Yang, Michael Madaio, and Qian Yang. “The Participatory Turn in AI Design: Theoretical Foundations and the Current State of Practice.” In: *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*. EAAMO ’23. Boston, MA, USA: Association for Computing Machinery, 2023. ISBN: 9798400703812. doi: [10.1145/3617694.3623261](https://doi.org/10.1145/3617694.3623261). URL: <https://doi.org/10.1145/3617694.3623261>.
- [63] Paul Röttger, Bertie Vidgen, Dirk Hovy, and Janet Pierrehumbert. “Two Contrasting Data Annotation Paradigms for Subjective NLP Tasks.” In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Ed. by Marine Carpuat, Marie-Catherine de Marneffe, and Ivan Vladimir

- Meza Ruiz. Seattle, United States: Association for Computational Linguistics, July 2022, pp. 175–190. doi: [10.18653/v1/2022.naacl-main.13](https://doi.org/10.18653/v1/2022.naacl-main.13). URL: <https://aclanthology.org/2022.naacl-main.13/>.
- [64] Gerd Bohner. “Writing about rape: Use of the passive voice and other distancing text features as an expression of perceived responsibility of the victim.” In: *British Journal of Social Psychology* 40.4 (2001), pp. 515–529. doi: <https://doi.org/10.1348/014466601164957>. eprint: <https://bpspsychub.onlinelibrary.wiley.com/doi/pdf/10.1348/014466601164957>. URL: <https://bpspsychub.onlinelibrary.wiley.com/doi/abs/10.1348/014466601164957>.
- [65] Christopher Hart. “Event-Frames Affect Blame Assignment and Perception of Aggression in Discourse on Political Protests: An Experimental Case Study in Critical Discourse Analysis.” In: *Applied Linguistics* 39.3 (June 2018), pp. 400–421. ISSN: 0142-6001. doi: [10.1093/applin/amw017](https://doi.org/10.1093/applin/amw017). eprint: <https://academic.oup.com/applij/article-pdf/39/3/400/25011650/amw017.pdf>. URL: <https://doi.org/10.1093/applin/amw017>.
- [66] Teun A. van Dijk. *News as Discourse*. 1st ed. Hillsdale, NJ: Routledge, 1988. ISBN: 9780805808285. doi: [10.4324/9780203062784](https://doi.org/10.4324/9780203062784). URL: <https://doi.org/10.4324/9780203062784>.
- [67] Gosse Minnema, Sara Gemelli, Chiara Zanchi, Viviana Patti, Tommaso Caselli, and Malvina Nissim. “Frame Semantics for Social NLP in Italian: Analyzing Responsibility Framing in Femicide News Reports.” In: *Proceedings of the Eighth Italian Conference on Computational Linguistics (CLiC-it 2021)*. Ed. by Elisabetta Fersini, Marco Passarotti, and Viviana Patti. Milan, Italy: CEUR Workshop Proceedings, June 2021, pp. 237–244. ISBN: 979-12-80136-94-7. URL: <https://aclanthology.org/2021.clicit-1.37/>.
- [68] Leonard Talmy. *Toward a Cognitive Semantics, Volume 1: Concept Structuring Systems*. The MIT Press, Sept. 2000. ISBN: 9780262284660. doi: [10.7551/mitpress/6847.001.0001](https://doi.org/10.7551/mitpress/6847.001.0001). URL: <https://doi.org/10.7551/mitpress/6847.001.0001>.
- [69] Ke Li and Qiang Zhang. “A Corpus-Based Study of Representation of Islam and Muslims in American Media: Critical Discourse Analysis Approach.” In: *International Communication Gazette* 84.2 (2022), pp. 157–180. doi: [10.1177/1748048520987440](https://doi.org/10.1177/1748048520987440). URL: <https://journals.sagepub.com/doi/10.1177/1748048520987440>.
- [70] Bonnie Stabile, Aubrey Grant, Hemant Purohit, and Kelsey Harris. “Sex, Lies, and Stereotypes: Gendered Implications of Fake News for Women in Politics.” In: *Public Integrity* 21.5 (2019), pp. 491–502. doi: [10.1080/10999922.2019.1626695](https://doi.org/10.1080/10999922.2019.1626695). URL: <https://doi.org/10.1080/10999922.2019.1626695>.
- [71] Klaus Krippendorff. *Content Analysis: An Introduction to Its Methodology*. 2nd ed. Thousand Oaks, CA: Sage Publications, 2004. ISBN: 9780761915447.
- [72] Hongwei Wang and Dong Yu. “Going Beyond Sentence Embeddings: A Token-Level Matching Algorithm for Calculating Semantic Textual Similarity.” In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Ed. by Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 563–570. doi: [10.18653/v1/2023.acl-short.49](https://doi.org/10.18653/v1/2023.acl-short.49). URL: <https://aclanthology.org/2023.acl-short.49/>.
- [73] Claire Kramsch. “Language and Culture.” In: *AILA Review* 27.1 (2014), pp. 30–55. doi: [10.1075/aila.27.02kra](https://doi.org/10.1075/aila.27.02kra). URL: <https://www.jbe-platform.com/content/journals/10.1075/aila.27.02kra>.

- [74] Tianyi Zhang*, Varsha Kishore*, Felix Wu*, Kilian Q. Weinberger, and Yoav Artzi. “BERTScore: Evaluating Text Generation with BERT.” In: *International Conference on Learning Representations*. 2020. URL: <https://openreview.net/forum?id=SkeHuCVFDr>.
- [75] Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. “M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation.” In: *Findings of the Association for Computational Linguistics: ACL 2024*. Ed. by Lun-Wei Ku, Andre Martins, and Vivek Srikumar. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 2318–2335. DOI: [10.18653/v1/2024.findings-acl.137](https://doi.org/10.18653/v1/2024.findings-acl.137). URL: <https://aclanthology.org/2024.findings-acl.137/>.
- [76] Nils Reimers and Iryna Gurevych. “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.” In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Ed. by Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan. Hong Kong, China: Association for Computational Linguistics, Nov. 2019, pp. 3982–3992. DOI: [10.18653/v1/D19-1410](https://doi.org/10.18653/v1/D19-1410). URL: <https://aclanthology.org/D19-1410/>.
- [77] Kabir Ahuja, Harshita Diddee, Rishav Hada, Millicent Ochieng, Krithika Ramesh, Prachi Jain, Akshay Nambi, Tanuja Ganu, Sameer Segal, Mohamed Ahmed, Kalika Bali, and Sunayana Sitaram. “MEGA: Multilingual Evaluation of Generative AI.” In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 4232–4267. DOI: [10.18653/v1/2023.emnlp-main.258](https://doi.org/10.18653/v1/2023.emnlp-main.258). URL: <https://aclanthology.org/2023.emnlp-main.258/>.
- [78] Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. “Fantastically Ordered Prompts and Where to Find Them: Overcoming Few-Shot Prompt Order Sensitivity.” In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Smaranda Muresan, Preslav Nakov, and Aline Villavicencio. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 8086–8098. DOI: [10.18653/v1/2022.acl-long.556](https://doi.org/10.18653/v1/2022.acl-long.556). URL: <https://aclanthology.org/2022.acl-long.556/>.
- [79] Yiming Zhang, Shi Feng, and Chenhao Tan. “Active Example Selection for In-Context Learning.” In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Ed. by Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, Dec. 2022, pp. 9134–9148. DOI: [10.18653/v1/2022.emnlp-main.622](https://doi.org/10.18653/v1/2022.emnlp-main.622). URL: <https://aclanthology.org/2022.emnlp-main.622/>.
- [80] Lean Wang, Lei Li, Damai Dai, Deli Chen, Hao Zhou, Fandong Meng, Jie Zhou, and Xu Sun. “Label Words are Anchors: An Information Flow Perspective for Understanding In-Context Learning.” In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 9840–9855. DOI: [10.18653/v1/2023.emnlp-main.609](https://doi.org/10.18653/v1/2023.emnlp-main.609). URL: <https://aclanthology.org/2023.emnlp-main.609/>.
- [81] Niklas Muennighoff, Thomas Wang, Lintang Sutawika, Adam Roberts, Stella Biderman, Teven Le Scao, M Saiful Bari, Sheng Shen, Zheng Xin Yong, Hailey Schoelkopf, Xiangru Tang, Dragomir Radev, Alham Fikri Aji, Khalid Almubarak, Samuel Albanie, Zaid Alyafeai,

- Albert Webson, Edward Raff, and Colin Raffel. “Crosslingual Generalization through Multitask Finetuning.” In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 15991–16111. doi: [10.18653/v1/2023.acl-long.891](https://doi.org/10.18653/v1/2023.acl-long.891). URL: <https://aclanthology.org/2023.acl-long.891/>.
- [82] Meta AI. *The Llama 4 Herd: The Beginning of a New Era of Natively Multimodal AI Innovation*. Blog post. Apr. 2025. URL: <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>.
- [83] Mistral AI. *Cheaper, Better, Faster, Stronger: Mixtral 8x22B*. Blog post. Apr. 2024. URL: <https://mistral.ai/news/mixtral-8x22b>.